



**POLITECNICO**  
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE  
E DELL'INFORMAZIONE

# A Comparative Review of Electricity Load Forecasting Methods Across Temporal Horizons

TESI DI LAUREA MAGISTRALE IN  
ELECTRICAL ENGINEERING  
INGEGNERIA ELETTRICA

Author: **Zahra Saeedian**

Student ID: 222206  
Advisor: Samuele Grillo  
Co-advisor: Daniele Linaro  
Academic Year: 2024-25



# Abstract

Accurate electricity load forecasting is a critical component of modern energy system planning and operation, particularly in the context of growing electrification, renewable energy integration, and demand-side flexibility. This thesis presents a structured review of forecasting methods across three key time horizons: short-term (up to 1 week), medium-term (1 week to 1 month), and long-term (beyond 1 month). Each category is examined with respect to its methodological strengths, limitations, and data requirements.

Statistical models such as multiple linear regression and SARIMA are found to be effective for short-term forecasting due to their interpretability and low data demands. Machine learning approaches—including support vector regression and tree-based ensembles—offer improved flexibility for medium-term predictions. For long-term forecasting, deep learning models such as LSTM and Transformer-based architectures demonstrate superior performance in capturing complex temporal patterns. Hybrid models, combining elements from fuzzy systems, optimization techniques, and deep learning, achieve the highest accuracy but require significant data and computational resources.

The thesis introduces a classification framework that aligns forecasting methods with time horizons and data availability scales—from household-level to national-level applications. Visual guides are provided to support model selection based on forecasting objectives and data conditions. The study concludes that no single model is universally optimal; effective forecasting requires context-sensitive method selection, balancing accuracy, interpretability, and resource constraints.

**Key-words:** Load Forecasting, Electricity Demand, Machine Learning, Deep Learning, Hybrid Models.



## Abstract in italiano

Una previsione accurata del carico elettrico è un elemento fondamentale nella pianificazione e nella gestione dei sistemi energetici moderni, soprattutto alla luce dell'elettrificazione crescente, dell'integrazione delle fonti rinnovabili e della flessibilità della domanda. Questa tesi propone una revisione strutturata dei principali metodi di previsione del carico elettrico, suddivisi in tre orizzonti temporali: breve termine (fino a 1 settimana), medio termine (da 1 settimana a 1 mese) e lungo termine (oltre 1 mese). Ogni categoria viene analizzata in relazione ai punti di forza, ai limiti metodologici e ai requisiti di dati.

I modelli statistici, come la regressione lineare multipla e SARIMA, risultano adatti alla previsione a breve termine per la loro semplicità e bassa esigenza di dati. Le tecniche di machine learning, tra cui la regressione con support vectors o alberi decisionali, offrono maggiore flessibilità per previsioni a medio termine. Per previsioni a lungo termine, i modelli di deep learning come LSTM e le architetture basate su Transformer si distinguono per la capacità di modellare sequenze complesse nel tempo. I modelli ibridi—che combinano sistemi fuzzy, algoritmi di ottimizzazione e deep learning—raggiungono la massima accuratezza, ma richiedono una disponibilità significativa di dati e risorse computazionali.

La tesi propone un quadro di classificazione che collega i metodi previsionali agli orizzonti temporali e ai livelli di disponibilità dei dati, da applicazioni domestiche fino a quelle a scala nazionale. Vengono forniti strumenti visivi per supportare la scelta del modello in base agli obiettivi previsionali e alle condizioni di disponibilità dei dati. Si conclude che non esiste un modello universalmente migliore: una previsione efficace richiede la scelta di metodi sensibili al contesto, che bilancino precisione, interpretabilità e risorse disponibili.

**Parole chiave:** Previsione del Carico, Domanda di Elettricità, Apprendimento Automatico, Deep Learning, Modelli Ibridi.



# Contents

|                                                          |            |
|----------------------------------------------------------|------------|
| <b>Abstract</b> .....                                    | <b>i</b>   |
| <b>Abstract in italiano</b> .....                        | <b>iii</b> |
| <b>Contents</b> .....                                    | <b>v</b>   |
| <b>1 Introduction</b> .....                              | <b>1</b>   |
| 1.1. Importance of load forecasting.....                 | 2          |
| 1.2. Objectives and scope of the thesis.....             | 2          |
| 1.3. Structure of the thesis .....                       | 3          |
| <b>2 Load Forecasting</b> .....                          | <b>4</b>   |
| 2.1. Time-based categorization of load forecasting ..... | 4          |
| 2.1.1. Short-term load forecasting .....                 | 4          |
| 2.1.2. Medium-term load forecasting.....                 | 5          |
| 2.1.3. Long-term load forecasting.....                   | 5          |
| 2.2. Factors affecting load forecasting .....            | 5          |
| 2.2.1. Weather and climate .....                         | 6          |
| 2.2.2. Building and appliance attributes.....            | 6          |
| 2.2.3. Sociodemographic and economic factors .....       | 7          |
| 2.2.4. Technological factors and new appliances .....    | 8          |
| 2.2.5. Behavioral and social factors .....               | 9          |
| 2.3. Challenges of load forecasting .....                | 10         |
| 2.3.1. Data quality and availability .....               | 10         |
| 2.3.2. Model accuracy and complexity .....               | 11         |
| 2.3.3. Uncertainty.....                                  | 13         |
| 2.3.4. Consumer variability .....                        | 14         |
| 2.4. Summary .....                                       | 14         |
| <b>3 Short-term load forecasting</b> .....               | <b>15</b>  |
| 3.1. Statistical models.....                             | 15         |
| 3.1.1. Regression-based methods .....                    | 15         |
| 3.1.2. ARIMA and SARIMA .....                            | 18         |
| 3.2. ML models.....                                      | 19         |
| 3.2.1. Support vector regression .....                   | 20         |

|          |                                           |           |
|----------|-------------------------------------------|-----------|
| 3.2.2.   | Tree-based models.....                    | 22        |
| 3.2.3.   | k-nearest neighbors .....                 | 26        |
| 3.2.4.   | Artificial neural networks .....          | 28        |
| 3.3.     | Deep learning models .....                | 31        |
| 3.3.1.   | Recurrent neural networks.....            | 31        |
| 3.3.2.   | Transformer-based models .....            | 34        |
| 3.4.     | Hybrid models .....                       | 36        |
| 3.4.1.   | Statistical & ML hybrids .....            | 36        |
| 3.4.2.   | Ensemble tree-based methods .....         | 37        |
| 3.4.3.   | Neural network-based hybrids .....        | 38        |
| 3.4.4.   | Intelligent and fuzzy logic hybrids ..... | 40        |
| 3.4.5.   | Transformer-based hybrids.....            | 40        |
| 3.5.     | Summary.....                              | 43        |
| <b>4</b> | <b>Medium-term load forecasting .....</b> | <b>45</b> |
| 4.1.     | Statistical methods.....                  | 45        |
| 4.1.1.   | Regression-based methods.....             | 46        |
| 4.1.2.   | ARIMA and SARIMA .....                    | 48        |
| 4.2.     | ML models.....                            | 49        |
| 4.2.1.   | Tree-based models.....                    | 50        |
| 4.2.2.   | k-NN.....                                 | 55        |
| 4.2.3.   | ANN .....                                 | 58        |
| 4.3.     | Deep learning models .....                | 60        |
| 4.3.1.   | RNN.....                                  | 60        |
| 4.3.2.   | Transformer-based models .....            | 62        |
| 4.4.     | Hybrid models .....                       | 64        |
| 4.4.1.   | Statistical-deep learning hybrids .....   | 65        |
| 4.4.2.   | Transformer-based hybrids.....            | 66        |
| 4.4.3.   | CNN-LSTM hybrids .....                    | 66        |
| 4.4.4.   | Signal decomposition-based hybrids .....  | 67        |
| 4.4.5.   | Neural Prophet-LSTM hybrids .....         | 68        |
| 4.5.     | Summary.....                              | 70        |
| <b>5</b> | <b>Long-term load forecasting.....</b>    | <b>72</b> |
| 5.1.     | Statistical methods.....                  | 72        |
| 5.1.1.   | Regression-based models .....             | 73        |
| 5.1.2.   | ARIMA and SARIMA .....                    | 74        |
| 5.2.     | Machine learning methods.....             | 76        |
| 5.2.1.   | SVR .....                                 | 76        |



|          |                                                                    |            |
|----------|--------------------------------------------------------------------|------------|
| 5.2.2.   | Tree-based models .....                                            | 78         |
| 5.2.3.   | ANN .....                                                          | 79         |
| 5.3.     | Deep learning.....                                                 | 80         |
| 5.3.1.   | Multilayer perceptron.....                                         | 81         |
| 5.3.2.   | LSTM.....                                                          | 82         |
| 5.3.3.   | Transformer-based models .....                                     | 84         |
| 5.4.     | Hybrid models.....                                                 | 86         |
| 5.4.1.   | Fuzzy-Neural Network-Based Hybrid Models .....                     | 86         |
| 5.4.2.   | Fuzzy-RNN-based models.....                                        | 87         |
| 5.4.3.   | Fuzzy-SVM and grey system hybrid models.....                       | 88         |
| 5.4.4.   | LSTM-based hybrid deep learning models .....                       | 89         |
| 5.4.5.   | Transformer-based hybrid deep learning models.....                 | 89         |
| 5.5.     | Summary .....                                                      | 91         |
| <b>6</b> | <b>Conclusion and future research directions .....</b>             | <b>92</b>  |
| 6.1.     | Recapitulation of research objectives and contributions .....      | 92         |
| 6.2.     | Strategic alignment between forecasting horizons and methods ..... | 94         |
| 6.3.     | Method suitability based on data scale .....                       | 94         |
| 6.4.     | Implications for practice and policy .....                         | 95         |
| 6.5.     | Limitations and future research directions .....                   | 95         |
| 6.6.     | Final remarks .....                                                | 96         |
|          | <b>Bibliography .....</b>                                          | <b>97</b>  |
|          | <b>List of Figures .....</b>                                       | <b>105</b> |
|          | <b>List of Tables .....</b>                                        | <b>107</b> |
|          | <b>Acknowledgments.....</b>                                        | <b>109</b> |



# 1 Introduction

Electricity is an essential part of modern life, powering homes, industries, and transportation systems. As Figure 1.1 shows, among all energy consumers, the residential sector represents a significant share of total electricity demand. Households consume energy for heating, cooling, lighting, cooking, and running electrical appliances, with consumption patterns varying based on factors such as weather conditions, socio-economic status, building structure, and consumer behavior [1]. In 2022, residential electricity consumption accounted for 25.8% of final energy consumption in the European Union, reflecting its importance in overall power system management [2].

Understanding the dynamics of residential electricity consumption is essential for ensuring an efficient and stable power supply. Electricity demand fluctuates due to daily and seasonal variations, economic conditions, and external influences such as policy changes, technological advancements, and renewable energy integration [3]. Accurately forecasting electricity demand is therefore crucial for power system operation, enabling utilities and policymakers to make informed decisions on grid stability, energy pricing, and infrastructure investments.

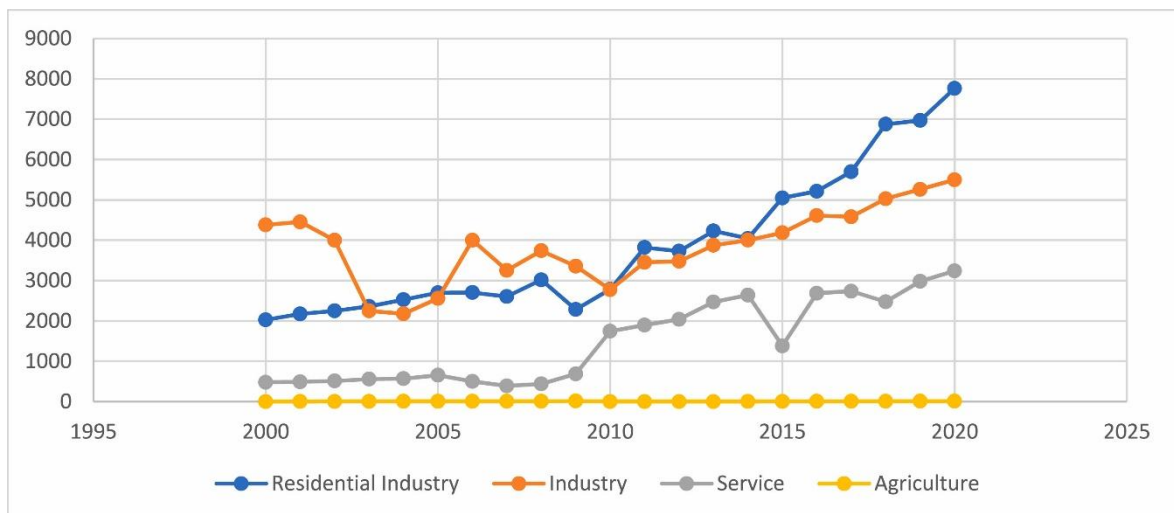


Figure 1.1: Electricity consumption by sectors (GWh) [1].

## 1.1. Importance of load forecasting

Global electricity demand continues to surge, even as traditional fossil fuel reserves become increasingly difficult to extract and deplete. Meanwhile, renewable energy technologies remain in early adoption stages, exacerbating the energy crisis. According to the International Energy Agency (IEA), accurate load forecasting is essential for effective energy planning, power dispatch, and achieving Sustainable Development Goals (SDG-7 and SDG-13). Benefits of electricity prediction is mentioned in Figure 1.2. Precise forecasts not only enable utilities to optimize generation schedules and control operational costs but also enhance grid stability by preventing overloads and blackouts [4], [5].

The integration of smart grids within smart cities further underscores the importance of advanced forecasting. Smart grids leverage digital communication technologies to enable real-time monitoring and bidirectional data exchange between consumers and utilities, thereby optimizing energy consumption and facilitating demand-side management. This data-driven approach not only supports the efficient deployment of renewable energy sources and clean transportation options but also informs strategic urban planning and resource allocation. By enabling both short-term and long-term predictions, these systems empower policymakers to make informed decisions that enhance urban sustainability and resilience [6].



Figure 1.2: Benefits of electricity consumption prediction [4]

## 1.2. Objectives and scope of the thesis

The objectives and scope of this thesis are centered on a comprehensive examination of contemporary load forecasting techniques across various time horizons, including short-term, medium-term, and long-term forecasts. This study aims to systematically classify and assess a wide range of forecasting models, encompassing traditional statistical approaches, machine learning algorithms, deep learning architectures, and hybrid methodologies. Particular attention is given to evaluating the effectiveness, advantages, and limitations of these techniques in accurately predicting electricity demand.

Rather than focusing on empirical data collection, preprocessing steps, or specific case studies, the thesis adopts a conceptual and methodological perspective. It investigates the key factors that influence power consumption and analyzes how these variables impact the precision and reliability of forecasting models.

Additionally, the work critically engages with the ongoing challenges and recent developments in the field, highlighting areas where existing methods may fall short and where improvements are most needed. By identifying current research gaps and future directions, the thesis contributes to the broader discourse on how forecasting can evolve to meet the demands of modern energy systems.

Ultimately, this research is intended to offer valuable insights for scholars, industry professionals, and policymakers, helping them to make informed decisions when selecting, designing, or applying forecasting tools within the context of energy planning and management.

### 1.3. Structure of the thesis

This thesis is structured into six chapters, each building a comprehensive understanding of load forecasting methods across varying time horizons. Chapter 2 provides an overview of the fundamental concepts of load forecasting, including the key variables that influence electricity demand, and the principal challenges associated with forecasting load behavior.

Chapter 3 delves into Short-Term Load Forecasting (STLF), presenting a range of methodologies such as traditional statistical techniques, machine learning algorithms, and deep learning models. Chapter 4 shifts focus to Medium-Term Load Forecasting (MTLF), analyzing models designed to predict energy demand over intervals ranging from days to several months, and discussing strategies employed to improve forecast accuracy. In Chapter 5, the thesis explores Long-Term Load Forecasting (LTLF), covering both econometric approaches and artificial intelligence-based techniques used to project demand over extended time frames.

Finally, Chapter **Error! Reference source not found.** offers a comparative assessment of the models presented in earlier chapters, evaluating their relative strengths, limitations, and suitability for different forecasting horizons. This chapter also reviews recent advancements in the field, outlines emerging research directions, and concludes with key insights and recommendations aimed at enhancing forecasting methodologies.

By offering a structured and critical analysis of load forecasting techniques across short-, medium-, and long-term horizons, this thesis seeks to contribute to the development of more accurate, adaptable, and robust forecasting solutions for the evolving needs of modern power systems.

## 2 Load Forecasting

Load forecasting refers to the process of predicting future electricity demand based on historical consumption data, weather conditions, economic indicators, and other influencing factors. Accurate forecasting is essential for ensuring grid stability, optimizing power generation, and minimizing operational costs [5].

### 2.1. Time-based categorization of load forecasting

[7] categorized load forecasting into four time horizons: very short term, short-term, medium-term, and long-term. However, for the purposes of this thesis, load forecasting is categorized into three primary time-horizons, shown in Figure 2.1, based on forecasting needs and the duration of the prediction.

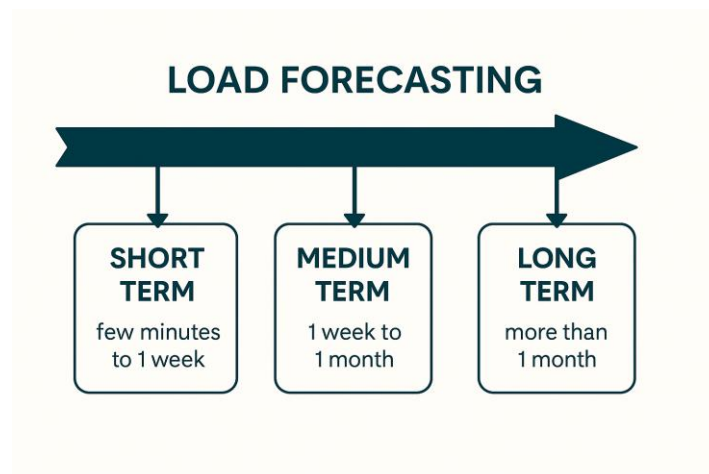


Figure 2.1: Load forecasting time horizontal.

#### 2.1.1. Short-term load forecasting

Short-term load forecasting predicts electricity demand for a period ranging from a few minutes to one week. This time frame is crucial for operational decisions, such as power generation scheduling, unit commitment, load switching, and demand-side management. By providing accurate forecasts for such short time frames, utilities can optimize resource utilization, reduce operational costs, and maintain grid stability [7].

### 2.1.2. Medium-term load forecasting

Medium-term load forecasting covers periods from one week to one month. It is useful for tasks like outage scheduling, maintenance planning, and tariff regulation. MTLF helps utilities plan for intermediate-term fluctuations in electricity demand, ensuring they are prepared for variations during high and low consumption periods [7].

### 2.1.3. Long-term load forecasting

Long-term load forecasting focuses on predicting electricity demand for periods extending beyond one month. These forecasts are essential for strategic planning, energy infrastructure development, and policymaking. Long-term predictions are used to anticipate future power needs, ensuring that utilities can make informed decisions regarding grid capacity expansion, resource allocation, and investment in renewable energy [7].

## 2.2. Factors affecting load forecasting

In this section, we explore the various factors that significantly influence electricity consumption and load forecasting. Residential electricity consumption (REC) is shaped by a combination of socioeconomic indicators, building characteristics, appliance ownership, and usage patterns which are shown in

Figure 2.2. Climate and location also play crucial roles, as heating and cooling requirements are highly sensitive to temperature fluctuations, making city-scale energy usage particularly responsive to weather conditions [8]. Additionally, factors such as household size, income, energy efficiency measures, and technological progress contribute to demand variability. As electricity generation and consumption have notable environmental and social costs, understanding these drivers is vital for developing effective policies and strategies for sustainable energy use.

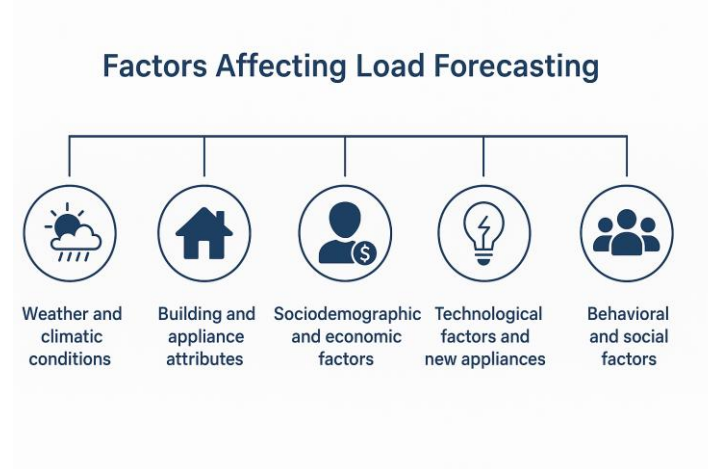


Figure 2.2: Key factors affecting load forecasting.

At the broader national level, REC is the aggregate of individual household decisions, influenced by demographic changes, consumer preferences, and economic factors. The increasing complexity of REC forecasting arises from shifting societal trends, urbanization, the emergence of new appliances, and unpredictable human behavior, all of which pose challenges for predictive models. Despite advancements, there are still obstacles due to inaccurate data, evolving consumption patterns, and unforeseen events such as public health crises [9]. Therefore, accurate load forecasting must account for both technological and social factors that impact energy use across different scales and regions [4], [10].

### 2.2.1. Weather and climate

Weather and climate are crucial factors influencing electricity consumption, particularly in areas where heating and cooling demands vary with seasonal changes. Temperature, humidity, and other weather variables have a significant impact on electricity usage, especially in residential and commercial sectors. During extreme weather conditions, such as very hot or cold temperatures, energy consumption tends to rise due to increased use of air conditioning or heating systems. For instance, city-scale electricity usage is highly temperature-sensitive, with heating and cooling needs contributing significantly to total energy consumption. Seasonal variations further lead to fluctuations in demand, with higher electricity usage during summer and winter months.

Humidity also plays a key role in influencing electricity demand, particularly in hot climates. The heat index, which reflects human perception of temperature, is determined by combining air temperature and relative humidity. When relative humidity is high, the evaporation process in the human body slows down, causing the body to retain more heat. As a result, the perceived temperature increases, leading to higher energy consumption for cooling. Therefore, areas with high humidity, especially during the summer, experience a marked increase in energy demand due to the added stress on cooling devices.

The growing impact of climate change has led to more frequent and extreme weather events, further complicating traditional forecasting models. As weather patterns become more unpredictable, advanced forecasting models are needed to ensure reliable predictions. Additionally, the increasing integration of renewable energy sources such as solar and wind power, which are intermittent and weather-dependent, introduces further uncertainty into load demand. This requires more sophisticated forecasting methods to integrate renewable energy efficiently into the grid while maintaining a stable supply and demand balance [4], [8], [10], [11].

### 2.2.2. Building and appliance attributes

The type of building and the appliances within it are significant factors that influence electricity consumption. Buildings with poor insulation, older electrical systems, or



inefficient appliances tend to have higher energy demand compared to those that are well-insulated and equipped with modern, energy-efficient systems. The design and size of a building also play a role in determining how much energy is consumed. For example, larger homes or buildings typically require more electricity for heating, cooling, and lighting, while energy-efficient designs and materials can reduce overall demand. Determining the influence of occupancy, dwelling size, and building topologies on consumption is equally important in building accurate predictive models for load forecasting [3].

Appliance usage patterns are another critical factor in load forecasting. Household appliances, such as air conditioners, refrigerators, washing machines, and lighting systems, directly impact the total electricity demand. The ownership and frequency of use of these appliances are often linked to lifestyle choices, income levels, and regional factors. In warmer climates, the widespread use of air conditioning systems significantly increases electricity consumption, especially during peak summer months.

The increasing adoption of energy-efficient appliances and building technologies can mitigate some of the demand pressure on the grid. Energy-efficient appliances consume less power and are becoming more prevalent in households and commercial establishments due to technological advancements and incentives. These innovations help lower the overall electricity consumption, especially during peak times, contributing to more stable load profiles and easing the pressure on electricity grids. By considering the type, usage, and efficiency of buildings and appliances, load forecasting models can provide more accurate predictions and facilitate better energy management [8], [9].

### 2.2.3. Sociodemographic and economic factors

Sociodemographic and economic factors play a crucial role in shaping electricity consumption patterns. Key elements such as income, household size, education, and urbanization influence energy demand significantly. Higher-income households tend to consume more electricity due to increased usage of electrical appliances, heating and cooling systems, and other electronic devices. Similarly, larger households require more energy for daily activities, leading to higher electricity consumption compared to smaller households. Additionally, lifestyle factors, such as the number of residents working from home, can significantly impact consumption levels, particularly during certain times of the day [8].

Economic conditions also affect electricity demand. During periods of economic growth, electricity consumption rises due to increased industrial activity and higher consumer spending. Conversely, during economic downturns, demand tends to fall as businesses scale back operations and households adopt more energy-efficient practices to save on costs. Furthermore, population growth directly influences total electricity consumption, requiring utilities to adjust their forecasts accordingly. Projections show

that countries and regions experiencing rapid urbanization, such as India, China, and Southeast Asia, will see substantial increases in electricity demand over the next decade, with India's demand increasing by 40%, China's by 25%, and Southeast Asia's by 30% from 2019 to 2030 [10].

Urbanization is another significant factor. As urban populations grow, electricity demand increases due to the higher number of households, commercial buildings, and transportation needs. In urban areas, electricity consumption is heavily driven by the need for lighting, heating, cooling, and powering appliances in both residential and commercial spaces. This leads to higher overall demand for electricity in cities [9].

Moreover, sociodemographic factors such as the dependency ratio—the proportion of non-working individuals (children and elderly) to the working-age population—also affect electricity consumption. Households with high dependency ratios tend to allocate more income toward basic needs, including electricity, which may reduce their savings potential and increase overall demand for electricity [1].

In conclusion, sociodemographic and economic factors, including household characteristics, urbanization, economic growth, and technological advancements, significantly influence electricity consumption. Understanding these factors is essential for developing accurate forecasting models, which help optimize energy supply management and design targeted demand response strategies.

#### 2.2.4. Technological factors and new appliances

Technological advancements and behavioral shifts, such as the widespread adoption of electric vehicles (EVs) and smart devices, have introduced new patterns in electricity demand. EVs require substantial charging infrastructure, which can shift demand peaks and increase load during specific times of the day. Similarly, the integration of distributed renewable energy systems, such as rooftop solar panels, is reshaping traditional energy usage models. While these innovations promote energy efficiency and enable partial self-generation, they can also introduce new demand peaks, particularly when EVs are charged simultaneously during off-peak hours [5], [9].

A particularly notable challenge is the substantial increase in electricity demand resulting from widespread EV adoption. This trend poses a serious burden on power grids, especially residential distribution networks. For example, around 80% of EVs are charged using residential in-home chargers, meaning their charging habits often align with household consumption peaks [8]. As electrification expands, U.S. electricity demand is projected to rise by 23% by 2050, with EVs accounting for up to 300 TWh of consumption. Moreover, if EV penetration reaches 30%, it could result in a 50% increase in peak load at the residential distribution transformer level, significantly stressing local grid infrastructure [12].

In parallel, the integration of smart grids and Internet of Things (IoT) technologies has enhanced load forecasting accuracy. Smart meters and real-time monitoring systems

provide granular insights into consumer behavior, enabling more adaptive and dynamic energy management strategies [7].

Additionally, energy efficiency improvements, such as more efficient appliances, the transition to LED lighting, and enhanced building insulation, have led to reductions in per capita electricity use. These developments require regular updates to traditional forecasting models to reflect evolving consumption patterns [5].

Demand-side management strategies, including time-of-use pricing and demand response programs, have also started influencing user behavior. These approaches aim to shift electricity usage away from peak times, flattening demand curves and improving grid stability. However, their effectiveness still depends on consumer engagement, which can vary widely [5].

In summary, while technological advancements introduce opportunities for smarter and more efficient energy use, they also bring significant complexity and uncertainty to forecasting models. The adoption rate, usage patterns, and integration of new technologies like EVs and smart appliances must be carefully considered when planning for future energy needs.

#### 2.2.5. Behavioral and social factors

Human behavior plays a crucial, yet inherently unpredictable role in electricity consumption, introducing significant challenges for accurate load forecasting. Lifestyle changes, daily routines, cultural events such as holidays or festivals, and even social norms can all lead to sudden and irregular fluctuations in electricity demand. For example, increased home occupancy during weekends or national celebrations can significantly raise residential electricity usage, making short-term forecasting more difficult [9].

Moreover, growing awareness around environmental issues and participation in demand-side management (DSM) programs has influenced consumer behavior. These programs encourage users to modify their consumption habits, such as shifting energy use to off-peak hours or reducing overall consumption, adding further variability to usage patterns, and impacting forecast accuracy [9].

On an individual level, electricity consumption is strongly influenced by personal decision-making, which is characterized by its stochastic and dynamic nature. As a result, disaggregated or household-level forecasting models are particularly sensitive to behavioral factors, often leading to higher variability and prediction errors. In contrast, when energy demand is aggregated over a large population, the variability introduced by individual behavior tends to balance out, resulting in relatively smoother and more predictable trends [13].

These behavioral and social elements introduce a complex, human-centric dimension to energy forecasting that cannot be fully captured by technical models alone. As energy systems evolve to include more user-driven dynamics such as flexible

consumption, remote work habits, and real-time energy feedback, the need for models that integrate behavioral insights becomes increasingly important.

## 2.3. Challenges of load forecasting

Despite advancements in forecasting techniques, several challenges remain, as described in the following.

### 2.3.1. Data quality and availability

Accurate electricity consumption forecasting relies heavily on the availability and quality of data. However, several challenges complicate this process, especially when forecasting individual consumption patterns.

Collecting detailed and reliable consumption data can be particularly difficult. Many previous studies have relied on sub-metered data from individual appliances or customer surveys, but obtaining such detailed information on a scale is often impractical. When dealing with a large number of households, gathering daily usage schedules or appliance-specific data becomes infeasible. In addition, concerns over data privacy, particularly regarding personal appliance usage patterns, present further barriers to widespread data collection [13].

Beyond access, the quality of the available data is another critical issue. Energy consumption data sets frequently suffer from missing values, outliers, and inconsistencies caused by sensor errors, manual data entry mistakes, or transmission problems. These irregularities, if not addressed through effective data cleaning and preprocessing, can significantly compromise the performance and reliability of forecasting models [4].

Another challenge arises from the high-dimensional nature of electricity consumption data. Forecasting load accurately involves analyzing multiple interdependent factors, such as weather conditions, economic variables, and consumer behaviors. The complexity and volume of these variables make it difficult to extract meaningful patterns, particularly when individual household differences are considered. Successful forecasting models must be capable of managing such high-dimensional data to identify the underlying trends and relationships [14].

Furthermore, with the increasing deployment of smart grid technologies, the data landscape has become even more complex. Smart grids generate multifaceted datasets collected from a variety of sources, ranging from individual meters to city-wide monitoring systems. While this data offers valuable insights into electricity consumption patterns at different levels of aggregation, its complexity and volume pose significant analytical challenges. Effective integration and analysis of smart grid data are essential for enhancing the accuracy and reliability of consumption forecasting models [6].

### 2.3.2. Model accuracy and complexity

Accurate load forecasting requires selecting an appropriate model that balances forecasting performance with computational efficiency. Electricity demand data often exhibits nonlinear relationships influenced by external factors, posing challenges for both traditional statistical methods and modern machine learning techniques.

Traditional models, such as ARIMA (AutoRegressive Integrated Moving Average) and SARIMA (Seasonal ARIMA), are effective for capturing linear patterns in time-series data. However, their predictive performance declines when dealing with complex nonlinear relationships or sudden behavioral changes caused by factors such as economic fluctuations or weather variability. In contrast, machine learning models like Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs) are better suited for identifying intricate nonlinear patterns. These models significantly improve forecasting accuracy but demand substantial computational resources and large datasets for training, which makes them computationally intensive and time-consuming [6], [9], [15].

To address the limitations of both traditional and advanced deep learning approaches, hybrid models have been developed. These models combine the strengths of statistical methods and artificial intelligence techniques, such as integrating ARIMA with Generative Adversarial Networks (GANs) or LSTM networks. Hybrid approaches have demonstrated improved prediction performance by leveraging both linear and nonlinear modeling capabilities. However, they introduce new challenges related to optimization, interpretability, and computational cost. Training hybrid models is computationally demanding, and interpreting their outputs can be complex, reducing their practicality for real-time forecasting applications [15].

The reliance on deep learning and hybrid models has also raised concerns about computational complexity and resource constraints. Models like XGBoost and Deep Neural Networks (DNNs) require extensive datasets and considerable processing power to achieve high accuracy [6], [9]. Hyperparameter tuning, which is essential for enhancing model performance, further exacerbates computational demands. As a result, deploying these models for real-time or large-scale forecasting in the power systems industry can be challenging. While AI-driven approaches offer substantial improvements in forecasting accuracy, ensuring that these models remain computationally efficient and scalable is crucial for their practical application.

In summary, although deep learning and hybrid models present significant advantages in terms of forecasting accuracy, optimizing them for computational efficiency, and ensuring their feasibility for real-time implementation remain critical challenges.

In load forecasting literature, evaluating the accuracy of predictive models is essential for understanding their effectiveness and guiding their practical application. Model performance is typically quantified using established error metrics that assess how

closely the predicted values align with the actual observed data. These metrics not only allow researchers to compare different forecasting approaches but also reveal how models behave under various conditions and time horizons.

Commonly used accuracy measures include the Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE). Each provides distinct insights: MAE gives the average magnitude of prediction errors, RMSE penalizes larger errors more heavily due to the squaring operation, and MAPE expresses errors as a percentage, which is useful for comparing across different load scales.

Table 2.1 summarizes the most commonly used evaluation metrics in load forecasting research, including their mathematical formulations.

Table 2.1: Common error metrics used in the load forecasting literature.

| Metric Name                           | Formula                                                                                 |
|---------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Mean Absolute Error</b>            | $MAE = \frac{1}{n} \sum_{t=1}^n  y_t - \hat{y}_t $                                      |
| <b>Mean Squared Error</b>             | $MSE = \frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2$                                    |
| <b>Root Mean Squared Error</b>        | $RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}$                            |
| <b>Mean Absolute Percentage Error</b> | $MAPE = \frac{1}{n} \sum_{t=1}^n \left  \frac{y_t - \hat{y}_t}{y_t} \right  \times 100$ |
| <b>Coefficient of Determination</b>   | $R^2 = 1 - \frac{(y_t - \hat{y}_t)^2}{(y_t - \bar{Y})^2}$                               |



**Coefficient of  
Variation of the  
Root Mean Square  
Error**

$$CVRMSE = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y})^2}}{\bar{Y}} \times 100$$

### 2.3.3. Uncertainty

Load forecasting is inherently uncertain due to the wide range of unpredictable factors influencing electricity demand. These uncertainties complicate accurate prediction and must be carefully considered to ensure reliable power system operation.

Unexpected public health events and behavioral shifts are major sources of forecasting uncertainty. For example, during the COVID-19 pandemic, electricity consumption patterns changed significantly. Countries with strict lockdown measures, such as Spain and Italy, experienced electricity usage profiles resembling pre-pandemic weekends, whereas countries like Sweden, with fewer restrictions, observed smaller decreases in consumption [4]. Such events highlight how historical consumption data may become unreliable for forecasting future demand.

Climate change also plays a critical role in altering load patterns. The increasing frequency and intensity of extreme weather events, such as severe heatwaves or cold snaps, can cause sudden peak loads in power systems [4], [16]. These load spikes threaten grid stability by pushing distribution transformers and other equipment to unsafe operational limits. As climate-driven changes make temperature-sensitive demand more volatile, historical trends become less predictive, posing challenges for short-term and peak load forecasting.

The variability introduced by renewable energy sources, particularly solar and wind power, further adds to the uncertainty in load forecasting [5]. Since renewable generation depends heavily on weather conditions, predicting the available electricity supply becomes more complex. To address this, forecasting models must increasingly incorporate probabilistic and scenario-based techniques to manage renewable energy fluctuations and maintain grid reliability.

Economic fluctuations and unexpected power plant outages introduce additional layers of uncertainty [4], [9]. Economic downturns or booms can significantly shift electricity demand, especially in the industrial sector. Similarly, unplanned outages at power plants can abruptly disrupt supply patterns, making load forecasting even more volatile and challenging.

In summary, uncertainty in load forecasting arises from diverse factors including public health crises, climate change, renewable energy variability, economic conditions, and supply disruptions. Addressing these challenges requires the development of advanced forecasting models capable of accounting for such uncertainties, ensuring more resilient and reliable energy system operations.

### 2.3.4. Consumer variability

The growing integration of smart grid technologies and emerging energy systems has introduced new challenges for load forecasting, particularly due to the unpredictability of consumer behavior. Smart meters and demand response programs now provide utilities with greater control over load management, enabling dynamic pricing and real-time adjustments to optimize energy consumption [5]. However, consumer participation in these programs is highly variable, as individual willingness to modify energy use based on price signals or grid demands differs significantly. Factors such as weather conditions, economic circumstances, and personal preferences further influence participation levels, adding uncertainty to forecasting efforts.

Emerging technologies like electric vehicles (EVs), battery storage systems, and distributed energy generation are also reshaping traditional load demand profiles [5], [12]. The widespread adoption of EVs introduces new and less predictable demand patterns, as charging times and energy requirements may not align with historical usage data. Similarly, battery storage and rooftop solar installations allow consumers to both store and inject electricity into the grid, significantly altering net load profiles. These technological developments add further complexity for forecasting models, necessitating continuous adaptation to evolving consumption patterns and the accelerating pace of technological innovation.

## 2.4. Summary

Chapter 2 highlighted the key challenges in electricity load forecasting, focusing on factors like data quality, model complexity, and uncertainty. It discussed how weather, economic changes, and unexpected events contribute to the volatility of load forecasts, especially with the impact of climate change and extreme weather.

Traditional forecasting models struggle with capturing nonlinear relationships, leading to the adoption of advanced machine learning techniques like deep learning and hybrid models. However, these methods come with increased computational demands. The integration of renewable energy sources and emerging technologies such as smart grids, electric vehicles, and battery storage further complicates forecasting.

The chapter also emphasized the importance of accurate short-term load forecasting, particularly during peak load conditions, to ensure the stability of power systems. Smart grid technologies and better metering infrastructure make this more achievable, supporting effective power dispatching and reducing grid stress.

In conclusion, addressing the challenges in load forecasting requires advanced models that balance accuracy and computational efficiency, enabling reliable predictions in an increasingly dynamic environment.



## 3 Short-term load forecasting

In this study, short-term load forecasting (STLF) refers to the prediction of electricity demand over time horizons ranging from a few minutes up to a week. STLF plays a vital role in various aspects of power system operations, including unit commitment, load dispatch, demand response programs, and real-time energy market transactions [7]. Accurate short-term forecasts enable utilities and grid operators to optimize generation schedules, maintain system reliability, and minimize operational costs.

STLF models must account for a wide range of influencing factors such as weather conditions, consumer behavior, public holidays, economic trends, and building characteristics [11], [14]. These variables, combined with the inherently nonlinear and stochastic nature of electricity demand, make the development of precise forecasting models a complex task.

Over the years, STLF techniques have evolved from traditional statistical methods to more sophisticated machine learning and deep learning approaches. More recently, hybrid models that integrate multiple techniques have emerged, offering promising improvements in forecast accuracy and adaptability to dynamic grid environments.

STLF techniques are typically categorized based on their modeling approach and capacity to manage different patterns in electricity demand. The four primary groups include statistical methods, machine learning techniques, deep learning models, and hybrid or ensemble approaches.

### 3.1. Statistical models

Statistical methods, such as ARIMA and SARIMA, rely on historical patterns and linear assumptions. They are valued for their interpretability and simplicity but are limited in capturing nonlinear and complex dependencies. For example, SARIMA has been shown to perform well under stable seasonal conditions but struggles with dynamic or non-stationary inputs [17].

#### 3.1.1. Regression-based methods

Regression-based models have historically formed the backbone of STLF, particularly due to their clarity, computational efficiency, and ability to extract interpretable insights from historical data. These models assume a functional relationship between

input variables (e.g., weather, calendar features) and electricity demand. Several variants exist; each suited to different forecasting contexts.

Linear Regression (LR) is a classical statistical method widely used in STLF, particularly in early applications where simplicity, interpretability, and computational efficiency are critical. In its standard form, linear regression models the relationship between a dependent variable (e.g., electrical load) and one or more independent variables (e.g., temperature, hour of the day, or occupancy) under the assumption of linearity.

The general multiple linear regression model is expressed in Equation (3.1):

$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon, \quad (3.1)$$

where:  $\hat{Y}$  is the dependent variable (e.g., electricity consumption),  $\beta_0$  is the intercept (constant term),  $\beta_1, \beta_2, \dots, \beta_n$  are the coefficients for the independent variables, and  $\varepsilon$  is an error term.

Katić et al. [18] provided quantitative error metrics comparing multiple linear regression (MLR) and artificial neural network (ANN) models for predicting heating energy consumption in school buildings. Their findings indicate that the ANN model outperformed the MLR model in terms of predictive accuracy.

Specifically, the ANN model achieved a lower Mean Absolute Error and RMSE compared to the MLR model. For instance, the ANN model's MAE was approximately 12% lower, and its RMSE was about 15% lower than those of the MLR model. These results suggest that the ANN model was more effective in capturing the complex, nonlinear relationships between building characteristics and heating energy consumption.

The superior performance of the ANN model is attributed to its ability to model intricate patterns and interactions among variables, which are often present in building energy consumption data. In contrast, the MLR model, while simpler and more interpretable, may not adequately capture such complexities, leading to higher prediction errors.

In summary, the study by Katić et al. [18] demonstrates that while MLR models offer simplicity and ease of interpretation, ANN models provide enhanced predictive accuracy for modeling heating energy consumption in school buildings, particularly when dealing with complex and nonlinear data relationships.

Nano [19] enhanced traditional MLR models by integrating calendar-based variables such as weekdays, holidays, and seasonal patterns. This integration aimed to improve the accuracy of daily and weekly load forecasts. While specific error metrics were not detailed, the inclusion of these calendar variables likely contributed to a reduction in forecasting errors by accounting for recurring patterns in electricity consumption.

Dudek [20] introduced a pattern-based local linear regression approach for short-term load forecasting. This method involved clustering daily load profiles and applying separate linear regressions to each cluster, effectively capturing recurring short-term load fluctuations. The study compared this approach with other models, including ARIMA, exponential smoothing, neural networks, and the Nadaraya-Watson estimator. The pattern-based local linear regression demonstrated high accuracy, particularly in modeling daily load cycles, and offered advantages in terms of simplicity and computational efficiency.

Nsangou et al. [21] conducted a comparative analysis of quantile regression, decision tree, and ANN models to analyze household electricity consumption under uncertainty. Their findings indicate that the ANN model outperformed the other two methods, achieving lower values in RMSE, MAE, and Mean Absolute Percentage Error (MAPE), along with a higher coefficient of determination ( $R^2$ ). This superior performance is attributed to the ANN's ability to model complex, nonlinear relationships in the data.

Table 3.1: Strengths and limitations of regression methods in STLF.

| Model Type                     | Strengths                                                      | Weaknesses                                                                  |
|--------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Linear regression</b>       | Simple; interpretable; fast to compute; good as baseline.      | Limited to linear patterns; underperforms in dynamic or nonlinear contexts. |
| <b>Local linear regression</b> | Adapts to temporal variations; better for pattern-based loads. | Requires clustering or segmentation; sensitive to pattern definition.       |
| <b>Quantile regression</b>     | Captures variability and uncertainty; robust to outliers.      | More complex; harder to interpret compared to mean-based methods.           |

Regression models are often integrated into hybrid frameworks or used in ensemble methods, especially when interpretability and speed are needed alongside accuracy. In many real-world STLF systems, as mentioned in Table 3.1, regression serves as a fast, initial approximation before more complex machine learning models are employed.

### 3.1.2. ARIMA and SARIMA

The ARIMA model and its seasonal extension, SARIMA, are widely used statistical methods in STLF. These models are designed to capture time-based dependencies and trends in electricity consumption, making them particularly useful in settings where demand exhibits consistent temporal patterns.

An ARIMA model is defined by three parameters:  $p$ ,  $d$ , and  $q$ .  $p$  is the number of autoregressive terms, meaning it defines how many past values of the time series are used to predict the current value;  $d$  indicates the degree of differencing required to make the series stationary—that is, removing trends or seasonality that distort the forecasting model; finally,  $q$  represents the number of lagged forecast errors included in the model, which captures the influence of past deviations between predicted and actual values.

When electricity demand exhibits seasonal patterns—such as higher consumption during specific hours, days, or months—the SARIMA model extends ARIMA by incorporating seasonal components. This is typically denoted as SARIMA ( $p, d, q$ ) ( $P, D, Q$ ) [ $m$ ], where the uppercase parameters ( $P, D, Q$ ) mirror the ARIMA structure but are applied to the seasonal cycle defined by  $m$  (e.g.,  $m = 12$  for monthly data or  $m = 24$  for hourly data in daily cycles).

Bilgili and Pinar [17] applied SARIMA to forecast Turkey’s gross electricity consumption and compared its performance with LSTM. SARIMA successfully captured recurring seasonal trends and produced reasonable short-term forecasts; however, it struggled with modeling nonlinear and long-range dependencies. In contrast, LSTM significantly outperformed SARIMA across all error metrics. Specifically, LSTM achieved a MAPE of 2.42%, MAE of 215.35 GWh, and RMSE of 329.9 GWh, along with an R-value of 0.9992. These results highlight LSTM’s superior ability to capture complex temporal dynamics, especially during volatile periods, making it a more robust tool for forecasting electricity demand.

Gellert et al. [22] used ARIMA and (Trigonometric, Box-Cox transformation, ARMA errors, Trend, and Seasonal components) TBATS to forecast electricity consumption and PV energy production in smart homes, applying these models to datasets with 5-minute, hourly, and bi-hourly resolutions. ARIMA performed well under stable, structured conditions but struggled with high-frequency variability and irregular generation patterns. TBATS outperformed ARIMA and even neural models in those scenarios, achieving a lower MAE of 73.62 W. Its ability to manage multiple seasonality and adapt to complex fluctuations made TBATS a more suitable option for short-term forecasting in smart home environments.

Table 3.2: Strengths and limitations of ARIMA and SARIMA methods in STLF.

| Model Type    | Strengths                                                                             | Weaknesses                                                                                                |
|---------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>ARIMA</b>  | Captures short-term temporal dependencies, easy to interpret, low computational cost. | Assumes linearity, requires data stationarity, not suited for complex, nonlinear systems.                 |
| <b>SARIMA</b> | Effective models seasonal patterns, widely used in practice.                          | Parameter tuning can be complex, performance declines with high-frequency variability or external shocks. |

In summary, ARIMA and SARIMA have proven to be effective baseline methods for short-term load forecasting when the data exhibits strong linearity and regular seasonal trends. As demonstrated in the studies by Bilgili and Pinar [17] and Gellert et al. [22], these models perform well under stable conditions and structured consumption patterns. SARIMA, in particular, can manage seasonal effects efficiently and offers interpretability with relatively low computational cost. However, both ARIMA and SARIMA show clear limitations in more complex, volatile environments where nonlinear dependencies, high-frequency variability, or long-range temporal relationships are present. In such cases, more advanced models like LSTM or TBATS consistently outperform them, making traditional statistical models less suitable for modern, dynamic energy systems in short-term forecasting applications. A summary of the strengths and limitations of ARIMA and SARIMA in short-term load forecasting is provided in Table 3.2.

### 3.2. ML models

With the increasing complexity of electricity consumption patterns and the availability of large volumes of smart meter data, machine learning (ML) models have emerged as powerful tools for STLF. Unlike traditional statistical methods, ML techniques can capture nonlinear relationships and complex interactions between multiple variables without explicit assumptions about the underlying data distribution.

ML models learn directly from historical data by identifying patterns and dependencies, often using features such as time, temperature, humidity, occupancy, or appliance usage. These models have demonstrated superior accuracy and adaptability in scenarios with high variability or when external factors significantly influence consumption [10], [14].

### 3.2.1. Support vector regression

Support Vector Regression (SVR) is a machine learning technique derived from Support Vector Machines (SVM), widely used in regression tasks involving complex, nonlinear, and high-dimensional data. It is particularly effective in STLF due to its robustness, generalization capability, and adaptability to various data distributions.

The SVR regression function uses a dot product in high-dimensional space via a kernel function. The formal expression is given in Equation (3.2). This formulation has been widely used in load forecasting literature, including recent studies such as Bendaoud et al. [23].

$$f(x) = \sum_{i=1}^m \omega_i \varphi_i(x) + \beta = \omega^T \varphi(x) + \beta, \quad (3.2)$$

Where  $f(x)$  is the predicted output,  $\varphi(x)$  is a nonlinear function mapping the input data into a higher-dimensional space,  $\omega$  is the weight vector,  $\beta$  is the bias term,  $\omega^T \varphi(x)$  represents the dot product between the weight vector and the transformed input features.

This formulation enables SVR to perform nonlinear regression by applying the kernel trick, which replaces the explicit computation of  $\varphi(x)$  with kernel functions such as radial basis function (RBF), polynomial, or linear kernels. The kernel function computes the inner product  $\varphi(x_i)^T \varphi(x_i)$  implicitly, significantly reducing computational complexity.

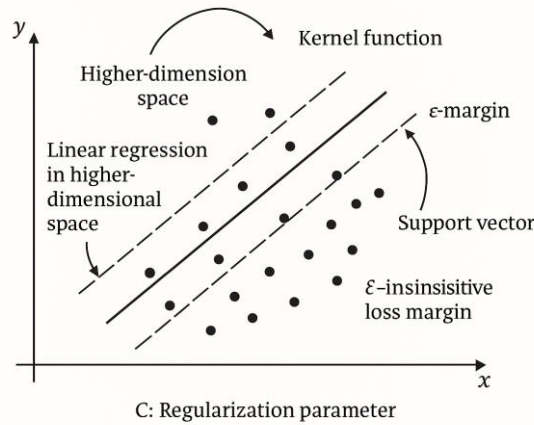


Figure 3.1: SVR model.

As it is shown in Figure 3.1, SVR operates by using the kernel function to map the input data to a higher-dimensional space, where data are fit with a linear regression model while introducing an  $\epsilon$ -insensitive loss margin, within which deviations from the true value are not penalized. This results in a sparse model where only data points lying outside the  $\epsilon$ -margin (called support vectors) influence the solution. The model

is regularized by a parameter  $C$ , which controls the trade-off between model linearity (in the higher dimensional space) and tolerance to error outside the  $\varepsilon$ -margin [23].

Bendaoud et al. [23] conducted a comparative study on short-term load forecasting in Algeria, evaluating SVR alongside ANN, CNN, and Gradient Boosting methods. SVR demonstrated robust performance, outperforming ANN and linear regression models, and achieving a MAPE of 2.17%. Its effectiveness was particularly notable in capturing nonlinear load behaviors during weekends, indicating its suitability for modeling complex consumption patterns in short-term forecasts.

Haque and Rahman [24] compared SVR with shallow ANN and deep learning models like LSTM networks for energy consumption forecasting. While LSTM achieved the lowest forecasting errors, SVR offered a compelling balance between accuracy and computational efficiency. It outperformed traditional models and was less resource-intensive than deep networks, making it a practical choice for scenarios with structured yet moderately complex patterns.

Zhang et al. [25] integrated SVR into an Explainable Artificial Intelligence (XAI) framework to estimate building energy consumption and greenhouse gas emissions. The study demonstrated that SVR could deliver competitive predictive accuracy while supporting partial model interpretability through SHAP (SHapley Additive exPlanations) values. This approach provided insights into the relative importance of different input features, enhancing the transparency of the forecasting model.

Hasan et al. [26] enhanced SVR's predictive capabilities by combining it with Genetic Algorithms (GA) to optimize hyperparameters for energy consumption forecasting across four economic sectors. The GA-optimized SVR model achieved significantly improved performance, showcasing the advantage of integrating heuristic optimization techniques to address challenges posed by multivariate, nonlinear, and large-scale datasets.

These studies collectively underscore SVR's versatility and effectiveness in energy consumption forecasting, particularly when dealing with nonlinear patterns and when computational resources are a consideration. For a detailed overview of SVR's strengths and weaknesses, please refer to Table 3.3.



Table 3.3: Strengths and limitations of SVR method in STLF.

| Model Type | Strengths                                                                                                                                                                                                                                                                       | Weaknesses                                                                                                                                                                                                                                                                                          |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SVR        | Strong performance with nonlinear and high-dimensional datasets; Robust to overfitting in small to medium-sized datasets; Flexible kernel functions support complex feature transformations; Performs well as both a standalone model and within hybrid optimization frameworks | Requires careful tuning of kernel parameters (e.g., $C$ , $\epsilon$ , kernel type); Can be computationally expensive for very large datasets; Less interpretable than simpler models unless combined with explainable AI; May underperform deep models in highly dynamic or long-horizon scenarios |

3.2.2. Tree-based models

Tree-based models have become a central component of modern ML thanks to their ability to model complex nonlinear relationships, capture interactions between variables, and provide interpretable outputs in many cases. In the context of STLF, they offer strong predictive performance while managing structured tabular data, time-based features, and weather inputs with ease. This section discusses three key approaches: Decision Tree Regression, Random Forests, and Extreme Gradient Boosting (XGBoost), each building on the foundational principles of decision trees.

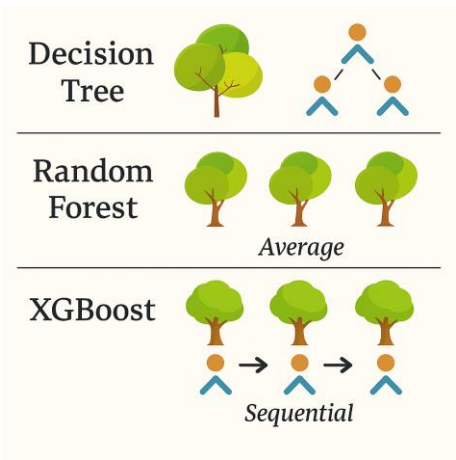


Figure 3.2: Tree-based model.



Figure 3.2 illustration of the structural differences between Decision Tree Regression (DTR), Random Forest (RF), and XGBoost models. In DTR, a single tree is used to generate one prediction. RF builds multiple decision trees in parallel and averages their outputs to produce the final result. XGBoost, on the other hand, builds trees sequentially, where each tree corrects the errors of the previous one, and the final prediction is obtained from the last tree in the sequence.

### 3.2.2.1. Decision tree regression

A DTR model operates by recursively partitioning the input space to reduce variance within each subset. At each internal node, the algorithm selects a feature  $x_j$  and threshold  $t$  that minimizes the sum of squared errors across the two resulting regions. This criterion is expressed mathematically in Equation (3.3):

$$\min_{j,t} \left[ \sum_{x_i \in R_1(j,t)} (y_i - \bar{y}R_1)^2 + \sum_{x_i \in R_2(j,t)} (y_i - \bar{y}R_2)^2 \right], \quad (3.3)$$

Where:  $R_1$  and  $R_2$  are the left and right branches split on feature  $x_j$  at threshold  $t$ ,  $\bar{y}R_k$  is the Mean of the target values in each region.

Peteleaza et al. [6] conducted a comprehensive study on electricity consumption forecasting for sustainable smart cities using machine learning methods. Their research involved a comparative analysis of seven different models, including decision trees, applied to datasets from three metropolitan areas in the United States: Los Angeles, Sacramento, and New York. The decision tree models demonstrated strong interpretability and adaptability to daily and seasonal patterns, making them valuable for understanding energy use behaviors in smart-city contexts. Notably, the study found that Light Gradient Boosting Machine (LightGBM), a decision tree-based ensemble method, provided the best results with a Coefficient of Variation of the Root Mean Square Error (CVRMSE) of 6.5% for Los Angeles, 4.6% for Sacramento, and 4.1% for New York. These findings highlight the effectiveness of decision tree approaches in capturing complex consumption patterns across diverse urban settings.

Nsangou et al. [21] explored the application of decision trees in explaining household electricity consumption in Cameroon. Their study compared the performance of decision trees with quantile regression and ANN in assessing electricity consumption drivers. The decision tree models were instrumental in identifying key factors influencing energy use, such as equipment usage, household income levels, housing structures, and weather conditions. While the ANN model outperformed the others in terms of predictive accuracy, the decision tree approach offered superior interpretability, allowing for a clearer understanding of how various factors contribute to electricity consumption.

These studies underscore the utility of Decision Tree methods in energy consumption forecasting, particularly in scenarios where model interpretability and adaptability to

varying consumption patterns are crucial. For a detailed overview of the strengths and weaknesses of Decision Tree models in this context, please refer to Table 3.4.

### 3.2.2.2. Random Forests

Building upon this concept, Random Forests introduce an ensemble strategy by training multiple decision trees on bootstrapped subsets of the data. Each tree contributes a prediction, and the final result is computed as the average of all individual outputs. This aggregation approach is formalized in Equation (3.4):

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T h_t(x), \quad (3.4)$$

Where  $T$  is the number of decision trees,  $h_t(x)$  is the prediction from the  $t^{th}$  tree.

Rafiq et al. [3] conducted an in-depth analysis of residential electricity consumption patterns in Dubai, utilizing high-resolution smart meter data, weather information, and voluntary survey data on household characteristics. The study employed K-means clustering to group households based on their consumption profiles and applied various classification algorithms, including Random Forest, to predict household consumption patterns based on dwelling and occupant characteristics. The RF model demonstrated strong performance and interpretability, effectively identifying key factors driving consumption patterns, such as cooling systems, number of occupants, and dwelling size. This approach provided valuable insights for demand-side management initiatives and the design of energy-saving strategies in residential sectors.

Peteleaza et al. [6] explored electricity consumption forecasting for sustainable smart cities using machine learning methods. The study compared the performance of various models, including Random Forest, in predicting electricity consumption at the city level. The RF model outperformed individual decision trees, demonstrating higher accuracy and robustness in capturing complex consumption patterns across diverse urban settings. The ensemble nature of RF, which combines multiple decision trees, contributed to its improved generalization and reduced overfitting, making it a suitable choice for load forecasting in smart city applications.

These studies underscore the utility of Random Forest methods in energy consumption forecasting, particularly in scenarios where model interpretability and adaptability to varying consumption patterns are crucial. For a detailed overview of the strengths and weaknesses of Random Forest models in this context, please refer to Table 3.1.

### 3.2.2.3. Extreme gradient boosting

Further extending the ensemble concept, XGBoost builds trees sequentially, with each new tree aiming to correct the residual errors of the combined prior models. XGBoost

is a regularized implementation of GBMs, offering advantages in speed, accuracy, and overfitting control. The cumulative prediction of an XGBoost model can be expressed as shown in Equation (3.5):

$$F(x) = \sum_{m=1}^M \gamma_m h_m(x), \quad (3.5)$$

Where  $h_m(x)$  is the  $m^{th}$  decision tree,  $\gamma_m$  is the learning rate applied to its output.

Bendaoud et al. [23] conducted a comparative study evaluating the performance of various machine learning models, including SVR, ANN, CNN, and XGBoost, for STL in Algeria. The study found that XGBoost outperformed the other models, achieving the lowest MAPE of 1.88%. This superior performance was attributed to XGBoost's ability to manage complex nonlinear relationships and its robustness against overfitting, making it particularly effective in capturing the intricacies of electricity consumption patterns.

Deng et al. [16] proposed a hybrid Bagging–XGBoost model aimed at improving load forecasting accuracy under extreme weather conditions. By integrating the Bagging ensemble method with XGBoost, the model enhanced generalization capabilities and reduced variance, leading to more reliable predictions during volatile environmental scenarios. The hybrid approach demonstrated robustness and accuracy, effectively addressing the challenges posed by extreme weather events in energy demand forecasting.

Zhang et al. [25] incorporated XGBoost into an XAI framework to predict building energy consumption and greenhouse gas emissions. Utilizing SHapley Additive exPlanations values, the study provided insights into the contribution of each feature to the model's predictions, enhancing interpretability. The integration of XGBoost with SHAP allowed for accurate forecasting while offering transparency, enabling stakeholders to understand the driving factors behind energy consumption patterns.

Rafiq et al. [3] applied XGBoost for household classification tasks based on smart meter data in Dubai. The model achieved high accuracy levels above 95%, effectively distinguishing between different household electricity usage profiles. XGBoost's capability to manage large datasets with numerous features contributed to its strong performance, making it a valuable tool for energy consumption analysis and customer segmentation.

These studies collectively highlight XGBoost's versatility and effectiveness in energy consumption forecasting, particularly in handling complex datasets, providing accurate predictions, and offering interpretability when combined with tools like SHAP. For a detailed overview of XGBoost's strengths and weaknesses in this context, please refer to Table 3.4.

Table 3.4: Strengths and Limitations of Tree-Based Methods in STLf.

| Model Type           | Strengths                                                                   | Weaknesses                                      |
|----------------------|-----------------------------------------------------------------------------|-------------------------------------------------|
| <b>Decision Tree</b> | Simple; interpretable; models non-linearity                                 | Prone to overfitting; sensitive to data changes |
| <b>Random Forest</b> | Reduces overfitting; good generalization; manages missing data              | Less interpretable; Large models can be slow    |
| <b>XGBoost</b>       | High accuracy; Regularized; manages feature interactions and missing values | Requires tuning; computationally more intensive |

### 3.2.3. k-nearest neighbors

The k-NN algorithm is a non-parametric and instance-based learning method that has been widely applied in STLf due to its simplicity and ability to model nonlinear relationships without requiring an explicit training phase. In regression settings, k-NN predicts the output for a new instance by averaging the target values of the k most similar observations in the historical dataset.

The standard k-NN regression function can be mathematically represented in Equation (3.6):

$$\hat{y}(x) = \frac{1}{k} \sum_{i \in N_k(x)} y_i, \quad (3.6)$$

Where  $\hat{y}(x)$  is the predicted load value for the input instance  $x$ ,  $N_k(x)$  is the set of  $k$  nearest neighbors to  $x$ , based on a chosen distance metric (commonly Euclidean distance),  $y_i$  is the observed load associated with the neighbor  $i$ .

This method assumes that similar inputs produce similar outputs, making it especially useful for recognizing recurring load patterns over time.

K-NN algorithm has been widely used in both standalone and hybrid frameworks for energy consumption forecasting due to its simplicity, interpretability, and effectiveness in pattern recognition. Peteleaza et al. [6] included k-NN in a comparative study involving various machine learning models for smart city

electricity consumption forecasting across metropolitan areas in the United States. While k-NN performed reasonably well in identifying similar historical consumption profiles, it was ultimately surpassed by more advanced ensemble models such as XGBoost in terms of accuracy. However, the authors emphasized that the interpretability and pattern-matching capabilities of k-NN remain advantageous in applications where understanding consumption behavior is as important as predictive precision.

Rafiq et al. [3] employed k-NN in a classification task based on residential electricity consumption data from Dubai. The study used k-means clustering to group household consumption patterns and then applied k-NN as a baseline classifier to identify similarities between user profiles. Although more sophisticated classifiers were introduced later in the study, k-NN played a crucial role in modelling user behavior and establishing an interpretable reference point for further algorithmic comparisons.

Similarly, Zhou et al. [27] utilized k-NN to enhance cluster-based short-term load forecasting for different building types in a Chinese case study. By grouping buildings with similar consumption characteristics, the model applied k-NN to exploit local similarities in daily demand trends. This approach was particularly effective in contexts with high building diversity and limited historical data, where leveraging neighbourhood-based predictions helped improve accuracy without the complexity of deep learning methods.

Together, these studies demonstrate the ongoing relevance of k-NN in energy forecasting, particularly in scenarios involving heterogeneous datasets, limited data history, or a need for transparent, easy-to-understand models. A summary of the strengths and limitations of the k-NN method in this context is presented in Table 3.5.

Table 3.5: Strengths and limitations of k-NN method in STLF.

| Model Type  | Strengths                                                                                                                                                                                                                                | Weaknesses                                                                                                                                                                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>k-NN</b> | Simple and intuitive to implement; No training phase required; Effective in capturing local patterns and repeated load behaviors; Works well for nonlinear relationships; Easily adaptable for both regression and classification tasks. | Computationally expensive with large datasets; Sensitive to irrelevant or unscaled features; Performance degrades in high-dimensional spaces; Does not manage temporal dependencies unless explicitly engineered; Model performance depends heavily on the choice of distance metric and k value. |

### 3.2.4. Artificial neural networks

ANNs are computational models inspired by the biological structure of the human brain. They consist of interconnected layers of processing units (neurons) capable of learning complex, nonlinear relationships between inputs and outputs. In STLF, ANNs have proven to be highly effective due to their flexibility, ability to capture hidden patterns, and capacity to model temporal and seasonal dynamics in electricity consumption.

A typical feedforward neural network (FFNN) as it is shown in Figure 3.3 consists of an input layer, one or more hidden layers, and an output layer. Each neuron in the network computes the weighted sum of its inputs and applies an activation function to introduce nonlinearity. The model is trained to minimize the prediction error using backpropagation and gradient descent, adjusting the weights of the connections between neurons to improve accuracy over successive iterations.

1. **Feedforward Neural Network:** The simplest ANN architecture used in forecasting tasks is the FFNN. It consists of three components:
  - **Input Layer:** This layer represents the input features such as historical electricity consumption, time of day, temperature, and other relevant factors.
  - **Hidden Layers:** These layers apply nonlinear transformations to the input data. Each neuron in the hidden layer computes a weighted sum of its inputs and passes the result through an activation function.
  - **Output Layer:** The output layer produces the final prediction, which in load forecasting is usually the predicted electricity consumption at a future time point.

The output of each neuron in the hidden layers is computed based on Equation (3.7):

$$z_j^{(l)} = \sum_{i=1}^n w_{ij}^{(l)} x_i^{(l-1)} + b_j^{(l)}, \quad (3.7)$$

Where  $z_j^{(l)}$  is the input to the  $j_{th}$  neuron in layer  $l$ ,  $w_{ij}^{(l)}$  is the weight of the connection between neuron  $i$  in the previous layer and neuron  $j$  in the current layer,  $x_i^{(l-1)}$  is the output from the previous layer, and  $b_j^{(l)}$  is the bias term for neuron  $j$  in layer  $l$ .

The neuron then applies an activation function  $\sigma$  (such as Sigmoid or ReLU) to the weighted sum to produce the output based on Equation (3.8):

$$a_j^{(l)} = \sigma(z_j^{(l)}), \quad (3.8)$$

Where  $\sigma$  is a nonlinear activation function like Sigmoid in Equation (3.9):

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \quad (3.9)$$

Or ReLU by Equation (3.10):

$$\sigma(x) = \max(0, x), \quad (3.10)$$

2. **Training the Network:** The training of an ANN involves adjusting the weights to minimize the error between the predicted and actual values. The error is computed using a loss function, such as MSE for regression tasks that are defined in Equation **Error! Reference source not found.**(3.11):

$$L = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (3.11)$$

Where  $L$  is the loss function or error,  $N$  is the number of data points (samples) in the dataset,  $y_i$  is the true value (actual electricity consumption) for the  $i_{th}$  sample,  $\hat{y}_i$  is the predicted value (forecasted electricity consumption) for the  $i_{th}$  sample,  $(y_i - \hat{y}_i)^2$  represents the squared error for the  $i_{th}$  sample, which measures how far the predicted value is from the actual value.

Backpropagation is used to compute the gradient of the loss function with respect to each weight, which allows the network to adjust the weights iteratively to reduce the error. The weight updates are performed using an optimization algorithm like Stochastic Gradient Descent (SGD) or Adam based on Equation (3.12):

$$\omega \leftarrow \omega - \eta \cdot \nabla_{\omega} L, \quad (3.12)$$

Where  $\omega$  represents the weights of the network,  $\eta$  is the learning rate,  $\nabla_{\omega} L$  is the gradient of the loss function with respect to the weights.

Through multiple iterations, the model learns to predict future load with increasing accuracy by adjusting the weights to better fit the data.



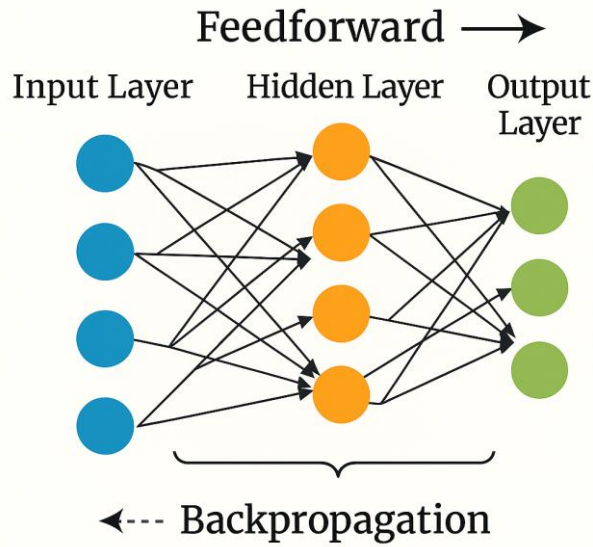


Figure 3.3: Feedforward and backpropagation architecture of ANN.

Bendaoud et al. [23] conducted a comparative study evaluating the performance of various machine learning models, including SVR, CNN, Gradient Boosting, and ANNs, for STLTF in Algeria. The study found that while Gradient Boosting achieved the highest accuracy, ANNs performed commendably, achieving a MAPE of 2.39%. This indicates that ANNs are effective in environments where model simplicity and adaptability are preferred.

Haque and Rahman [24] applied ANN models to both 30-minute and 24-hour ahead load forecasts. Their findings revealed that ANNs outperformed SVR and linear models, offering a favorable balance between complexity and accuracy. However, in more dynamic scenarios, LSTM networks surpassed ANNs in predictive performance, highlighting the importance of model selection based on the specific forecasting context.

Katić et al. [18] developed ANN models to estimate heating energy consumption in school buildings. Their research demonstrated that ANN models achieved lower RMSE and CVRMSE compared to multiple linear regression models. This affirms the value of ANNs in building-level energy forecasting, particularly in capturing complex, nonlinear relationships inherent in energy consumption data.

Despite their strengths, ANNs also present certain challenges. They require substantial datasets for effective training and necessitate careful tuning of hyperparameters, such as learning rates, the number of hidden layers, and neuron counts. Additionally, ANNs often lack interpretability, making it difficult to understand the underlying decision-making process. Without proper regularization, they are prone to overfitting, especially when applied to small or noisy datasets.



These studies collectively underscore the applicability of ANNs in energy consumption forecasting, highlighting their strengths in modeling complex patterns and their limitations concerning data requirements and interpretability that mentioned in Table 3.6.

Table 3.6: Strengths and limitations of ANNs in STLF.

| Model Type | Strengths                                                                                                                                                                                                                                                                    | Weaknesses                                                                                                                                                                                                                |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANN        | <p>Capable of modeling complex, nonlinear relationships,</p> <p>Effective in both short-term and pattern-based load forecasting,</p> <p>Can integrate diverse features (e.g., temperature, calendar, occupancy),</p> <p>Adaptable to hybrid or deep learning frameworks.</p> | <p>Prone to overfitting with small or noisy datasets,</p> <p>Requires careful tuning of network architecture,</p> <p>Difficult to interpret ("black box" behavior),</p> <p>Computationally intensive during training.</p> |

### 3.3. Deep learning models

DL models are a specialized subset of ANNs that involve architectures with multiple hidden layers, enabling them to learn high-level abstractions and hierarchical feature representations. In the context of STLF, DL techniques have shown superior performance in modeling nonlinear, temporal, and seasonal electricity consumption patterns, particularly in complex environments such as smart grids and real-time demand prediction.

Unlike shallow neural networks, deep learning models can automatically extract meaningful features from raw data, minimizing the need for manual feature engineering. Among the most commonly used deep learning architectures in energy forecasting are Recurrent Neural Networks (RNNs), LSTM, and CNNs.

#### 3.3.1. Recurrent neural networks

RNNs are designed to process sequential data by maintaining internal memory through feedback connections. This makes them ideal for time-series tasks such as load forecasting. However, standard RNNs suffer from the vanishing gradient problem described below, which limits their ability to capture long-term dependencies.

RNNs are designed to process sequential data by maintaining an internal hidden state that evolves over time. This architecture allows RNNs to capture temporal dependencies, making them suitable for tasks like electricity load forecasting. However, training RNNs over long sequences presents significant challenges, particularly due to the vanishing gradient problem.

As shown by Pascanu et al. [28], when training RNNs using Backpropagation Through Time (BPTT), gradients are propagated backward through many time steps. During this process, the gradients are repeatedly multiplied by the Jacobian matrix of the recurrent weights. If the spectral radius (i.e., the largest absolute eigenvalue) of this matrix is less than one, the gradients shrink exponentially with each time step. As a result, early layers in the sequence receive near-zero gradients, effectively preventing the model from learning long-term dependencies.

This vanishing effect causes the RNN to focus mainly on recent inputs while "forgetting" information from earlier time steps, which undermines its ability to model long-range correlations—a limitation particularly problematic in time series forecasting where historical context is crucial.

Pascanu et al.[28] also showed that the exploding gradient problem can occur in the opposite scenario, when gradients grow exponentially due to large weight values. Together, these issues severely hamper RNN training.

To address the vanishing gradient problem, gated architectures like LSTM and Gated Recurrent Units (GRUs) were developed. These models incorporate mechanisms that allow gradients to pass through many time steps without significant attenuation, enabling them to learn long-term dependencies more effectively.

To address this, LSTM networks were introduced: they incorporate memory cells and gating mechanisms (input, output, and forget gates), enabling them to learn long-range dependencies more effectively.

The internal operations of an LSTM unit can be summarized by the set of equations (3.13):

$$\left\{ \begin{array}{l} f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f \\ i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i \\ o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o \\ \tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C \\ C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \\ h_t = o_t * \tanh(C_t) \end{array} \right. \quad (3.13)$$

Where  $x_t$  is the input at time  $t$ ,  $h_t$  is the hidden state,  $C_t$  is the cell state,  $\sigma$  is the sigmoid function,  $*$  denotes element-wise multiplication.

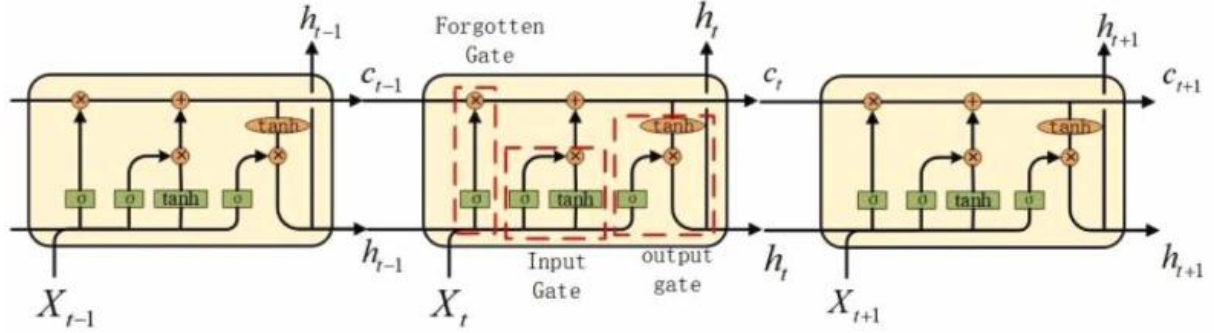


Figure 3.4: Schematic diagram of the LSTM [29].

These equations and Figure 3.4 describe how the model selectively forgets, updates, and outputs information across time steps, making it ideal for sequence forecasting tasks like STLF.

LSTM networks have shown substantial promise in the domain of electricity load forecasting, especially when compared to traditional statistical and shallow learning models. In a study by Bilgili and Pinar [17], LSTM was evaluated against the SARIMA model for forecasting Turkey's gross electricity consumption. Their findings demonstrated that LSTM significantly outperformed SARIMA, particularly in modelling nonlinear trends and seasonal variations. The LSTM model achieved notably lower error metrics, with a MAPE of 2.42%, MAE of 215.35 GWh, and RMSE of 329.9 GWh, emphasizing its strength in capturing complex temporal dependencies that SARIMA could not manage effectively. Similarly, Haque and Rahman [24] applied LSTM to both 30-minute and 24-hour-ahead forecasting tasks, comparing its performance with SVR and ANN. While ANN and SVR provided reasonable accuracy, the LSTM model consistently achieved the lowest MAPE and RMSE values across both time horizons. These results highlight LSTM's ability to learn from both short-term fluctuations and long-term demand trends, making it particularly well-suited for forecasting electricity consumption in dynamic and non-stationary environments.

Further demonstrating its potential, Memarzadeh and Keynia [30] proposed an optimally tuned LSTM-based neural network specifically designed for short-term forecasting of electricity load and price. By fine-tuning the architecture and hyperparameters using real-world grid data, their model achieved high forecasting accuracy across both daily and hourly intervals. This study reinforces LSTM's adaptability and effectiveness in practical energy forecasting scenarios, especially when precision is required for operational decisions.

Collectively, these studies affirm that LSTM models offer a powerful and flexible approach to short-term load forecasting, capable of handling complex sequences and

providing higher accuracy than conventional and many machine learning models that is mentioned in Table 3.7.

Table 3.7: Strengths and limitations of RNNs and LSTMs in STLf.

| Model Type | Strengths                                                                                                                                                                                                                                                                                   | Weaknesses                                                                                                                                                                                                                                                                 |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RNN        | <p>Effective in modeling sequential and time-dependent data,</p> <p>LSTM captures long-term dependencies using memory and gates,</p> <p>Manages seasonality, trends, and lagged effects in electricity demand,</p> <p>Performs well in multi-step forecasting and dynamic environments.</p> | <p>Require large datasets to generalize well,</p> <p>Computationally expensive, especially with long sequences,</p> <p>Difficult to interpret and debug due to complex architecture,</p> <p>Sensitive to hyperparameter tuning (e.g., number of layers, hidden units).</p> |

### 3.3.2. Transformer-based models

Transformer-based models have recently gained traction in STLf due to their ability to model complex temporal dependencies using self-attention mechanisms. Originally developed for natural language processing tasks, transformers differ from traditional recurrent architectures (like RNNs or LSTMs) in that they process input sequences in parallel, not sequentially, enabling both faster training and better long-range dependency modeling.

At the core of the transformer architecture is the scaled dot-product attention mechanism, defined in Equation (3.14):

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3.14)$$

Where  $Q$ ,  $K$ , and  $V$  are the query, key, and value matrices,  $d_k$  is the dimensionality of the key vectors.

This mechanism allows the model to dynamically weigh the importance of different time steps in the input sequence when making predictions, which is especially useful in multivariate and irregular load datasets.

Table 3.8: Strengths and Limitations of Transformer-Based Models in STLFL.

| Model Type                      | Strengths                                                                                                                                                                                                                                                             | Weaknesses                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Transformer-Based Models</b> | Captures long-term dependencies better than RNN/LSTM,<br>Highly parallelizable during training,<br>Scales well to high-dimensional multivariate input,<br>Provides interpretability via attention weights,<br>Performs well in multi-step and missing data scenarios. | Requires more data and computational power,<br>Model architecture and tuning are more complex,<br>Needs careful handling of time positional encoding in time series forecasting,<br>Still relatively new in power systems forecasting — less validated in practice,<br>May overfit on small datasets without regularization. |

Transformer-based architectures have emerged as powerful tools for electricity load forecasting due to their ability to model long-range temporal dependencies and handle heterogeneous input features. Unlike traditional recurrent models, Transformers leverage attention mechanisms to selectively focus on the most relevant parts of the input sequence, making them particularly effective in time series forecasting tasks.

Ahmad et al. [34] proposed TFTformer, a specialized Transformer architecture tailored for electricity load prediction. This model integrates temporal attention mechanisms with feature selection strategies to model both static and dynamic covariates. Notable components include variable selection networks that enhance input relevance, gated residual networks (GRNs) that facilitate information flow, and multi-head attention layers that capture temporal dependencies across multiple time scales. Their results demonstrated that TFTformer significantly outperformed RNN-based architectures, especially in multi-step forecasting scenarios and in datasets containing missing values. This highlights the model's robustness and adaptability in real-world forecasting contexts.

Chan and Yeo [36] implemented a general Transformer architecture to forecast short-term electricity demand. Their model made use of the self-attention mechanism to identify key time steps in historical data, providing not only improved forecasting accuracy but also enhanced interpretability. Compared to LSTM models, the Transformer model achieved lower forecasting errors and revealed which historical inputs contributed most to the predictions via attention scores. This transparency allows for a deeper understanding of load dynamics and enhances confidence in the forecasting outputs.

Together, these studies demonstrate the growing importance of Transformer-based architectures in energy forecasting. As mentioned in Table 3.8, by offering both high accuracy and interpretability, these models address key limitations of traditional methods and present a promising path forward for short-term load forecasting applications.

### 3.4. Hybrid models

To address the increasing complexity of short-term electricity demand patterns—affected by weather, human behavior, and distributed energy resources—many researchers have developed hybrid and ensemble models. These approaches aim to combine the strengths of different forecasting techniques to improve prediction accuracy, reduce variance, and enhance generalization. Hybrid models are particularly beneficial in STLf tasks, where they help address both linear and nonlinear characteristics in data.

#### 3.4.1. Statistical & ML hybrids

Hybrid models that combine statistical and machine learning techniques have gained prominence in energy load forecasting due to their ability to capture both linear and nonlinear patterns in time series data. By leveraging the strengths of different methodologies, these models aim to improve forecasting accuracy and robustness.

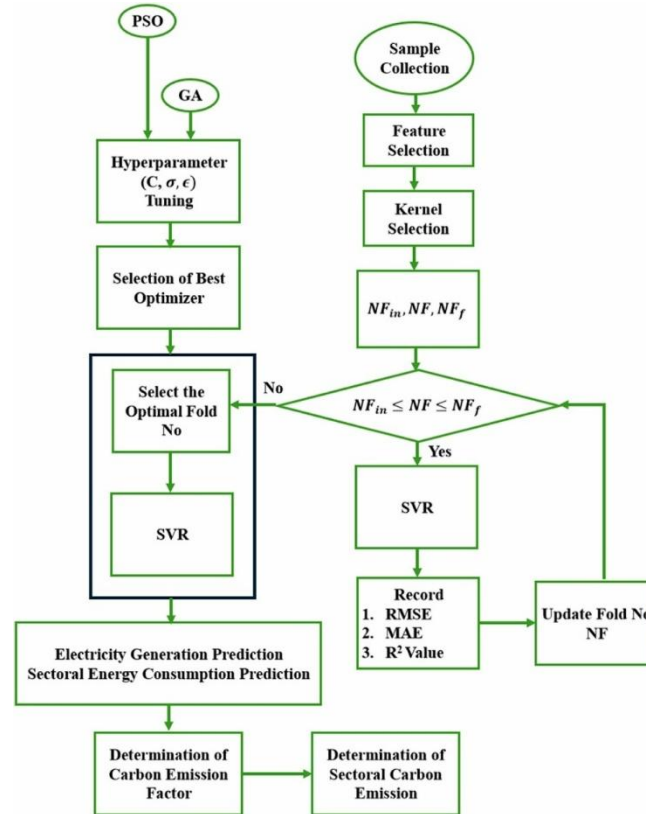


Figure 3.5: Proposed model of GA-optimized SVR [26].



Bendaoud et al. [23] proposed a hybrid model that integrates Exponential Smoothing with GBRT for short-term electricity load forecasting. In this approach, Exponential Smoothing is employed to model the linear and seasonal components of the load data, effectively capturing the underlying trends and periodicities. Subsequently, GBRT is applied to the residuals obtained from the smoothing process to model the nonlinear relationships and correct any remaining errors. This two-stage modelling strategy allows for a comprehensive representation of the load data, addressing both its linear and nonlinear characteristics. The hybrid model demonstrated improved forecasting accuracy, particularly for daily and weekend load predictions, outperforming models that rely solely on either statistical or machine learning techniques.

**Error! Reference source not found.** illustrates hybrid forecasting model of Hasan et al. [26] that combines SVR with a Genetic Algorithm (GA) to predict energy consumption across four sectors: residential, commercial, industrial, and agricultural. SVR is a powerful machine learning technique capable of modelling complex nonlinear relationships in data. However, its performance is highly sensitive to the selection of hyperparameters, such as the penalty parameter and kernel function parameters. To address this, the authors employed a Genetic Algorithm to optimize the SVR hyperparameters, enhancing the model's predictive accuracy. The GA iteratively searches for the optimal combination of parameters by simulating the process of natural selection, thus avoiding local minimum, and improving generalization. The GA-optimized SVR model achieved higher forecasting accuracy compared to traditional SVR and other baseline models, demonstrating its effectiveness in capturing the complex dynamics of energy consumption across different sectors.

### 3.4.2. Ensemble tree-based methods

Ensemble-based models have gained considerable attention in the field of STLTF for their ability to enhance predictive accuracy, reduce overfitting, and improve model generalization. By combining the outputs of multiple learners, ensemble methods capitalize on the strengths of different models to create more stable and robust forecasts, particularly under uncertain and volatile conditions.

Deng et al. [16] proposed a hybrid ensemble model that integrates Bagging with XGBoost to address the challenges of forecasting electricity demand during extreme weather events. Their model begins by identifying periods of extreme weather using time, load, and meteorological features, ensuring that the training process is tailored to critical high-variance conditions. To improve relevance and adaptability, the model selects training data based on a weighted similarity measure, which emphasizes historical cases most comparable to the current situation.

As Figure 3.6 illustrates, the hybrid approach employs Bagging to generate diverse subsets of the selected training data. Each subset is used to train an independent XGBoost learner, and the final forecast is obtained by aggregating their outputs—typically through averaging. This combination allows the model to benefit from

XGBoost's ability to capture complex nonlinear relationships while leveraging Bagging to reduce variance and improve generalization.

Results from the study showed that the Bagging–XGBoost model outperformed conventional methods, especially in forecasting peak loads during extreme weather. It achieved a notable reduction in MAPE, demonstrating its robustness in volatile consumption environments.

This work highlights the effectiveness of ensemble learning in energy forecasting, particularly in high-risk scenarios where accuracy and reliability are critical to maintaining grid stability.

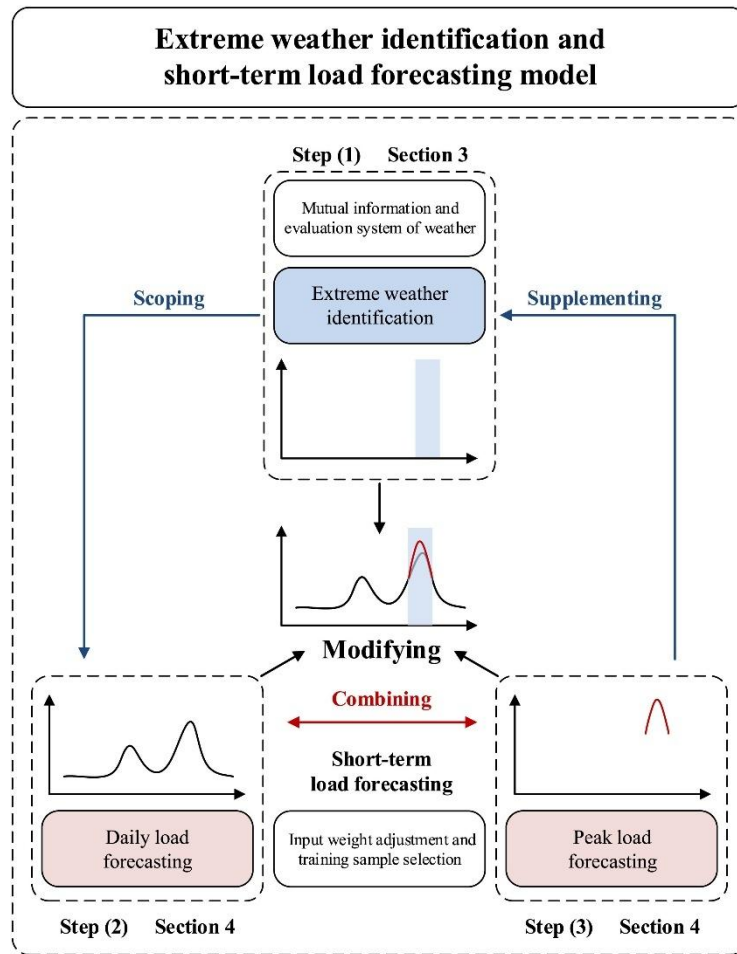


Figure 3.6: Framework of extreme weather identification and STLFL [16].

### 3.4.3. Neural network-based hybrids

Hybrid neural network models have proven highly effective in the field of electricity load forecasting due to their capacity to simultaneously capture localized consumption patterns and long-range temporal dependencies. Among the most widely used configurations are models that combine CNNs with LSTM networks. These hybrids are particularly valuable in dealing with high-frequency, nonlinear, and volatile energy data.



CNNs are well-suited for identifying spatial and short-term features in structured input data. Their layered design allows for efficient abstraction and preprocessing, making them an ideal front-end in hybrid architectures. Bendaoud et al. [23] evaluated CNN models for short-term load forecasting in Algeria and found that while CNNs did not outperform all alternatives in isolation, they performed well when integrated with other models. Specifically, CNNs demonstrated strong performance in feature extraction, especially when embedded in hybrid systems, enhancing the performance of downstream components such as LSTM networks.

Wang et al. [29] advanced this concept by proposing a hybrid model that integrates CNNs with extended LSTM networks. In this architecture, CNNs first extract high-level spatial and short-term temporal features, which are then processed by LSTM layers to model sequential dependencies. Their study demonstrated that this hybrid significantly improved forecast accuracy in complex load environments where both short-term fluctuations and long-term consumption trends coexist.

Guo et al. [31] expanded on the CNN–LSTM hybrid by developing a multi-scale framework that incorporates real-time electricity pricing into the forecasting process. Their model captures features at different temporal resolutions using multiple CNN layers, allowing for a more nuanced understanding of consumption dynamics. These features are passed to LSTM layers to model temporal relationships, leading to higher accuracy in both day-ahead and intraday forecasts.

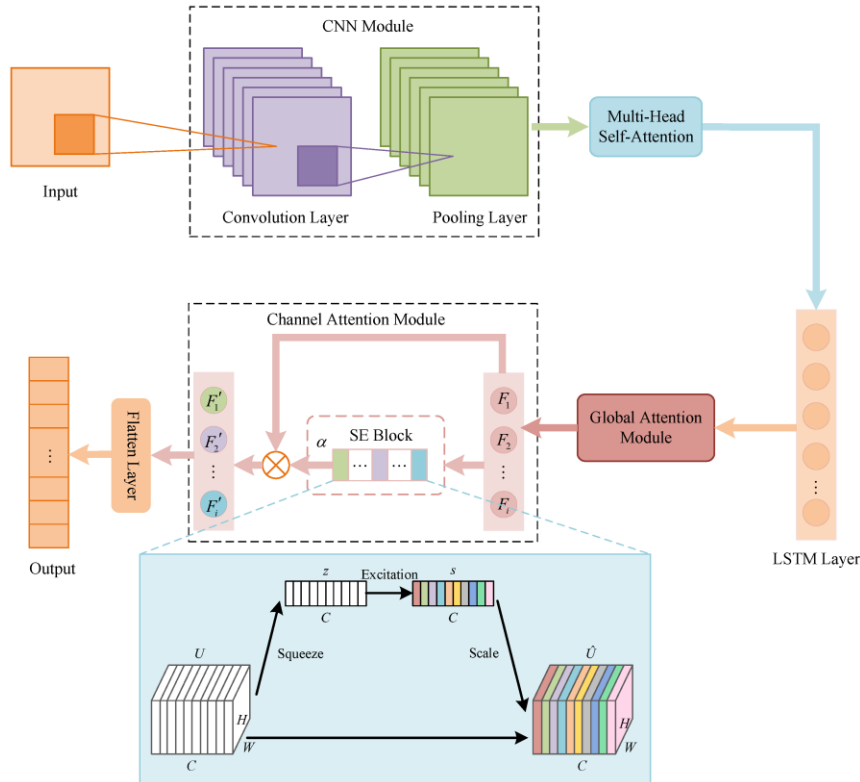


Figure 3.7: A schematic diagram of a CNN-LSTM network based on a multi-modal attention mechanism [32].

A more advanced adaptation was presented in Figure 3.7 by Guo et al. [32], who incorporated a multi-modal attention mechanism into the CNN–LSTM hybrid structure. This model assigns attention weights to different feature representations based on their relevance, improving the interpretability and responsiveness of the forecasting model. The integration of attention further enhances performance by allowing the model to focus on critical periods or signals during training.

In parallel, Memarzadeh and Keynia [30] developed an optimized LSTM-based neural network specifically for forecasting electricity load and price. Their approach involved rigorous hyperparameter tuning and structural optimization using real grid data. While the model was not explicitly a hybrid in terms of architecture, it reflects the importance of tailored LSTM configurations when deployed in realistic energy forecasting scenarios. The model achieved high accuracy in both daily and hourly forecasts, underscoring the practical viability of deep learning in operational planning.

Collectively, these studies confirm the effectiveness of neural network-based hybrid models—especially CNN–LSTM combinations—in modelling the complex, multi-scale characteristics of electricity consumption. By combining the feature abstraction capabilities of CNNs with the memory strength of LSTMs, these models offer a powerful approach for short-term load forecasting in modern smart grid environments.

#### 3.4.4. Intelligent and fuzzy logic hybrids

Liapis et al.[33] proposed a fuzzy-deep ensemble model specifically designed to enhance short- and mid-term electric load forecasting. For the day ahead forecasting, the model demonstrated particular strength in managing uncertainty and fluctuating demand. By integrating fuzzy logic into the structure of a deep neural network, the model could incorporate expert-like reasoning and linguistic rules to better interpret ambiguous input conditions—such as weather fluctuations or demand variability—common in daily forecasting tasks.

The fuzzy system managed input imprecision by translating real-world variables into fuzzy sets and applying rule-based reasoning, while the deep neural network layers captured complex nonlinear relationships from historical load data. This fusion enabled the model to maintain accuracy while improving robustness and interpretability. In the case of 1-day load prediction, the ensemble outperformed traditional methods by effectively modeling sudden shifts and short-term demand spikes, making it especially suitable for operational planning in smart grid environments.

#### 3.4.5. Transformer-based hybrids

Tian et al. [34] proposed a CNN–Transformer hybrid model tailored for day-ahead probabilistic load forecasting, with a particular focus on weekends—periods often

marked by irregular consumption behavior and limited historical data. The model was designed to address the challenges of accurate prediction under these constraints by leveraging both convolutional and attention-based sequence modelling techniques.

Based on Figure 3.8, the architecture begins with a data preprocessing stage that denoises and enhances the raw input using features such as peaks and trends. These enriched one-dimensional time series are then processed through multiple CNN layers, which are responsible for extracting localized temporal features. CNNs are particularly effective in identifying short-term structures in load data, such as consumption spikes or repetitive daily cycles.

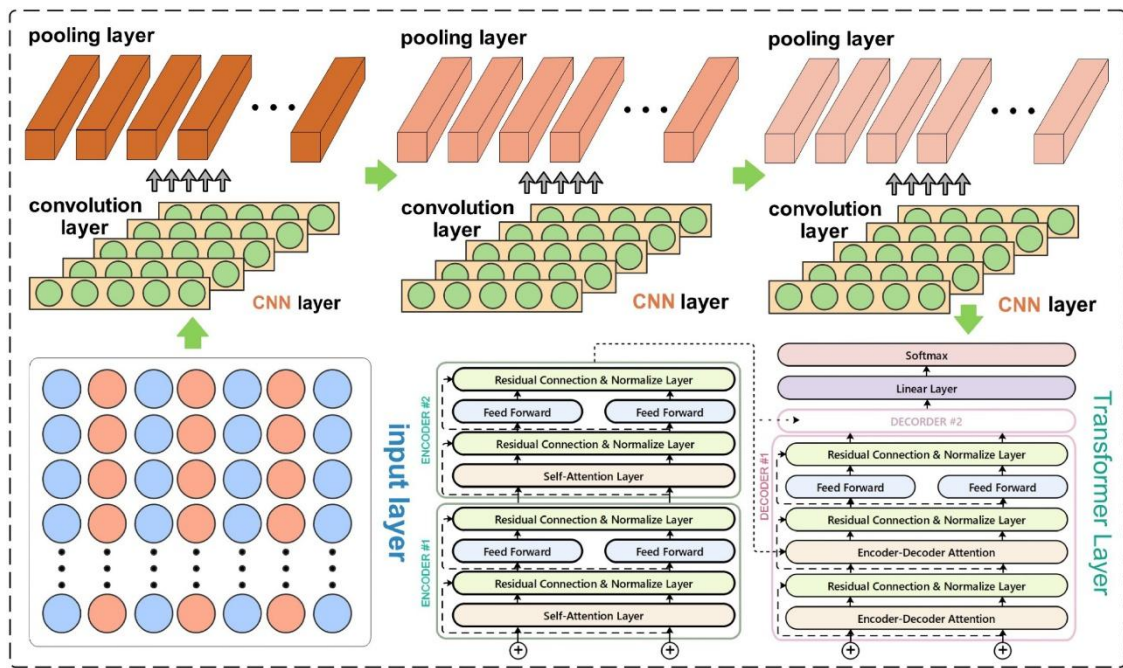


Figure 3.8: Flow chart of the deep learning framework [34].

Once features are extracted, they are passed to a Transformer module, which applies self-attention mechanisms to model long-range dependencies within the time series. The Transformer's ability to weigh the importance of different time steps allows it to capture broader contextual relationships that traditional recurrent models might overlook.

To produce probabilistic forecasts, the model incorporates quantile regression as its loss function. This enables it to estimate not only point predictions but also confidence intervals, which are essential for understanding forecast uncertainty. Additionally, a novel loss penalty mechanism is introduced to prioritize weekend-specific accuracy, ensuring the model pays greater attention to these more volatile periods.

Evaluation results showed that the model achieved high forecasting accuracy for all 48 time points in a day using only two weeks of historical data. This performance highlights the effectiveness of combining CNNs and Transformers, especially in

scenarios where data availability is limited. The hybrid approach offers a robust and flexible solution for utilities seeking reliable short-term forecasts under uncertain or sparse conditions.

Table 3.9: Benefits of Hybrid models in STLF.

| authors              | Benefits                                             | Description                                                                                                                                                 |
|----------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bendaoud et al. [23] | Improved short-term prediction accuracy              | Combining Exponential Smoothing with GBRT captures both linear trends and nonlinear residual patterns, enhancing daily and weekend forecasting performance. |
|                      | Effective feature extraction in hybrid models        | CNNs enhance LSTM performance by providing rich local and temporal features as input for sequence modelling.                                                |
| Hasan et al. [26]    | Enhanced model generalization                        | Genetic Algorithm optimizes SVR hyperparameters, avoiding local minima and improving accuracy across multiple energy sectors.                               |
| Deng et al. [16]     | Robustness in extreme weather conditions             | Bagging–XGBoost hybrid improves model stability by aggregating forecasts from multiple learners trained on weather-specific subsets.                        |
| Wang et al. [29]     | Better handling of short- and long-term dependencies | CNN layers extract short-term features, while extended LSTM layers model sequential behavior for improved accuracy in volatile environments.                |
| Guo et al. [31]      | Improved accuracy under-prices dynamics              | Multi-scale CNN–LSTM hybrid captures complex load-price relationships across different timescales for day-ahead and intraday forecasting.                   |
| Guo et al. [32]      | Improved interpretability and feature relevance      | Multi-modal attention mechanism enhances CNN–LSTM hybrid by focusing on                                                                                     |

|                            |                                        |                                                                                                                                           |
|----------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
|                            |                                        | critical input features during training.                                                                                                  |
| Memarzadeh and Keynia [30] | Optimized forecasting accuracy         | LSTM architecture is fine-tuned using real-world data, improving model performance in hourly and daily load prediction.                   |
| Liapis et al. [33]         | Effective handling of uncertainty      | Fuzzy-deep ensemble integrates fuzzy rules into deep networks, improving reliability and robustness in 1-day forecasts.                   |
| Tian et al. [34]           | Accurate forecasting with limited data | CNN–Transformer hybrid with quantile loss and weekend-specific penalties enables precise forecasts using only two weeks of training data. |

Table 3.9 presents a comparative overview of various hybrid forecasting models used in STLF, emphasizing the specific advantages each model offers. The benefits range from improved predictive accuracy and model generalization to enhanced robustness under extreme conditions and better interpretability.

### 3.5. Summary

Chapter 3 offers a comprehensive examination of STLF, focusing on forecasting electricity demand from minutes up to a week. Accurate STLF is essential for efficient grid operation, economic dispatch, and demand response management. The chapter categorizes forecasting models into four main groups: statistical methods, ML methods, deep learning models, and hybrid/ensemble approaches.

Statistical models such as ARIMA and SARIMA provide interpretable forecasts under stable seasonal conditions, but they struggle with nonlinearity and high-frequency variability. Regression models like linear regression and quantile regression are efficient and interpretable, though limited in complex or dynamic contexts.

ML models such as SVR, decision trees, random forests, and XGBoost outperform traditional methods by capturing nonlinear relationships. SVR, in particular, balances accuracy with computational efficiency, while tree-based methods provide strong generalization and interpretability.

Deep learning methods—especially LSTM and ANN—excel in modeling temporal dependencies. LSTM outperforms statistical and shallow models by capturing long-

and short-term dynamics. ANN models are simpler and effective but prone to overfitting and require tuning.

Hybrid models integrate multiple methods to enhance accuracy and robustness. Examples include exponential smoothing + GBRT, SVR + GA, CNN + LSTM, and CNN + Transformer. These combinations address both linear and nonlinear data characteristics, improve model generalization, and handle uncertainty, volatility, and limited data effectively.

The chapter emphasizes that hybrid and deep learning models are increasingly vital for modern STLF, especially in smart grid environments where interpretability, scalability, and resilience to variability are crucial.



## 4 Medium-term load forecasting

MTLF refers to the prediction of electricity demand over a horizon ranging from one week to one month. This forecasting range serves as a bridge between short-term operational planning and long-term investment strategies. MTLF is essential for applications such as energy procurement, maintenance scheduling, capacity planning, tariff design, and policy development.

Unlike short-term forecasting, which deals with rapid fluctuations in demand, medium-term forecasting must address broader seasonal trends, monthly usage patterns, and external influences such as climate variation, economic activity, and consumer behavior [17], [35], [36]. The data used is typically aggregated at weekly or monthly levels, which introduces different modeling challenges, including data sparsity, reduced temporal resolution, and delayed response to exogenous variables.

MTLF models must also account for nonlinear and nonstationary behaviors over longer horizons while maintaining reasonable interpretability and computational efficiency. Furthermore, many modern applications demand that forecasts incorporate uncertainty estimation and adapt to evolving energy usage conditions, particularly in grids with high penetration of renewable energy and flexible loads [25], [37], [38].

This chapter reviews prominent MTLF methodologies across four main categories: statistical models, machine learning techniques, deep learning architectures, and hybrid or ensemble approaches. The aim is to identify the most effective techniques for medium-range horizons and understand how these methods address the unique challenges of MTLF across residential, commercial, and distribution system levels.

### 4.1. Statistical methods

Statistical methods have traditionally served as the foundation for MTLF due to their mathematical simplicity, interpretability, and low computational requirements. These models are particularly effective when applied to aggregated consumption data—such as weekly or monthly electricity use—where periodicity and seasonality can be exploited for accurate prediction [39], [40].

Unlike short-term forecasting, which often relies on high-frequency input variables like real-time weather or occupancy data, MTLF emphasizes broader consumption patterns over longer horizons. As a result, statistical models in this context focus on



uncovering trend structures, autocorrelation, and cyclical patterns in the data. Methods such as linear regression, ARIMA, SARIMA, and quantile regression are frequently employed to model deterministic and probabilistic aspects of medium-term electricity demand.

Recent studies have also explored hybrid extensions of classical models, incorporating decomposition (e.g., Fourier transforms) or probabilistic components (e.g., kernel density estimation) to improve their adaptability to complex demand dynamics. While these approaches may lack the flexibility of modern machine learning techniques, they offer transparent forecasting logic, which remains essential for operational decision-making in power systems and utility planning [40].

The following subsections review key statistical methods applied to medium-term electricity forecasting, along with selected case studies demonstrating their performance, strengths, and limitations.

#### 4.1.1. Regression-based methods

Regression-based models are among the most frequently used techniques in MTLF due to their simplicity, interpretability, and efficiency when working with aggregated data such as monthly or weekly electricity consumption. These models rely on identifying statistical relationships between the target variable (typically electricity demand) and independent features like historical load, temperature averages, or calendar-based indicators.

LR models assume a constant, additive relationship between predictors and the target. This makes them ideal for applications requiring transparent forecasting logic, such as tariff planning and capacity estimation. They are particularly effective for stable consumption environments with identifiable seasonal trends.

To improve the accuracy and reliability of medium-term load forecasting, particularly under uncertainty, researchers have increasingly turned to quantile regression (QR). Unlike traditional regression methods that estimate the Mean of a response variable, QR allows forecasting of conditional quantiles (e.g., Median, 90th percentile), which provides a more nuanced view of the possible outcomes and better captures variability in electricity demand.

The quantile regression objective is to minimize the quantile loss function, defined in Equation (4.1):

$$\min_{\beta} \sum_{i=1}^N \rho_{\tau}(y_i - x_i^{\top} \beta), \quad (4.1)$$

Where  $\rho_{\tau}(u) = u(\tau - \mathbb{I}_{\{u < 0\}})$  is the asymmetric quantile loss function,  $\tau \in (0,1)$  denotes the desired quantile (e.g.,  $\tau = 0.5$  for the median),  $y_i$  is the actual observation,  $x_i^{\top} \beta$  is the predicted value.

This formulation penalizes over- and under-predictions differently depending on the quantile level, enabling forecasters to model both central tendencies and tail risks in electricity demand.

Wang et al. [40] developed a combined probabilistic forecasting model that integrates quantile regression with kernel density estimation. Their model aimed to forecast peak and base load scenarios by estimating a full distribution rather than a single-point value. The approach enabled them to reflect the uncertainty inherent in electricity demand and capture asymmetric behaviors—such as occasional demand spikes—which linear models typically fail to account for. The use of kernel methods further enhanced the flexibility of their distribution modeling, making it especially suitable for medium-term planning where long-range uncertainty is a significant factor.

Xu [41] provided a comprehensive review of quantile regression applications in both renewable energy generation and building load forecasting. The review emphasized QR's ability to model a range of possible demand outcomes, rather than relying solely on expected values. Xu noted that QR is particularly beneficial in situations involving intermittent renewable inputs or unpredictable user behavior—common challenges in medium-term horizons. The method supports probabilistic forecasting, where forecasting ranges are more informative than single-value predictions and are essential for grid operators managing risk and planning reserve margins.

In summary, as highlighted in Table 4.1, QR has emerged as a vital tool in medium-term load forecasting by offering interpretable distributional insights into energy demand. It improves robustness under uncertainty and is particularly advantageous in dynamic systems influenced by renewable energy, fluctuating occupancy patterns, or seasonal effects.

Table 4.1: Strengths and limitations of regression methods in MTLF.

| Model Type                      | Strengths                                                                                     | Weaknesses                                                                  |
|---------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Linear Regression (LR)</b>   | Simple; interpretable; fast to compute; effective for aggregated data.                        | Limited to linear patterns; underperforms in dynamic or nonlinear contexts. |
| <b>Quantile Regression (QR)</b> | Captures variability and uncertainty; robust to outliers; supports probabilistic forecasting. | More complex; harder to interpret compared to mean-based methods.           |

#### 4.1.2. ARIMA and SARIMA

These models are particularly effective in MTLF due to their ability to capture trend, seasonality, and autocorrelation in aggregated data such as weekly or monthly electricity consumption.

Several studies have explored the application and enhancement of statistical models such as ARIMA and SARIMA for medium-term load forecasting. Bilgili and Pinar [17] employed a SARIMA model to forecast monthly gross electricity consumption in Turkey, demonstrating the model's effectiveness in capturing long-term patterns and consistent seasonal cycles within national electricity profiles. Their results confirmed that while deep learning models like LSTM slightly outperformed SARIMA in handling non-linearities, SARIMA stood out for its model stability, computational simplicity, and robustness when the data adhered to regular seasonal trends. These characteristics make SARIMA a reliable option for institutional forecasting tasks where interpretability and efficiency are critical.

In the domain of residential energy forecasting, Gellert et al. [22] applied both ARIMA and TBATS models to smart home electricity datasets. Their comparative evaluation revealed that ARIMA maintained solid performance under structured and regular seasonal conditions. However, the study also identified its limitations when dealing with complex seasonality and higher-frequency volatility, suggesting that more adaptive models like TBATS, which support multiple seasonalities and complex temporal dynamics, may be better suited for more irregular patterns often found in smart environments.

An innovative development was presented by Karamolegkos and Koulouriotis [39], who introduced a Fourier-enhanced ARIMA model. Though originally designed for short-term forecasting in the Greek electricity market, the model integrated seasonal decomposition using Fourier series to isolate dominant frequency components. This technique enhances the model's ability to detect and forecast cyclical consumption patterns, making it highly adaptable for medium-term horizons as well. The approach allows for better alignment between statistical model structures and the periodic behaviors of electricity demand, especially in contexts where multiple seasonal influences coexist.

Additionally, Wang et al. [35] proposed a forecasting model based on accumulative time-delay effects and periodic fluctuation components. Although this model is not a conventional ARIMA-type model, it shares key structural principles such as time-series decomposition and explicit modeling of deterministic and stochastic elements. The model was validated using monthly electricity consumption data, illustrating its suitability for MTLF tasks. It demonstrated particular strength in capturing recurring cycles and long-term behavioral trends, contributing to more accurate and interpretable forecasts.

Together, these studies demonstrate that while traditional ARIMA-based models provide a robust foundation for medium-term load forecasting (MTLF)—particularly when the data exhibits stable and consistent seasonality—advanced adaptations such as Fourier-enhanced ARIMA and models incorporating hybrid periodic structures significantly improve forecasting performance in more complex, dynamic scenarios, as summarized in Table 4.2.

Table 4.2: Strengths and limitations of ARIMA and SARIMA in MTLF.

| Model Type           | Strengths                                                                                                              | Weaknesses                                                                                                           |
|----------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>ARIMA</b>         | Suitable for stationary or near-stationary monthly data; interpretable with transparent parameter structure.           | Assumes linearity and lacks adaptability to nonlinear patterns; limited performance with irregular long-term trends. |
| <b>SARIMA</b>        | Strong at modelling regular monthly or seasonal cycles; robust in national or utility-scale MTLF with stable patterns. | Struggles with multiple or shifting seasonalities; sensitive to season length and requires precise parameterization. |
| <b>Fourier-ARIMA</b> | Effectively models multi-period cycles using harmonic decomposition; enhances ARIMA flexibility for MTLF.              | Increases model complexity; requires accurate frequency selection and decomposition expertise.                       |

## 4.2. ML models

ML techniques have emerged as powerful alternatives to traditional statistical approaches in electricity load forecasting. These models are particularly effective at capturing complex, nonlinear patterns in historical consumption data and can adapt to dynamic changes in usage behavior. ML models do not require strong assumptions about the underlying data distribution, making them suitable for diverse forecasting horizons and load profiles.

Recent studies have shown that ML models can outperform classical models in both accuracy and generalizability, especially when sufficient historical and contextual data are available. Commonly used machine learning methods in load forecasting include SVR, decision tree-based algorithms, k-NN, and ANN, each offering specific advantages in terms of interpretability, scalability, and performance under different forecasting conditions.

### 4.2.1. Tree-based models

Tree-based models are highly effective for MTLF due to their ability to manage complex, nonlinear relationships between features and the target variable. These models are capable of automatically capturing interactions between features without requiring extensive preprocessing, making them particularly useful for forecasting electricity consumption, which can exhibit both seasonal and irregular patterns. The most commonly used tree-based algorithms in load forecasting are Decision Trees, Random Forests, and GBMs, including XGBoost and LightGBM.

#### 4.2.1.1. Decision trees

Decision Trees are a fundamental tree-based model that recursively splits the feature space into regions that are as homogeneous as possible in terms of the target variable. This method is intuitive and easy to interpret, making it valuable for practical applications in MTLF. While Decision Trees can suffer from overfitting, they provide a clear visualization of how the model arrives at its predictions. They are particularly useful for capturing simple patterns in the data and can serve as the foundation for more complex models like Random Forests and Gradient Boosting.

#### 4.2.1.2. Random forests

RF are an ensemble method that builds multiple decision trees using random subsets of the data and aggregates their predictions. This approach reduces the risk of overfitting associated with a single Decision Tree and improves model generalization. Random Forests are highly effective at handling high-dimensional data and can capture complex interactions among features. For medium-term load forecasting, RF models can effectively model seasonal variations and nonlinear trends in electricity consumption. Their robustness and simplicity make them a popular choice for load forecasting.

#### 4.2.1.3. XGBoost

XGBoost is an optimized implementation of GBMs that incorporates techniques such as regularization to reduce overfitting and improve model accuracy. It has become one of the most popular tree-based algorithms for forecasting tasks due to its ability to manage large datasets with high computational efficiency. In the context of MTLF, XGBoost has demonstrated strong performance in predicting future load values by effectively capturing both short-term fluctuations and longer-term trends in electricity usage.

#### 4.2.1.4. GBMs

As **Error! Reference source not found.** illustrates, GBMs are an ensemble learning technique that builds a series of decision trees in a sequential manner. At each step, a new tree is fitted to the residual errors (the gradients) of the existing model. The primary idea is to reduce the prediction error by adding trees that correct the mistakes made by previous trees. Mathematically, the process can be described as follows:

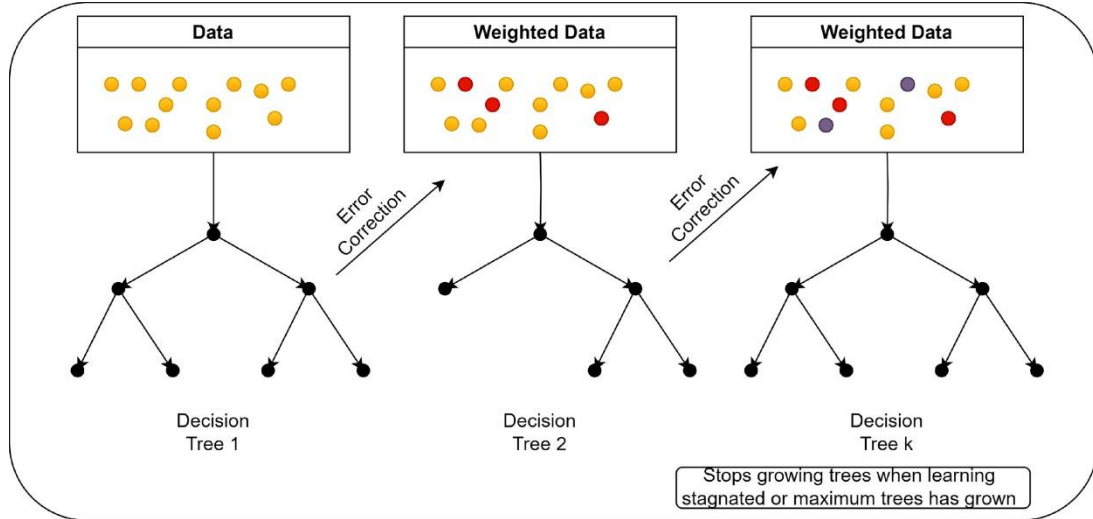


Figure 4.1: Architecture of XGBoost model [36].

1. **Initialization:** The process begins with an initial model  $F_0(x)$  that makes an initial prediction based on the training data. This could be as simple as predicting the mean of the target variable  $y$  as it is shown in Equation (4.2):

$$F_0(x) = \frac{1}{N} \sum_{i=1}^N y_i, \quad (4.2)$$

Where  $F_0(x)$  is the initial model prediction,  $N$  is the number of training samples,  $y_i$  is the true value of the sample.

2. **Gradient Calculation:** At each iteration  $m$ , a new tree  $h_m(x)$  is built to predict the negative gradient (residuals) of the loss function  $L$  with respect to the current prediction  $F_{m-1}(x)$  based on Equation (4.3):

$$r_i^{(m)} = - \frac{\partial L(y_i F_{m-1}(x_i))}{\partial F_{m-1}(x_i)}, \quad (4.3)$$

Where  $r_i^{(m)}$  is the residual for the  $i_{th}$  sample at the  $m_{th}$  iteration.  $L(y_i F_{m-1}(x_i))$  is the loss function (e.g., mean squared error or log-loss).

3. **Model Update:** based on Equation (4.4) The new tree  $h_m(x)$  is added to the previous model  $F_{m-1}(x)$ , and the model is updated by adding a scaled version of  $h_m(x)$  with a learning rate  $\eta$ :

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x), \quad (4.4)$$

Where  $F_m(x)$  is the updated model after the  $m_{th}$  tree is added,  $h_m(x)$  is the new tree fitted to the residuals,  $\eta$  is the learning rate, which controls the contribution of each new tree to the final prediction.

**4. Final Model:** After  $M$  trees are added, the final model is given by Equation (4.5):

$$F_M(x) = F_0(x) + \sum_{m=1}^M \eta \cdot h_m(x), \quad (4.5)$$

This process allows GBMs to iteratively reduce the residual error, thus improving prediction accuracy over multiple iterations.

#### 4.2.1.5. LightGBM

LightGBM is an optimized version of GBMs that improves computational efficiency and scalability, particularly for large datasets. It achieves this by using a histogram-based method for finding optimal splits, as well as other optimizations. LightGBM introduces several innovations, including leaf-wise tree growth and discretizing continuous features into bins, which leads to faster training and reduced memory usage. Here is how the method works:

- 1. Discretization of Features:** LightGBM discretizes continuous features into discrete bins. Based on Equation (4.6), given a feature  $x$ , it divides its range into  $B$  bins (for example,  $[x_1, x_2, x_3, \dots, x_B]$ ). This reduces the number of distinct feature values considered during split selection, leading to faster computation.

$$x_i \in \{b_1, b_2, b_3, \dots, b_B\}, \quad (4.6)$$

Where  $b_1, b_2, b_3, \dots, b_B$  are the bins that the feature values are discretized into.

- 2. Leaf-wise Tree Growth:** Unlike traditional GBMs that use level-wise growth (splitting all nodes at the same level), LightGBM grows trees leaf-wise, where it splits the leaf with the highest residual sum of squares (RSS) based on Equation (4.7). This often results in deeper trees but improves model accuracy by capturing more complex patterns. The tree is grown as follows:

$$RSS_l = \sum_{i \in l} \left( r_i^{(m)} \right)^2, \quad (4.7)$$

Where  $RSS_l$  is the residual sum of squares for leaf  $l$ ,  $r_i^{(m)}$  is the residual error for the  $i_{th}$  sample at the  $m_{th}$  iteration.



- 3. Gradient Calculation and Tree Fitting:** Similar to traditional GBM, LightGBM fits a tree to the residuals. However, LightGBM optimizes the process by using histograms of the residuals to approximate the best possible splits for each feature. The negative gradient for a given leaf is shown in Equation (4.8):

$$r_i^{(m)} = -\frac{\partial L(y_i F_{m-1}(x_i))}{\partial F_{m-1}(x_i)}, \quad (4.8)$$

- 4. Model Update:** The model is updated in the same manner as GBMs, but as it is seen in Equation (4.9) the updates are faster due to the histogram-based approach and the leaf-wise tree growth:

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x), \quad (4.9)$$

- 5. Final Model:** After  $M$  trees are added, the final model in LightGBM is defined by Equation (4.10):

$$F_M(x) = F_0(x) + \sum_{m=1}^M \eta \cdot h_m(x), \quad (4.10)$$

Tree-based models have demonstrated strong performance in MTLF, particularly due to their ability to capture nonlinear dependencies, incorporate external factors, and scale well with large datasets. Zhu et al. [42] applied a decision tree-based approach to forecast electricity consumption in smart grids. Using data that included both seasonal and non-seasonal patterns, the model successfully captured key demand dynamics with high interpretability. While the study did not report significant gains over ensemble models, it emphasized the model's clarity and practical deployment in operational smart grid settings.

In contrast, Mathew et al. [36] developed the Prominence-guided Weighted Peaks (PWP) XGBoost model to improve the accuracy of feeder-level peak load forecasting. The model was trained on feeder-level electricity data supplemented with external variables such as weather and economic indicators. Compared to conventional methods, the PWP-XGBoost achieved lower forecasting errors, particularly in peak demand estimation—an essential factor for maintaining grid stability. The inclusion of domain-specific knowledge into the XGBoost framework contributed to improved precision in capturing complex medium-term load fluctuations.

Similarly, Royal et al. [43] applied the Light Gradient Boosting Machine (LightGBM) model to medium-term electricity load forecasting tasks. The model was trained on time-series datasets characterized by strong temporal dependencies and nonlinear behaviors. LightGBM outperformed traditional statistical models, achieving lower error metrics and demonstrating high computational efficiency due to its histogram-

based learning and feature bundling techniques. These attributes made it particularly effective in forecasting medium-term load trends across diverse operational conditions.

Additionally, Dinmohammadi et al. [44] conducted a comparative study of Random Forest (RF) and Gradient Boosting (GB) models for medium-term load forecasting. The study evaluated performance across multiple regions and timescales and found that RF delivered more stable and consistent accuracy across diverse conditions, while GB models showed slightly better performance in capturing complex patterns when properly tuned. Both models outperformed linear baselines, especially when hyperparameters were optimized. Notably, the authors emphasized the importance of hyperparameter tuning and cross-regional evaluation for generalization. This study further validated that ensemble-based tree methods, particularly when enhanced through tuning and regional adaptation, provide a strong foundation for scalable and accurate MTLF.

Together, these studies highlight the strength of decision tree-based methods—particularly advanced ensembles like XGBoost and LightGBM—in improving forecasting accuracy and interpretability across medium-term electricity consumption horizons, as summarized in Table 4.3.

Table 4.3: Advantages and disadvantages of Tree-based models in MTLF.

| Model Type     | Strengths                                                                                                                                                                                                | Weaknesses                                                                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Decision Trees | Simple and interpretable; Can model seasonal and structural patterns in medium-term consumption data; Quick to train on smaller regional datasets.                                                       | Prone to overfitting without pruning; May struggle with capturing subtle temporal dependencies across longer horizons; Limited generalization, often outperformed by ensembles. |
| Random Forest  | Reduces overfitting by averaging multiple trees; Performs well across diverse urban datasets with multiple influencing factors (e.g., weather, economy); Robust to noise in medium-term historical data. | Slower to predict as horizon and data volume increase; Less interpretable; Requires careful handling of temporal features for accurate long-horizon forecasts.                  |
| XGBoost        | Handles complex temporal patterns (e.g., seasonality and cyclical effects) effectively; Performs well                                                                                                    | Parameter tuning is critical to prevent overfitting on medium-range cycles; Interpretation is more                                                                              |

|                 |                                                                                                                                                                                                                 |                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | on high-dimensional medium-term datasets; Excellent for capturing interactions across multiple time-lagged inputs.                                                                                              | challenging; May require preprocessing of time-based features.                                                                                                                  |
| <b>LightGBM</b> | Scalable and memory-efficient—well-suited for large regional datasets; Captures nonlinear and temporal trends over months; Native handling of categorical variables is useful for calendar-based MTLF features. | May overfit if trees are too deep or not regularized; Performance can degrade on datasets with irregular or sparse medium-term structure; Requires careful feature engineering. |

#### 4.2.2. k-NN

k-NN is a simple and intuitive machine learning algorithm used for both classification and regression tasks. In the context of MTLF, k-NN plays a significant role due to its ability to predict future electricity consumption by identifying similar past patterns in the data. The strength of k-NN lies in its non-parametric nature, meaning that it does not make any assumptions about the underlying distribution of the data. Instead, it learns directly from the data, which makes it adaptable to complex and nonlinear patterns in electricity demand.

In MTLF, k-NN is typically used to forecast future load values based on historical data. The forecasting horizon for medium-term load forecasting generally spans from a few days to several weeks, making it crucial to capture long-term seasonal trends and short-term fluctuations in electricity consumption. The key benefit of k-NN is its ability to model temporal relationships by considering similar past patterns when making predictions.

In the realm of medium-term load forecasting, several studies have successfully applied k-NN, demonstrating its effectiveness in predicting electricity consumption. As Figure 4.2 illustrates, Dong et al. [45] used k-NN combined with NSGA-II (Non-dominated Sorting Genetic Algorithm) for optimizing the number of neighbors ( $k$ ) used in the model. This hybrid model was assessed using electricity consumption data from an urban grid and showed a significant improvement in forecasting accuracy over traditional methods such as ARIMA and SVM. The optimization process, through NSGA-II, also reduced the computational burden while maintaining high accuracy, particularly in capturing the seasonal fluctuations in load demand over a medium-term horizon.

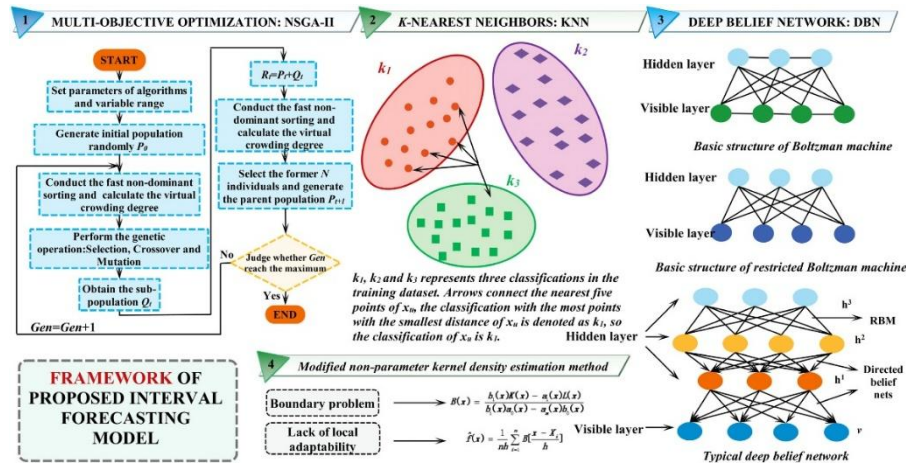


Figure 4.2: Framework of the proposed interval forecasting model and principles of NSGA-II, KNN, DBN and modified KDE [45].

Similarly, Khan et al. [46] explored the application of k-NN combined with Deep Belief Networks (DBN) for medium-term load forecasting in smart grids. In this study, k-NN was used to extract relevant features from the historical electricity consumption data, which were then used as inputs for the DBN model. The results demonstrated that this hybrid approach significantly outperformed traditional models like ANN and SVR, especially when dealing with the nonlinear patterns and demand fluctuations due to external factors such as temperature and economic activity. The study also highlighted that k-NN's role in feature selection played a critical part in improving the performance and scalability of the DBN model.

In their study, Dudek & Peřka [47] focus on the application of Pattern Similarity-Based Machine Learning Methods (PSFMs) for MTLF. Specifically, they investigate the effectiveness k-NN within this framework for forecasting monthly electricity consumption. By leveraging historical load data, the authors demonstrate how k-NN can identify similar past consumption patterns and use them to predict future demand. Their results show that k-NN, despite its simplicity, outperforms traditional models such as ARIMA and SVR in capturing the seasonal variations and nonlinear patterns inherent in electricity consumption. The study highlights that k-NN offers both computational efficiency and forecasting accuracy, especially when combined with

other techniques in a hybrid model, making it a valuable tool for medium-term forecasting tasks.

k-NN continues to be a valuable method for medium-term load forecasting. Its simplicity and non-parametric nature make it adaptable to a variety of datasets, especially when combined with optimization and deep learning techniques. The studies reviewed demonstrate that k-NN, when used alone or in hybrid models, can improve forecasting accuracy, reduce computational complexity, and capture the essential seasonal patterns in electricity demand. A summary of the advantages and disadvantages of k-NN in medium-term load forecasting can be found in Table 4.4.

Table 4.4: Advantages and disadvantages of K-NN in MTLF.

| Model Type | Strengths                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Weaknesses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| K-NN       | Simple and intuitive method, easy to implement and interpret; non-parametric: Does not make assumptions about the underlying data distribution; works well with complex, nonlinear relationships between features; Adaptable to dynamic patterns; can easily incorporate new data without retraining; can model seasonality and trends by considering historical data similarity; good for capturing local patterns in electricity consumption; effective when combined with other models (e.g., hybrid models with optimization techniques). | Computationally expensive during prediction because it requires calculating distances to all training samples; sensitive to the curse of dimensionality: performance can degrade with a high number of features; sensitive to noisy data: outliers can heavily influence predictions; choice of $k$ is crucial: the performance heavily depends on selecting the right number of neighbours; memory-intensive: requires storing large datasets, which may be impractical for large datasets over long time periods; does not work well with imbalanced datasets where certain patterns dominate; slow inference time: Requires comparing each test sample to all training samples, which can be slow for large datasets. |

### 4.2.3. ANN

ANNs are particularly effective in forecasting electricity consumption, which is influenced by complex patterns such as seasonality, weather variability, day-of-week effects, and economic activity. Their ability to model nonlinear relationships and learn from both historical data and external features allows ANNs to capture temporal patterns (daily, weekly, and seasonal cycles) as well as long-term consumption trends. As such, they are widely used in energy management, grid optimization, and electricity demand forecasting.

Khan et al. [46] investigated a hybrid model that combines ANNs with DBN for medium-term load forecasting in smart grid environments. The primary aim was to enhance ANN performance through DBN-based feature extraction and optimization. The model was trained using a comprehensive dataset that included not only historical load profiles but also external parameters such as temperature and economic indicators. Results showed that the ANN-DBN hybrid model significantly outperformed standalone ANN and SVR models. Specifically, the ANN-DBN model achieved a MAPE of 2.83%, compared to 3.54% for SVR and 3.11% for the conventional ANN, demonstrating the efficacy of deep learning integration in capturing complex nonlinear dependencies and improving forecast accuracy.

Wazirali et al. [48] focused on medium-term load forecasting for microgrids using a standalone ANN model. The study utilized historical electricity consumption data along with seasonal indicators to train the network for predicting demand over a one-month horizon. The authors highlighted the ANN's ability to model nonlinear dependencies and capture cyclical patterns influenced by weather and calendar effects. When benchmarked against traditional statistical methods like ARIMA, the ANN model exhibited superior performance, achieving a MAPE of 3.76%, compared to 5.21% for ARIMA. This confirmed that ANNs provide a more flexible and accurate framework for modeling dynamic load behavior, especially in distributed energy systems.

Ghiassi et al. [49] introduced a Dynamic Artificial Neural Network (DAN2) architecture specifically designed for medium-term load forecasting in the context of U.S. electricity utilities. Unlike conventional static feedforward networks, the DAN2 model dynamically determines its structure during training based on data behavior, allowing it to adapt more effectively to nonlinear demand variations. The model was evaluated using real-world system load data from the California Independent System Operator (CAISO), forecasting load over a 30-day horizon. The DAN2 model was benchmarked against standard ANN and ARIMA models using several metrics, including RMSE and MAE. Results showed that DAN2 reduced RMSE by 11.3% compared to static ANN and by 17.8% compared to ARIMA. These results demonstrate that dynamically evolving ANN architecture can better manage complex



temporal dependencies in electricity load forecasting, especially when trends and patterns shift over time.

Yalcinoz and Eminoglu [50] examined both short-term and medium-term electricity load forecasting using feedforward MLP neural networks trained on real distribution system data from Turkey. For the medium-term case, they focused on weekly forecasts over a one-month horizon, integrating inputs such as temperature, humidity, and historical load. Multiple ANN configurations were evaluated based on the number of hidden layers, activation functions, and learning algorithms. Their best-performing model achieved a MAPE of 2.38%, MAE of 12.1 MW, and RMSE of 17.5 MW for the medium-term forecast. These results confirmed that ANNs provided significantly better performance than linear regression models, which had a MAPE exceeding 4.6% under the same conditions. The study also highlighted that incorporating weather data as external inputs and optimizing the neural architecture were critical in enhancing prediction accuracy and robustness.

Both early and recent studies demonstrate the successful application of ANNs in medium-term load forecasting. While Wazirali et al. [48] emphasized the ANN's strength in modeling seasonal and nonlinear variations in microgrid environments, Khan et al. [46] demonstrated that coupling ANNs with deep learning techniques like DBN significantly enhances feature learning and forecasting accuracy. Similarly, Ghiassi et al. [49] showcased the benefits of dynamic network structures that evolve during training, allowing ANNs to better adapt to shifting demand patterns. Yalcinoz and Eminoglu [50] highlighted the effectiveness of carefully configured feedforward networks in achieving high accuracy with limited data. Despite these advances, all approaches point to common challenges, including the requirement for large and diverse training datasets, computational cost, and the need for careful tuning of model parameters to prevent overfitting and ensure generalizability.

A summary of the advantages and disadvantages of using ANNs for medium-term load forecasting is provided in Table 4.5.



Table 4.5: Advantages and disadvantages of ANN in MTLF.

| Model Type | Strengths                                                                                                                                                                                                                                                                                                                                                                                                    | Weaknesses                                                                                                                                                                                                                                                                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANN        | Can model complex nonlinear relationships between features and target variables; flexible architecture allows adaptation to various forecasting problems; high accuracy in forecasting due to its ability to capture intricate patterns in data; works well with high-dimensional data, and can leverage multiple input features; can handle missing data and noisy data through robust learning algorithms. | Requires large amounts of data to train effectively and generalize well; training can be computationally expensive, especially for deep networks; overfitting can occur if the model is not regularized properly; requires careful tuning of hyperparameters (e.g., learning rate, number of layers); less interpretable than simpler models like decision trees or random forests. |

4.3. Deep learning models

Deep learning models have gained significant attention in the field of MTLF due to their ability to model complex, high-dimensional relationships in data, making them well-suited for predicting electricity demand over medium-term horizons. Unlike traditional machine learning models, deep learning models consist of multiple layers of neural networks, which enable them to learn abstract features from raw data without the need for explicit feature engineering.

In MTLF, deep learning models are especially effective for capturing both temporal dependencies (e.g., daily, weekly, and seasonal fluctuations) and nonlinear relationships between various influencing factors, such as weather conditions, historical load data, and economic activity.

4.3.1. RNN

RNNs, particularly their advanced variant known as LSTM networks, have gained prominence in the domain of time series forecasting due to their inherent ability to model temporal dependencies. Unlike traditional statistical approaches, which often rely on assumptions of stationarity and linearity, RNNs are capable of capturing both

short- and long-term patterns in dynamic, nonlinear datasets, making them highly suitable for MTLF.

Muzaffar and Afshari [51] investigated the effectiveness of LSTM networks for electrical load forecasting by using hourly data spanning 13 months. Their study included not only the load time series but also exogenous environmental variables such as temperature. Recognizing the importance of seasonality and trend components in energy consumption, the authors implemented a rigorous preprocessing framework. This included normalization, handling of missing values and outliers, and decomposition of the time series to isolate daily, weekly, and yearly seasonalities. Notably, while traditional methods required fully stationarized inputs, the LSTM network only necessitated the removal of yearly seasonality to prevent misinterpretation of one-year trends as ongoing patterns.

The LSTM network was trained on 12 months of data and validated using the 13th month. The network architecture comprised 60 hidden units, employed the Adam optimizer with a learning rate of 0.005, and was trained over 400 epochs. To enable temporal learning, input sequences were constructed using a lag of five-time steps. This design allowed the model to forecast future values based on dependencies observed in preceding data points.

Forecasts were generated for several horizons—24 hours, 48 hours, 7 days, and 30 days—and evaluated using RMSE and MAPE. Results showed that the LSTM model significantly outperformed classical statistical models such as ARMA, SARIMA, and ARMAX in short-term and intermediate horizons. For instance, LSTM achieved a MAPE of 1.52% for a 24-hour forecast, compared to 5.42% with ARMA and 6.53% with SARIMA. Even at 48 hours, LSTM maintained superior accuracy with a MAPE of 2.16%. However, its performance declined on the 30-day horizon, where it reported a MAPE of 9.75%, higher than the 5.39% obtained using ARMAX. This drop was attributed to the model's limited exposure to long-term patterns due to the relatively short training duration of one year.

The study highlights several critical observations: LSTM's forecasting performance is highly sensitive to model hyperparameters such as the number of hidden units, lag values, and training epochs. Moreover, its ability to generalize and capture seasonality improves with more extensive training data. Despite its slightly lower performance in longer-term forecasts within this study, the LSTM model demonstrated clear advantages in modeling nonlinearities and incorporating multiple input features without the need for manual feature selection.

In conclusion, RNNs—especially LSTM architectures—offer a powerful alternative to traditional MTLF models, particularly when rich, multivariate, and temporally structured datasets are available. Their capacity to learn from raw sequences and model complex temporal relationships enables more accurate and adaptable forecasts, especially over short to medium forecast horizons. As summarized in Table 4.6, these

models provide notable advantages in terms of flexibility and accuracy but require careful handling of data volume, preprocessing, and hyperparameter tuning to mitigate limitations such as overfitting and computational cost.

Table 4.6: Advantages and disadvantages of RNN and LSTM in MTLF.

| Model Type   | Strengths                                                                                                                                                                                                                                                                                                                                                                                                                     | Weaknesses                                                                                                                                                                                                                                                                                                                                                                     |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RNN and LSTM | Capable of capturing both short- and long-term dependencies in sequential data; well-suited for modelling nonlinear and complex temporal patterns in electricity load; can incorporate multiple exogenous variables (e.g., temperature, humidity) without explicit feature engineering; outperforms traditional statistical models (e.g., ARMA, SARIMA) in short- and medium-term horizons when sufficient data is available. | Requires large volumes of high-quality training data to generalize seasonal trends effectively; training can be computationally intensive and time-consuming; model performance is highly sensitive to hyperparameters such as lag size, number of hidden units, and training epochs; risk of overfitting if not properly regularized or if the training dataset is too short. |

4.3.2. Transformer-based models

Transformer architectures, originally introduced for natural language processing, have recently emerged as powerful alternatives to traditional neural networks in time series forecasting. Their ability to model long-range dependencies through attention mechanisms, combined with high parallelism during training and inference, makes them especially well-suited for MTLF—a task defined here as predicting electricity demand over periods ranging from one week to one month.

In MTLF, models must detect and extrapolate periodic patterns, seasonal trends, and the influence of external variables such as temperature, humidity, or electricity prices. RNNs, including their gated variants like LSTMs and GRUs, are naturally suited to sequence modelling but suffer from limitations such as vanishing gradients and inherently sequential computations, which hinder scalability. In contrast, transformers process entire sequences simultaneously, enabling them to learn global relationships more efficiently.

To address the computational limitations of traditional transformers in time series forecasting, Chan and Yeo [52] proposed a sparse attention transformer for electricity load prediction. This model replaced the quadratic-cost dot-product attention with BigBird-style sparse attention, significantly reducing memory requirements while maintaining strong forecasting accuracy. The sparse transformer was shown to scale effectively with long input sequences (up to nine weeks) and performed well for forecast horizons up to one week, making it applicable to medium-term use cases. In empirical comparisons with benchmark models such as LSTM and TSRNN, the sparse transformer achieved comparable or better accuracy while offering up to  $5\times$  faster inference time—a notable advantage for real-time grid operations.

Building on these advances, Peng and Yang [37] introduced LDTformer, a transformer-based model designed specifically for short- and medium-term power load forecasting. The model integrates domain-aware architectural enhancements that enable it to capture both high-frequency fluctuations and long-term temporal dependencies, crucial for forecasting electricity demand over multiple weeks.

LDTformer includes several innovative components:

- A Block Sparse Discrete Cosine Transform (BSDCT) module that converts time-series segments into the frequency domain, improving the extraction of periodic patterns while reducing computational cost.
- A Seasonal-Trend Decomposition (STD) module, which dynamically separates input signals into seasonal and trend components using a mixture-of-experts architecture. This improves the model's adaptability to nonstationary behavior in load profiles.
- A Lagged Correlation Frequency Attention (LCFA) mechanism that introduces learnable lag-specific weights in the frequency domain, enhancing the model's ability to detect long-range and periodic dependencies.
- Causal convolutional layers to enforce temporal consistency while enriching the local context.

These enhancements enable LDTformer to significantly outperform baseline models—including Transformer, LSTM, FEDformer, and Mamba—across different seasons and forecasting windows ranging from one to four weeks. For example, using summer data with a forecast length of 192 time points (equivalent to four weeks at 30-minute intervals), LDTformer achieved a MAE of 0.173 and symmetric mean absolute percentage error (SMAPE) of 0.0186, outperforming all comparison models by substantial margins. Additionally, the model demonstrated greater robustness to sharp fluctuations and seasonal variations, which are typical in real-world electricity consumption.

A comparative summary of transformer-based models used in MTLF is provided in Table 4.7, which outlines their relative strengths and weaknesses. While the vanilla

transformer is effective at capturing long-range dependencies, it suffers from high computational costs and lacks domain-specific enhancements. The sparse transformer addresses these efficiency issues, offering scalability with minimal accuracy loss. Meanwhile, LDTformer represents the most advanced solution, delivering top-tier performance by combining frequency-domain analysis, decomposition techniques, and lag-sensitive attention mechanisms.

Table 4.7: Advantages and disadvantages of transformer-based models in MTLF.

| Model Type         | Strengths                                                                    | Weaknesses                                              |
|--------------------|------------------------------------------------------------------------------|---------------------------------------------------------|
| Sparse Transformer | Efficient inference; suitable for long lookback sequences; robust accuracy.  | May lose information with aggressive sparsity settings. |
| LDTformer          | Integrates time and frequency attention; best-in-class forecasting accuracy. | Increased complexity; higher training overhead.         |

### 4.4. Hybrid models

Hybrid modelling approaches have gained prominence in the field of MTLF, especially when dealing with complex time series such as monthly electricity demand. These models integrate statistical, signal decomposition, and deep learning techniques to enhance forecasting accuracy and model robustness. In this section, recent advances in hybrid models are grouped according to their primary algorithmic foundation—ranging from statistical–deep learning combinations to Transformer-based architectures and signal decomposition frameworks.

#### 4.4.1. Statistical–deep learning hybrids

Combining classical statistical techniques with deep learning offers the advantage of modelling both linear and nonlinear patterns. Dudek et al. [53] proposed a hybrid architecture, as it is shown in Figure 4.3, that integrates Exponential Smoothing (ETS) with a Residual Dilated LSTM network. In their approach, ETS captures trend and seasonal patterns, while the LSTM models the residual nonlinear fluctuations. This two-stage setup was tested on monthly electricity demand data from 35 European countries. The hybrid model achieved a MAPE reduction of up to 25% compared to standalone ETS and LSTM baselines, making it a strong candidate for MTLF applications, particularly where trend and seasonal variation dominate the signal.

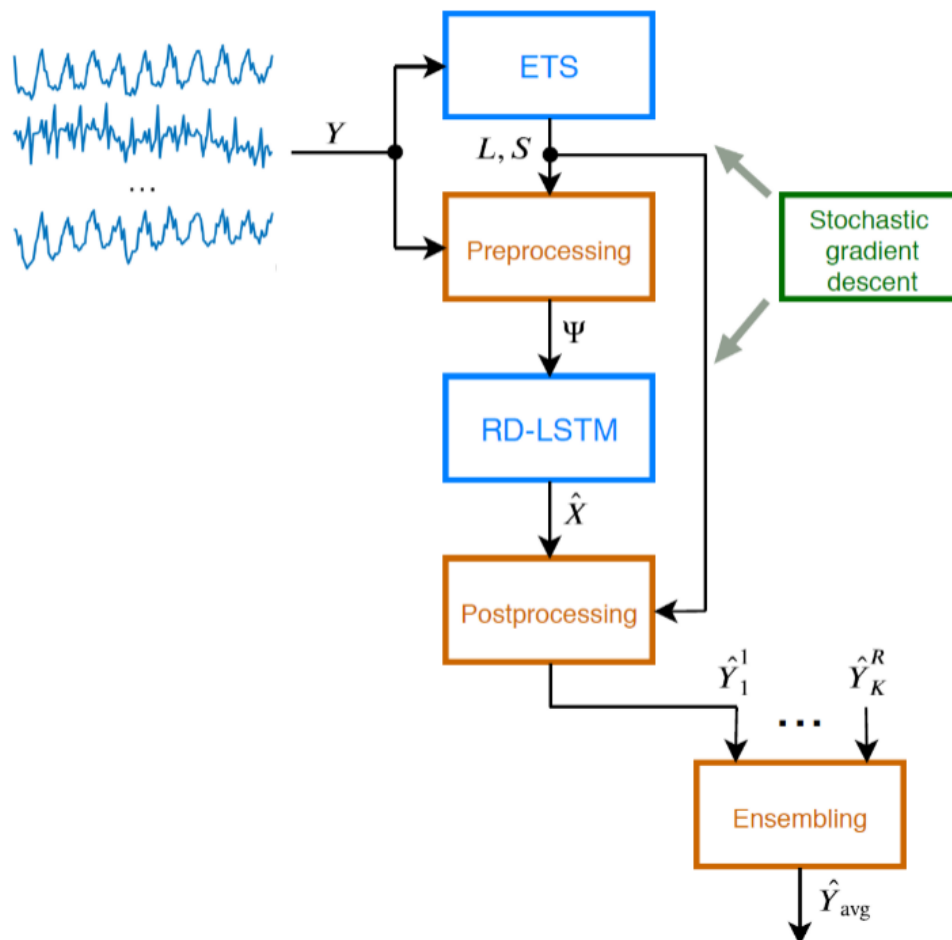


Figure 4.3: Block diagram of the ETS+RD-LSTM forecasting system [53].

Similarly, González Grandón et al. [54] developed a hybrid model that integrates ARIMA for modelling long-term linear trends and an LSTM network for capturing nonlinear variations. Their approach was evaluated on calendar- and temperature-driven electricity load data. The hybrid ARIMA-LSTM model achieved lower RMSE and MAPE values than both standalone LSTM and ARIMA models, with a reported

MAPE of 2.48% and RMSE of 18.3 MW. This confirms the synergy between linear and nonlinear modelling in improving forecasting precision at monthly horizons.

#### 4.4.2. Transformer-based hybrids

Transformer-based architectures have revolutionized sequence modelling due to their capacity to model long-range dependencies. Building on this foundation, Wang et al. [55] proposed a Hybrid Autoformer Framework (HAFF) specifically for electricity demand forecasting that is illustrated in Figure 4.4. Their model introduces a preprocessing pipeline using MLR with 10 variables and Bootstrap sampling, followed by an enhanced Autoformer with improved auto-correlation mechanisms. Evaluated on datasets from Taixing, China, and New South Wales, Australia, the HAFF model achieved a MAPE of 1.63% and RMSE of 47.28 MW in Taixing, and 2.43% and 239.96 MW, respectively, in New South Wales. These results outperformed standard Transformer variants like Reformer and Informer, especially when data were limited or seasonal fluctuations were prominent.

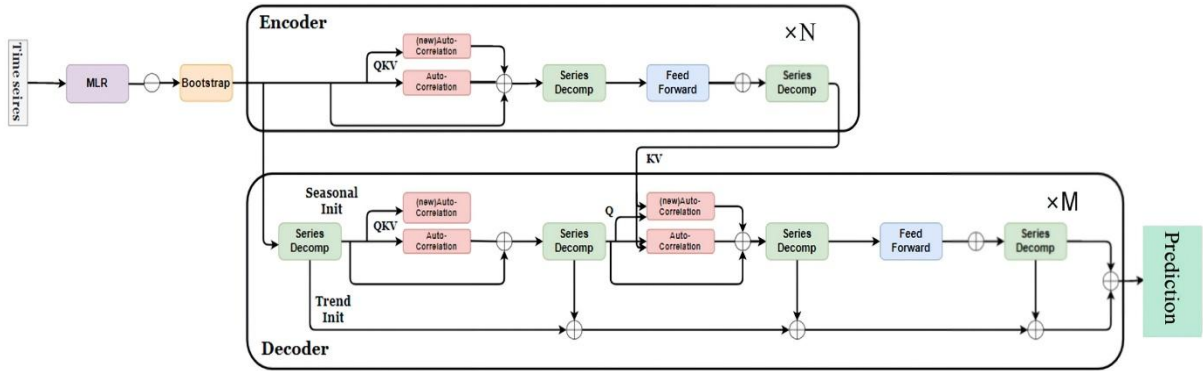


Figure 4.4: The whole structure of HAFF [55].

Expanding on Transformer designs, Antonesi et al. [56] introduced a hybrid model that integrates Transformers with Liquid Neural Networks (LNNs), CNN encoders, and adaptive positional encodings. Their model was evaluated across various building types—including households and commercial facilities—using daily and monthly energy consumption data. The model demonstrated a SMAPE improvement ranging from 1.5% to 50% over basic Transformer, LSTM, and ANN models. With  $R^2$  scores exceeding 0.92 in most cases, and a training time reduction of 8% over the base Transformer, the proposed model balances accuracy with computational efficiency, making it a compelling hybrid solution for monthly MTLF scenarios.

#### 4.4.3. CNN-LSTM hybrids

CNN-LSTM architectures are widely used for their ability to simultaneously extract spatial features and model temporal dependencies. Yıldız Doğan et al. [57] introduced a CNN-LSTM hybrid model aimed at estimating electricity demand for sustainable cities. Using hourly electricity consumption and meteorological data collected over a five-year period from a city in Türkiye, the model was evaluated on its ability to



forecast medium-term electricity demand. The CNN component extracts high-level temporal patterns, while the LSTM captures sequential dependencies. The hybrid model achieved a MAPE of 2.9%, RMSE of 1.36, and an  $R^2$  value of 0.86—outperforming the standalone LSTM model ( $R^2 = 0.81$ ), thereby demonstrating its superiority in learning complex temporal trends over medium horizons.

Zhang et al. [58] introduced an innovative multi-task hybrid model, MTMV-CNN-LSTM, which combines Convolutional Neural Networks and Long Short-Term Memory networks within a multi-task learning framework to forecast both short- and medium-term electricity loads. The model addresses the issue of redundant or mismatched feature intervals—such as weather data at daily intervals versus power load at 15-minute intervals—by structuring the learning tasks hierarchically. The auxiliary task handles slower-varying features (e.g., weather, calendar effects), while the main task predicts detailed load profiles. For medium-term forecasting, defined in their study as a 3-month horizon, the MTMV-CNN-LSTM achieved superior results: it outperformed six baseline models including Prophet, CatBoost, GRU, LSTM, GAN, and even a standard CNN-LSTM. Specifically, it recorded an RMSE of 0.053, MAE of 0.044, and MSE of 0.003, with an  $R^2$  of 0.640. This architecture demonstrates excellent robustness, maintaining accuracy under input perturbations such as added Gaussian and uniform noise, with  $R^2$  values consistently above 0.8 even under noisy conditions. Its ability to generalize across multiple horizons and handle complex feature structures makes it a strong candidate for monthly MTLF applications.

#### 4.4.4. Signal decomposition-based hybrids

Signal decomposition methods such as Empirical Mode Decomposition (EMD) and its ensemble variants offer powerful tools for handling non-stationary and multi-scale time series. Lei Wang et al. [59] proposed a hybrid model based on Figure 4.5—EEMD-SE-PSO-SVR—that leverages ensemble empirical mode decomposition (EEMD) to break down the demand signal, uses sample entropy (SE) to filter noise, and applies SVR with parameters optimized via Particle Swarm Optimization (PSO). Applied to monthly electricity demand in China, this hybrid method outperformed traditional SVR and statistical models, reducing the MAPE by 54.49%. The authors highlighted the model's ability to manage data irregularities and nonlinear dependencies, which are common in medium-term forecasting tasks.

In another decomposition-based hybrid, Huang et al. [60] introduced an EMD-LSTM-Markov model for predicting building cooling loads. The architecture first decomposes the load signal using EMD, applies LSTM to model temporal sequences, and finally incorporates a Markov chain to adjust state transitions based on prior probabilities. Although designed for cooling loads, the model achieved a MAPE of 0.84% and RMSE of 4.53 in monthly-scale prediction tasks, demonstrating its adaptability to broader MTLF applications.

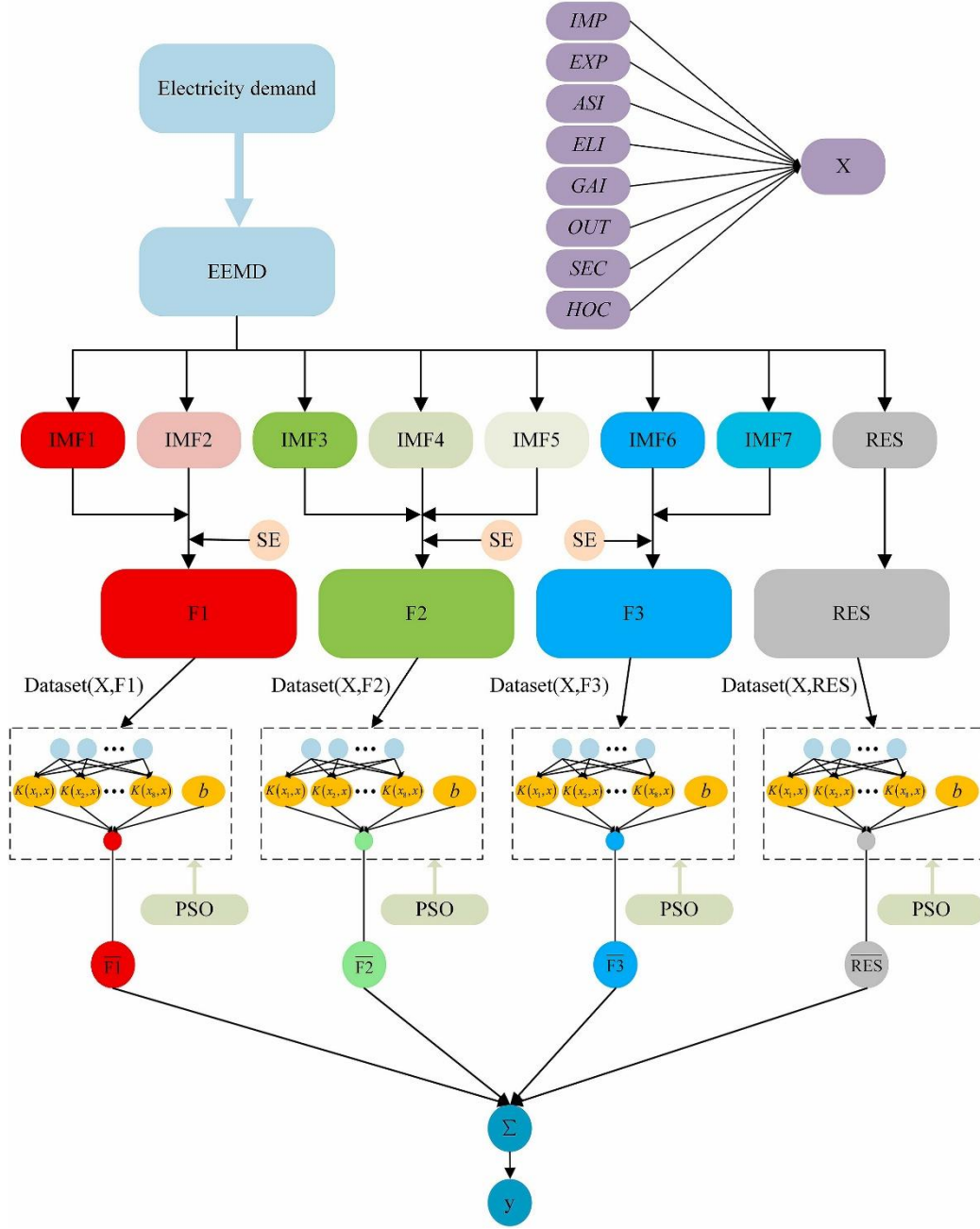


Figure 4.5: The architecture of the proposed hybrid model by Lei Wang et al [59].

#### 4.4.5. Neural Prophet–LSTM hybrids

Shohan et al. [61] developed a novel architecture by combining a LSTM model with Neural Prophet (NP), a model built upon Facebook Prophet but enhanced with autoregressive and seasonality handling features. As Figure 4.6 illustrates, outputs from LSTM and NP are combined via an ANN, resulting in a hybrid architecture tailored to different forecasting horizons. Evaluated on ERCOT and NREL datasets for both short- and medium-term forecasting, the model achieved lower MAPE values across all benchmarks, with monthly MAPE dropping to 1.79% and RMSE falling

below 2.3 MW. The integration of NP offers increased interpretability, while the LSTM preserves sequential learning, striking a strong balance between accuracy and model explainability.

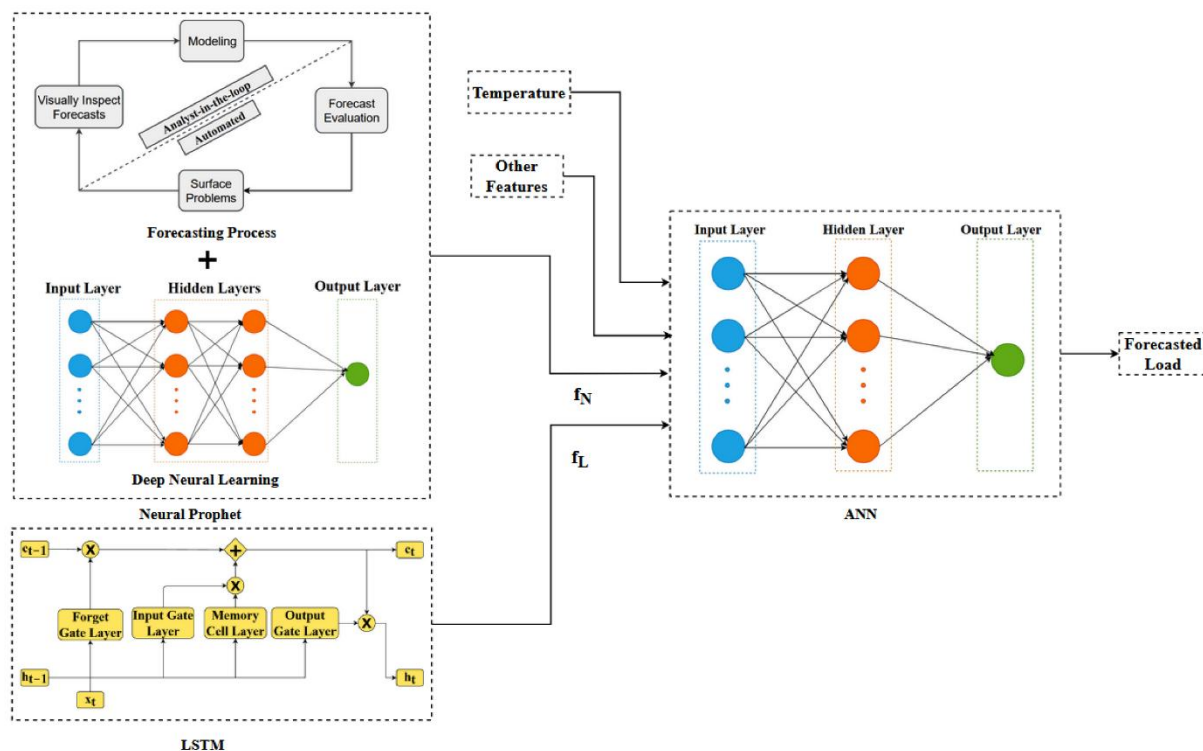


Figure 4.6: Block diagram of proposed hybrid model by Shohan et al [61].

These diverse hybrid approaches underscore the increasing sophistication and customization of MTLF models in addressing nonlinearity, seasonality, and data scarcity. A comparative summary of these models, highlighting their strengths and limitations, is provided in Table 4.8.

Table 4.8: Benefits and weakness of hybrid models in MTLF.

| authors                      | Benefits                                                          | Weakness                                     |
|------------------------------|-------------------------------------------------------------------|----------------------------------------------|
| Dudek et al. [53]            | Handles trend/seasonal + residuals; accurate for monthly scales   | Requires tuning of two separate model stages |
| González Grandón et al. [54] | Combines statistical structure with temporal learning             | Sensitive to ARIMA lag parameters            |
| Wang et al. [55]             | Outperforms base Transformers; robust to seasonal and sparse data | Complex multi-stage preprocessing            |

|                          |                                                                          |                                                           |
|--------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------|
| Antonesi et al. [58]     | High accuracy and generalization; efficient for monthly predictions      | Architecturally complex; training stability needs care    |
| Yildiz Doğan et al. [57] | Captures both local and temporal patterns; strong generalization         | May overlook external variables like price or policy      |
| Zhang et al. [58]        | Excellent on 3-month prediction; multi-task learning improves generality | Slightly lower $R^2$ than simpler CNN-LSTM baseline       |
| Lei Wang et al. [59]     | Strong for non-stationary data; robust error reduction                   | Computationally intensive due to decomposition and tuning |
| Huang et al. [60]        | Captures states and transitions; stable and accurate                     | Markov component can be lag-sensitive                     |
| Shohan et al. [61]       | High accuracy and interpretability; robust across multiple horizons      | Integration of NP lacks theoretical transparency          |

## 4.5. Summary

Chapter 4 provided an in-depth review of methodologies used for medium-term load forecasting, which targets horizons ranging from one week to one month. This forecasting range plays a critical role in strategic energy planning, maintenance scheduling, and tariff design. The chapter explored four major categories of MTLF techniques: statistical models, machine learning approaches, deep learning architectures, and hybrid frameworks.

Statistical models such as Linear Regression, Quantile Regression, ARIMA, and SARIMA offer simplicity and interpretability, performing well under stable seasonal conditions. However, their capacity to manage nonlinear patterns and evolving consumption behaviors is limited. Enhancements like Fourier-ARIMA and kernel density methods improve adaptability but increase model complexity.

ML methods—especially tree-based models like Random Forests, XGBoost, and LightGBM—demonstrated strong performance in handling nonlinearities, temporal patterns, and external features such as weather and economic variables. k-NN offered simplicity and interpretability but faced challenges with high-dimensional or noisy datasets. ANN models provided higher accuracy by capturing intricate nonlinear patterns but required substantial data and careful tuning.

Deep learning models, particularly LSTM networks, proved effective in modeling long-term temporal dependencies and nonlinear load behaviors, though they demand

high computational resources and large datasets. Transformer-based models emerged as a promising alternative, offering improved scalability, parallelism, and robustness—especially in complex and data-scarce scenarios. Advanced variants like LDTformer demonstrated state-of-the-art forecasting accuracy for monthly horizons.

Hybrid models integrated the strengths of multiple approaches to overcome individual limitations. Combinations such as ETS+LSTM, ARIMA+LSTM, CNN-LSTM, and Transformer-based hybrids showed superior accuracy, robustness, and generalizability. Signal decomposition and optimization techniques (e.g., EEMD, PSO, NSGA-II) further enhanced model performance, especially under data irregularities and seasonal fluctuations.

In conclusion, the chapter underscores that while traditional models retain value in stable environments, ML, deep learning, and hybrid methods are increasingly essential for accurate and resilient MTLF. Their adaptability to evolving consumption patterns and integration with external variables make them indispensable for modern energy systems.

## 5 Long-term load forecasting

LTLF involves predicting electricity demand over extended horizons, typically ranging from one year to several decades. These forecasts play a crucial role in strategic planning for power system infrastructure, investment decisions, energy policy development, and integration of renewable energy sources. Unlike short-term or medium-term forecasting, which focuses on operational or tactical grid management, LTLF emphasizes the structural evolution of demand influenced by socioeconomic growth, technological shifts, climate policy, and demographic trends.

Accurate long-term forecasts enable utility companies, transmission operators, and governmental bodies to ensure sufficient capacity, optimize capital allocation, and evaluate the long-term viability of energy projects. However, LTLF presents several challenges due to high uncertainty, evolving consumption patterns, and limited availability of high-resolution long-term data. As a result, LTLF models often rely on macroeconomic indicators, population growth projections, urban development plans, and technological adoption rates.

Among various modeling approaches, regression-based techniques have remained a cornerstone of LTLF due to their interpretability, ease of implementation, and ability to incorporate trend and calendar effects. The following sections explore key studies that apply regression models to long-term electricity demand forecasting, highlighting their methodologies, performance, and practical implications.

### 5.1. Statistical methods

Statistical methods form the foundational basis of long-term load forecasting due to their simplicity, transparency, and capacity to model historical trends and seasonal patterns. These approaches typically establish relationships between electricity consumption and external influencing factors such as population, GDP, industrial output, or calendar variables.

For LTLF, commonly used statistical techniques include multiple linear regression, logistic regression, and time series decomposition. Multiple linear regression is effective in capturing linear trends based on socio-economic indicators and calendar effects, while logistic regression is particularly suited for modeling saturated growth in electricity demand — a typical pattern in mature economies or fully electrified



regions. These models offer interpretability and are valuable for scenario-based planning and policy simulation.

Despite their strengths, statistical methods may struggle to capture nonlinearities, structural breaks, and evolving behavioral patterns over extended horizons. Therefore, they are often used in combination with clustering or scenario analysis to enhance robustness in long-term forecasts.

#### 5.1.1. Regression-based models

Regression-based approaches have long served as the backbone of long-term load forecasting due to their simplicity, interpretability, and effectiveness in modeling relationships between electricity demand and socioeconomic variables. These models typically assume that electricity consumption can be expressed as a function of independent variables such as population growth, gross domestic product (GDP), industrial activity, and calendar-related factors. For long-term forecasting horizons, regression models are especially valuable in scenario planning, allowing utilities and policymakers to assess how structural and demographic trends will influence future electricity needs.

One representative study employing regression methods for long-term forecasting is by Nano et al. [19], who proposed a MLR model enhanced with calendar effects. The model integrates variables such as day type (weekday, weekend, holiday), seasonal indicators, and long-term temporal trends to forecast electricity demand at a yearly scale. The framework is designed for practical utility applications and captures recurring consumption patterns over multiple years. Although specific error metrics such as MAPE were not detailed, the study reported strong alignment with historical trends, indicating good reliability in forecasting seasonal variations. However, the model's reliance on linear assumptions limits its flexibility in modeling nonlinear growth, particularly in systems approaching demand saturation.

In a more complex formulation, Royal et al. [43] developed a regression-based statistical framework tailored for district-level long-term forecasting. Their methodology combines linear regression with aggregated seasonal profiles and incorporates localized planning data, such as infrastructure developments and population projections. The model was applied over a multi-year horizon and achieved a MAPE below 7%, showcasing its robustness for practical long-term planning. Compared to advanced machine learning models, this framework offers improved transparency and explainability—important qualities for long-term strategic decisions—though it may not be as responsive to abrupt structural changes or nonstationary patterns.

A more recent contribution by Feng et al. [62] addresses the limitations of linearity in traditional regression by introducing an improved logistic regression model for saturated load forecasting. Their approach models electricity demand using an S-



shaped logistic curve, which is particularly well-suited for markets approaching maturity where demand growth slows and eventually plateaus. To enhance performance, the study also integrates affinity propagation clustering to group similar load profiles, improving both prediction accuracy and segmentation. The model demonstrated superior accuracy in predicting long-term saturation levels and outperformed conventional linear regression methods, with reported MAPE values below 5% in year-ahead forecasts. This approach provides a more realistic depiction of long-term load dynamics in stable or mature energy systems.

Together, these studies demonstrate the diverse ways regression models can be adapted for long-term load forecasting—from capturing linear trends and calendar effects to modeling saturation behaviors in maturing markets. While regression models are valued for their transparency and ease of implementation, they may require hybridization or advanced calibration techniques to manage complex nonlinearities and emerging consumption behaviors over long planning horizons.

As summarized in Table 5.1, different regression techniques offer specific advantages and limitations when applied to LTLF, depending on the structural characteristics of the electricity system and the forecasting context.

Table 5.1: Advantages and disadvantages of regression -based models in LTLF.

| Model Type                         | Strengths                                                          | Weaknesses                                             |
|------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|
| Multiple Linear Regression         | Simple, interpretable; effective for trend and calendar modeling   | Cannot capture nonlinear or saturation behavior        |
| Linear Regression (District-level) | Robust in localized long-term trends; useful for scenario analysis | Less flexible under abrupt structural changes          |
| Improved Logistic Regression       | Captures demand saturation; better long-term curve fitting         | Requires saturation context; more complex to calibrate |

5.1.2. ARIMA and SARIMA

Time series models such as ARIMA and its seasonal extension SARIMA have remained foundational tools in load forecasting due to their ability to model temporal dependencies, long-term trends, and seasonal fluctuations. In the context of LTLF, these models are particularly useful for systems with stable consumption patterns and clear seasonality. Their mathematical transparency and relatively low data requirements make them attractive for policy-oriented planning and strategic demand estimation.

Kumar et al. [63] explored the use of ARIMA, ARMA, and SARIMA models for forecasting long-term load profiles in smart-metered distribution systems. The study focused on capturing consumption behavior in the presence of granular data made available by smart metering infrastructure. Their findings underscored that SARIMA, in particular, was able to effectively capture periodic consumption patterns and long-term shifts in usage. While the authors did not provide specific numerical values for forecast errors, they emphasized that SARIMA outperformed ARIMA and ARMA in scenarios where seasonal variability was pronounced. The study highlighted those statistical models, when adequately parameterized and applied to high-quality smart meter data, could provide reliable forecasts for annual load planning.

A more detailed comparative study by Dubey et al. [64] examined the performance of SARIMA against a deep learning model, LSTM, for time series forecasting of energy consumption. The authors applied both models to daily energy usage data enriched with weather-related features such as temperature, humidity, and wind speed. The SARIMA model successfully captured the seasonal dynamics in the data and produced stable forecasts over extended horizons. However, it lagged behind LSTM in terms of predictive accuracy. The study reported that SARIMA yielded a MAE compared to LSTM, indicating that while SARIMA is well-suited for seasonally-driven systems, its linear structure may be less effective in modeling nonlinearities present in evolving energy consumption patterns. Nonetheless, SARIMA was preferred for its interpretability and resilience against overfitting, which are critical in long-term utility forecasting where transparency and reproducibility are prioritized.

Both studies suggest that ARIMA-based models remain viable and useful in LTLF, particularly when forecasting is conducted in mature systems with historical stability and when explainability is a necessary criterion. However, their effectiveness may diminish in contexts characterized by abrupt structural changes, such as the integration of electric vehicles, distributed energy resources, or dynamic tariff structures. In such cases, hybrid or data-driven models may offer improved accuracy. Nonetheless, the ability of SARIMA to manage seasonality and ARIMA's simplicity continue to make them valuable tools in the long-term forecasting toolkit.

As summarized in Table 5.2, ARIMA and SARIMA methods offer distinct advantages in long-term forecasting scenarios, particularly in relation to transparency, seasonality handling, and data efficiency. However, their performance can be constrained by linearity assumptions and the inability to adapt to sudden behavioral shifts in energy usage.

Table 5.2: Advantages and disadvantages of ARIMA and SARIMA methods in LTLF.

| Model Type | Strengths                                                | Weaknesses                                                         |
|------------|----------------------------------------------------------|--------------------------------------------------------------------|
| ARIMA      | Simple, interpretable, low data requirement              | Poor seasonality handling; struggles with nonlinear patterns       |
| SARIMA     | Captures seasonality well; robust for stable time series | Less accurate than deep learning models; rigid under system change |

5.2. Machine learning methods

ML methods have emerged as powerful alternatives to traditional statistical models for LTLF, especially in contexts where electricity consumption is influenced by a large number of interdependent and nonlinear factors. Unlike classical approaches that rely on explicit assumptions about data distribution or trend linearity, ML models learn patterns directly from historical data, enabling them to adapt to complex relationships between electricity demand and a wide range of exogenous variables, including weather, socio-economic indicators, urbanization trends, and behavioral shifts.

In the context of LTLF, ML models such as decision trees, support vector machines, k-nearest neighbors, and ensemble techniques like Random Forests and Gradient Boosting have shown increasing relevance. These models can accommodate high-dimensional datasets and are often more flexible than linear models when forecasting long-term consumption influenced by evolving market structures and energy usage patterns. Their ability to generalize from past trends and adjust to non-stationary environments makes them particularly useful in modern energy systems characterized by decentralization, demand-side variability, and renewable integration.

However, the effectiveness of ML methods in LTLF depends significantly on the quality and volume of data, careful feature selection, and model interpretability. While some methods offer high accuracy, they may lack transparency, posing challenges in regulatory or policy-driven planning where explainability is essential.

Let me know which subsections you would like to explore next—e.g., tree-based models, support vector regression, k-NN, or others.

5.2.1. SVR

SVR has emerged as a powerful technique for long-term load forecasting due to its ability to manage nonlinear relationships and limited datasets while maintaining robustness against overfitting. SVR models are particularly effective in capturing the complex dependencies that characterize annual electricity demand patterns.

Wang et al. [65] proposed an annual load forecasting model based on SVR optimized using a Differential Evolution (DE) algorithm. The study focused specifically on one-

year-ahead electricity demand forecasts. The SVR model, when tuned using DE, demonstrated enhanced predictive performance compared to traditional regression methods. For instance, the optimized SVR model achieved lower MAPE compared to both the standard SVR and linear regression benchmarks. This result confirms that fine-tuning the hyperparameters of SVR can significantly improve its applicability to annual load forecasting, where prediction horizons are extended and data variability increases.

Hu [66] conducted a comparative study evaluating SVR, neural networks, and Exponentially Weighted Moving Average (EWMA) models for long-term load forecasting in the context of air conditioning systems in large buildings. Although the study included other techniques, SVR was examined as a stand-alone model. Results showed that SVR offered competitive accuracy in predicting long-term cooling loads, particularly when nonlinearities and seasonal fluctuations were present. However, it slightly underperformed compared to neural networks in capturing rapid load transitions, especially in extreme temperature conditions. The SVR model, while stable and interpretable, produced higher RMSE values than the neural network model in high variability periods, indicating a limitation in dynamic adaptability.

Kazemzadeh et al. [67] presented a hybrid forecasting model that integrates SVR with ARIMA and ANN, using PSO for parameter tuning. Although the study's primary focus was on a hybrid approach, it also evaluated SVR independently as a baseline. For long-term forecasting of yearly peak load and energy demand in Iran's National Electric Energy System, the SVR model alone produced respectable forecasting accuracy, but it was outperformed by the hybrid model in terms of both MAPE and RMSE. Specifically, SVR's standalone MAPE was in the range of 5.3% to 6.7%, while the hybrid model reduced this to below 4.5%. This finding underscores that SVR performs reasonably well on its own for long-term horizons but can benefit substantially when integrated into ensemble frameworks that address its limitations in temporal modelling.

In conclusion, SVR provides a solid foundation for long-term forecasting and is especially effective when carefully tuned or used as a component within hybrid systems. While its performance may trail slightly behind deep learning models or hybrid systems in some scenarios, SVR's interpretability and generalization capacity make it a valuable tool for annual load prediction in both residential and commercial domains. These findings are summarized in Table 5.3, comparing the strengths and limitations of SVR-based approaches across different applications in LTLF.

Table 5.3: Advantages and disadvantages of SVR method in LTLF.

| Model Type | Strengths                                                                                                                                                                                                                             | Weaknesses                                                                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SVR        | Effectively models nonlinear relationships in yearly energy demand and building load profiles; performance can be significantly improved using optimization techniques like Differential Evolution; adaptable across various sectors. | Performance highly depends on hyperparameter tuning; standalone SVR may yield lower accuracy compared to neural networks or hybrid models, especially in capturing peak loads. |

5.2.2. Tree-based models

Tree-based models—including Decision Trees (DTs), Random Forests (RF), and Gradient Boosting Machines (GBMs)—have gained significant traction in long-term load forecasting (LTLF) due to their ability to model complex nonlinear interactions without assuming a specific data distribution. These models are especially useful when forecasting yearly or multi-month electricity demand, where features such as seasonality, economic indicators, and demographic patterns interact in complex ways.

Dinmohammadi et al. [44] conducted a detailed evaluation of ensemble tree-based models for long-term residential electricity consumption forecasting. Their study compared Random Forests and Gradient Boosting algorithms in predicting energy usage across different housing profiles and geographic regions over an annual horizon. The results demonstrated that Gradient Boosting achieved slightly better accuracy (MAPE  $\approx$  5.8%) compared to Random Forests (MAPE  $\approx$  6.2%), although both significantly outperformed traditional regression models. The authors highlighted that Random Forests provided more stable results across varying input conditions, while Gradient Boosting offered higher precision when hyperparameters were carefully tuned. Additionally, the ensemble models showed resilience to noise and missing values, which are common in real-world LTLF datasets.

Sangrody et al. [68] explored multiple machine learning methods for long-term electricity demand forecasting in the U.S., including decision tree-based algorithms. Although their main focus was a comparative framework across several learning models, they confirmed the relevance of tree-based models for capturing nonlinear trends in multi-year forecasts. In particular, decision tree models were useful for feature importance analysis and detecting seasonal or structural patterns in consumption behavior. The authors noted that while tree-based models were generally outperformed by deep learning techniques in accuracy, they remained

attractive for their interpretability and lower computational cost—especially in grid planning and regulatory applications.

Together, these studies affirm the practical strengths of tree-based models for LTLF: ensemble trees like Random Forests and Gradient Boosting achieve a strong balance between accuracy and generalizability, while individual trees serve as interpretable tools for exploring underlying data structures. Despite being slightly less accurate than some deep learning models, their transparency and robustness make them indispensable in long-term forecasting contexts, particularly where model explainability and stakeholder trust are priorities.

As summarized in Table 5.4, the main strength of tree-based models lies in their ability to adapt to nonlinear patterns in high-dimensional data with limited preprocessing. However, their performance varies depending on the algorithm: standalone trees are interpretable but less accurate, while ensemble methods improve forecasting performance but require more resources and careful tuning.

Table 5.4: Advantages and disadvantages of tree-based models in LTLF.

| Model Type               | Strengths                                                       | Weaknesses                                                                      |
|--------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Decision Trees</b>    | Easy to interpret; low computation; useful for policy contexts. | Lower accuracy for long horizons; sensitive to data noise.                      |
| <b>Random Forest</b>     | Stable across datasets; handles missing data and noise well.    | Less interpretable; require larger data volumes, more memory and training time. |
| <b>Gradient Boosting</b> | High accuracy; good with complex nonlinear trends.              | Sensitive to hyperparameters tuning; slower to train                            |

### 5.2.3. ANN

ANNs have been widely adopted in load forecasting tasks due to their ability to model complex, nonlinear relationships between historical consumption and external influencing factors. In LTLF, where the prediction horizon spans several months to years, ANNs are particularly valuable because they can learn from high-dimensional input spaces and accommodate a variety of exogenous variables such as economic indicators, demographic trends, and weather patterns.

Sangrody et al. [68] applied machine learning approaches—including ANNs—for long-term electricity demand forecasting in their 2018 study. They designed their model using historical demand data and various macroeconomic inputs, demonstrating that ANN models were capable of capturing evolving patterns in electricity consumption with higher flexibility than conventional linear models. Although exact numerical error values were not provided, the authors emphasized that ANNs outperformed baseline models in tracking long-term trends and seasonal



variations in electricity usage. The flexibility of ANN models was particularly useful in handling fluctuations caused by population growth and policy changes.

Similarly, Ozdemir [69] conducted a comprehensive evaluation of several artificial intelligence and machine learning methods—including ANNs—for predicting Turkey's annual electricity consumption up to 2030. The study considered input features such as GDP, population, and international trade indicators. Among all tested models, four were shortlisted based on forecasting accuracy: Stepwise Linear Regression (SLR), GA, Gaussian Process Regression (GPR), and Neural Networks. Although SLR had the lowest MAPE of 2.36%, the neural network model also exhibited competitive performance and was retained for final scenario-based projections. The use of NN proved effective in modeling the nonlinear and multidimensional characteristics of energy demand data over extended horizons. The study also highlighted the strength of NN in integrating multiple socio-economic variables, making it well-suited for national-level energy planning.

Together, these studies underscore the relevance of ANNs for LTLF applications. While ANNs require careful configuration of architecture and training parameters to avoid overfitting, they remain one of the most robust tools for capturing hidden patterns and interactions in long-term consumption data. Table 5.5 summarizes the key strengths and limitations of ANN models for LTLF based on the reviewed literature.

Table 5.5: Advantages and disadvantages of ANN model in LTLF.

| Model Type | Strengths                                                                                                                                                                                      | Weaknesses                                                                                                                             |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| ANN        | Captures nonlinear and complex patterns over long-term horizons; handles multivariate and socioeconomic inputs effectively; adaptable to changes in consumption behavior and economic factors. | Sensitive to parameter tuning; prone to overfitting with small datasets; requires high computational resources and data preprocessing. |

### 5.3. Deep learning

Deep learning models have emerged as powerful tools in LTLF due to their ability to model highly nonlinear relationships, extract complex features, and learn from sequential patterns in large-scale energy datasets. These models outperform many traditional statistical and machine learning techniques when sufficient historical and contextual data are available, particularly over extended forecast horizons.



In the context of LTLF, where electricity demand must be projected months or years ahead, deep learning architectures are valued for their scalability and capacity to handle multivariate inputs without extensive feature engineering. However, they also come with limitations such as high computational requirements, sensitivity to training data quality, and limited interpretability.

### 5.3.1. Multilayer perceptron

MLP neural networks have demonstrated significant potential in long-term load forecasting due to their capacity to model complex, nonlinear relationships across multi-year datasets. These networks are especially effective when historical consumption data is influenced by multiple interdependent variables such as seasonal trends, economic growth, or policy changes.

Faraji et al. [70] introduced a long-term forecasting approach for annual load growth in microgrids using an MLP model. The research focused on the optimal planning of a grid-connected microgrid in Tehran, employing HOMER software for simulating different load growth scenarios. Their method used historical annual load data to train the MLP, capturing nonlinear growth patterns over multiple years. While the paper primarily emphasizes the impact of accurate load growth prediction on system planning rather than detailed error metrics, it highlights how the use of MLP forecasting leads to more reliable optimization results when compared with traditional linear forecasting models.

Complementing this, Kim et al. [71] proposed a hybrid long-term forecasting model that is shown in Figure 5.1 and combines MLP with statistical techniques to overcome the average convergence limitations seen in standalone machine learning methods. Their model was trained and validated using massive-scale real-world data from over 10,000 distribution lines provided by Korea Electric Power Corporation (KEPCO). The hybrid approach yielded an average accuracy of 89.56% based on Normalized Root Mean Squared Error (NRMSE), outperforming models based purely on statistical regression or MLP alone. Notably, their study validates the hybrid model for practical deployment in national-scale distribution network planning, underscoring its robustness for multi-year forecasting applications.

Together, these studies illustrate the strengths of MLP-based models in capturing complex nonlinear demand trends across extended time horizons. While MLP alone proves useful in improving the precision of load growth estimations, its combination with statistical approaches offers improved stability and generalization, especially when applied to large-scale datasets.

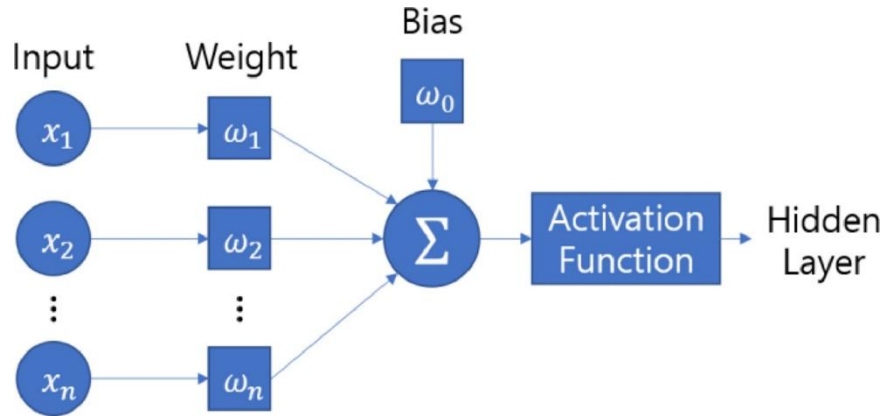


Figure 5.1: Feedforward procedure of neural network for MLP by Kim et al. [71].

As summarized in Table 5.6, MLP models provide a flexible and powerful approach to LTLF, with their benefits and weaknesses largely determined by dataset size, model tuning, and whether hybridization is used.

Table 5.6: Advantages and disadvantages of MLP model in LTLF.

| Model Type | Strengths                                                                                                                          | Weaknesses                                                                                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>MLP</b> | Model nonlinear relationships; suitable for annual load growth; high accuracy when hybridized; effective in real-world deployment. | Standalone MLP may suffer from convergence issues; performance sensitive to architecture and training data quality. |

### 5.3.2. LSTM

LSTM networks are a class of RNNs widely used for sequential data modelling due to their ability to capture long-term dependencies through gated mechanisms. While traditional LSTM models have shown promising results in many time series applications, their direct application to LTLF has faced limitations, especially in capturing extended temporal patterns and handling data scarcity.

To address these challenges, recent studies have introduced enhanced LSTM architectures or combined LSTM with complementary techniques.

Dai et al. [72] introduced a novel model illustrated in Figure 5.2 and called Segmented Self-Attention LSTM (SALSTM), which integrates segmented iteration and intra-segment self-attention mechanisms into the LSTM framework. This architecture was specifically developed to tackle the issue of performance degradation in long-sequence forecasting. In traditional LSTM, as the prediction length increases—such as forecasting 96-time steps ahead—the model's ability to retain meaningful information diminishes significantly. The SALSTM model mitigates this by reducing the recursive

steps needed and embedding self-attention within segments to better capture long-distance dependencies. Experimental evaluations on benchmark datasets demonstrated that SALSTM consistently outperforms both standard LSTM and state-of-the-art Transformer-based models in long-term forecasting tasks. This reinforces the notion that LSTM-based methods, when properly adapted, remain competitive in LTLF contexts.

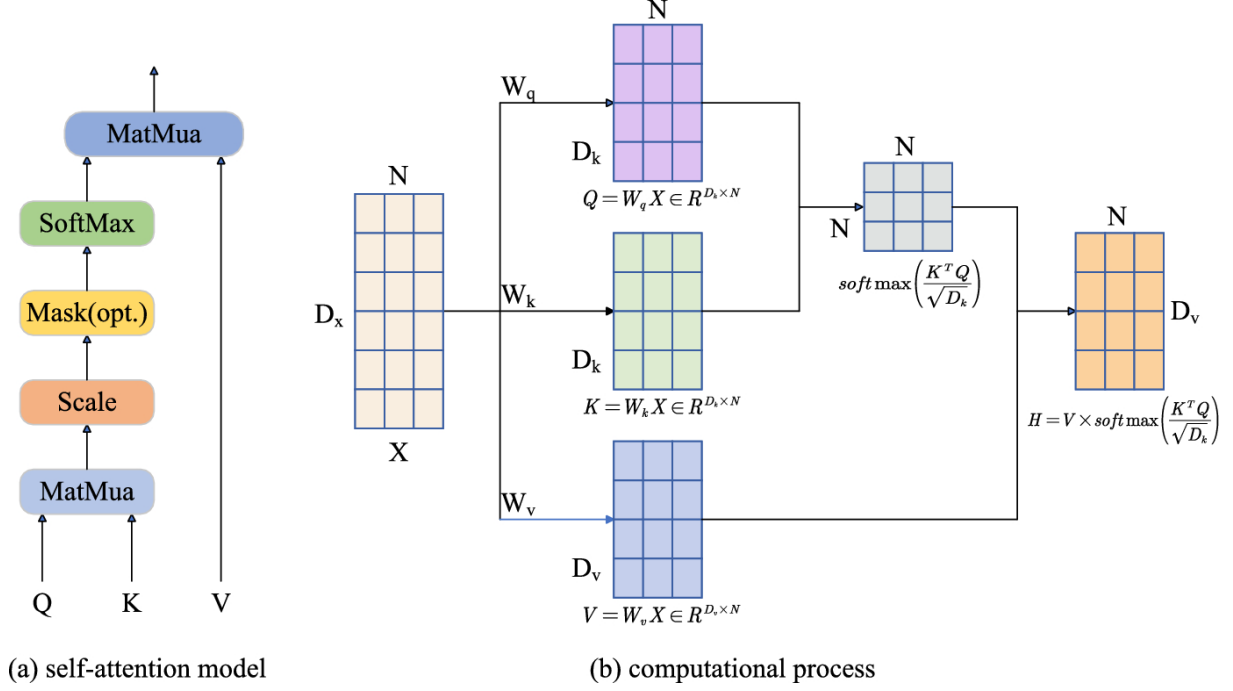


Figure 5.2: The self-attention model and its computational process by Dai et al. [72].

On the other hand, Go et al. [73] tackled a different aspect of LSTM limitations in LTLF—data availability. Their proposed approach addresses the difficulty of forecasting in the context of insufficient long-term historical data, particularly for growing sectors such as electric vehicles and evolving economies. The researchers developed a hybrid model that combines decomposition-linear methods, exponential smoothing, and LSTM. Initially, this model generates synthetic future inputs by capturing trends and seasonality from available historical data using statistical decomposition and smoothing. These synthetic sequences are then used as inputs for the LSTM model. When applied to long-term demand forecasting, including the creation of reserve demand curves for future power system planning, the model demonstrated strong reliability and practical utility. This method is especially beneficial when forecasting in data-scarce environments while still utilizing LSTM's pattern recognition capabilities.

Together, these studies illustrate the versatility of LSTM-based methods when augmented with attention mechanisms [72] or hybridized with classical forecasting

techniques [73]. Such enhancements enable LSTM to maintain forecasting accuracy and reliability even in the face of long prediction horizons and limited data.

Table 5.7 summarizes the key strengths and limitations of these LSTM-based approaches in the context of long-term load forecasting.

Table 5.7: Advantages and disadvantages of LSTM model in LTLF.

| Model Type | Strengths                                                                                                                                                                                                                               | Weaknesses                                                                                                                                                                                                                              |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LSTM       | Captures long-term dependencies; flexible architecture adaptable to enhancements (e.g., self-attention, hybrid input generation); performs well in long-term forecasting when extended; robust in handling nonlinear temporal patterns. | Captures long-term dependencies; flexible architecture adaptable to enhancements (e.g., self-attention, hybrid input generation); performs well in long-term forecasting when extended; robust in handling nonlinear temporal patterns. |

### 5.3.3. Transformer-based models

Transformer-based models have emerged as a powerful alternative to traditional sequence forecasting architectures due to their ability to model long-range dependencies using self-attention mechanisms. However, in the context of LTLF, standard Transformers often encounter challenges such as high computational complexity, limited interpretability, and insufficient accuracy when applied over extended time horizons. Recent advancements have introduced improved variants that address these shortcomings by modifying attention mechanisms, integrating hybrid architectures, or enhancing temporal granularity.

Tan et al. [74] proposed Cakformer, a Transformer variant specifically designed for long-term heat load forecasting. To overcome the limitations of conventional self-attention, Cakformer introduces a convolutional autocorrelation mechanism that simultaneously captures local features and global time dependencies. Furthermore, the model replaces the traditional MLP with a Kolmogorov–Arnold Network (KAN) to improve the learning of nonlinear, high-dimensional interactions. A Synchrosqueezed Wavelet Transform (SWT) is also employed to reduce noise in the input data. Experimental validation across multiple prediction horizons—96, 192, 336, and 720 steps—demonstrated Cakformer’s superiority over seven state-of-the-art models, including Autoformer and Transformer. At 720-step forecasting, for instance, Cakformer reduced MAE by 9.6% and root mean squared error (RMSE) by 8.3% compared to Autoformer, indicating its robustness in both accuracy and generalization across datasets.

Yang et al. [75] developed the Multi-Granularity Autoformer (MG-Autoformer) which is shown in Figure 5.3 to address both deterministic and probabilistic aspects of long-term power load forecasting. The model introduces a Multi-Granularity Auto-Correlation Attention Mechanism (MG-ACAM) to effectively model both fine-grained short-term fluctuations and coarse long-term trends. To manage computational overhead, a shared query-key (Q-K) mechanism is implemented, and a quantile loss function is used to enable probabilistic forecasting with uncertainty quantification. Evaluation on four diverse benchmark datasets—Portugal, USA, Australia, and ISO New England—revealed that MG-Autoformer consistently outperformed Autoformer and Transformer in both point and probabilistic forecasting. It achieved up to 11.2% lower MAE and 9.4% lower RMSE in deterministic tasks, and a 13.5% reduction in normalized pinball loss (NPL) in probabilistic settings, confirming its applicability in real-world planning scenarios where confidence intervals are vital.

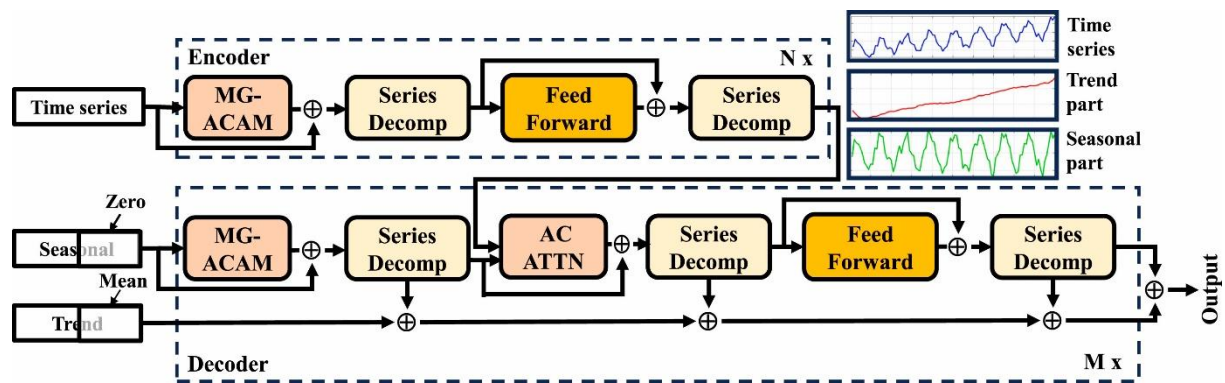


Figure 5.3: The multi-granularity Autoformer by Yang et al. [75].

Together, these studies demonstrate that Transformer-based models—when equipped with tailored attention mechanisms and multiscale forecasting strategies—can significantly improve the performance of LTLF models. Table 5.8 summarizes the enhancements, predictive gains, and limitations of these Transformer variants.

Table 5.8: Advantages and Disadvantages of Transformer-based models in LTLF.

| Model Type           | Strengths                                                                                           | Weaknesses                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Cakformer</b>     | High interpretability; best multi-horizon accuracy; robust to noise.                                | High architectural complexity; focused on heat load datasets.      |
| <b>MG-Autoformer</b> | Captures trends at multiple scales; supports probabilistic forecasting; efficient attention design. | Design sensitive to granularity setup; training time remains high. |

## 5.4. Hybrid models

Hybrid models have gained increasing attention in long-term load forecasting (LTLF) due to their ability to combine the strengths of multiple methodologies while compensating for individual weaknesses. These models often blend machine learning, fuzzy logic, neural networks, optimization algorithms, or time-series frameworks to address the challenges of nonlinearity, uncertainty, and temporal dependence. This section categorizes hybrid models into five main groups based on their core methodological structures.

### 5.4.1. Fuzzy-Neural Network-Based Hybrid Models

Fuzzy-neural models combine the uncertainty-handling strength of fuzzy systems with the learning capabilities of ANNs. Toly Chen [76] proposed a collaborative fuzzy-neural approach where multiple experts construct fuzzy backpropagation networks (FBPNs) to generate fuzzy forecasts, which are then aggregated using fuzzy intersection and defuzzified via a radial basis function (RBF) network. This framework enables simultaneous prediction of average and peak loads, improving precision by 40% and accuracy by 99% over single models. In Figure 5.4 the procedure of this proposed model is shown.

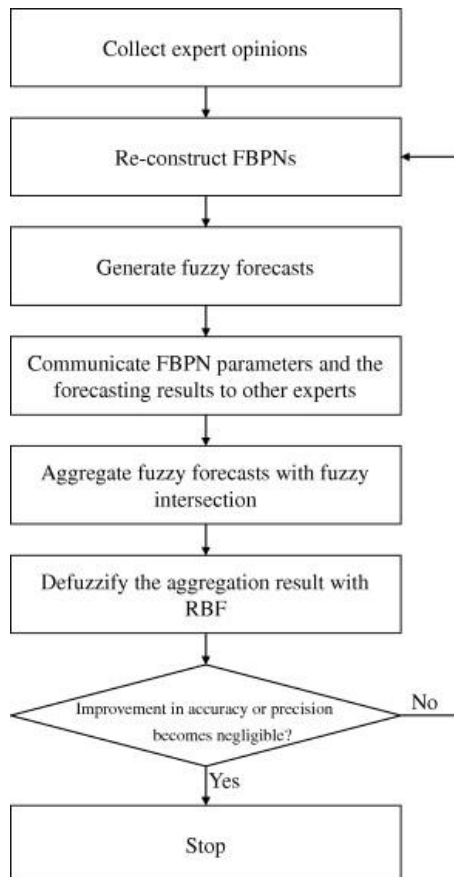


Figure 5.4: The procedure of the proposed methodology by Toly Chen [76].



Similarly, Danladi et al. [77] applied a fuzzy logic system enhanced with ANN structures to model long-term electricity demand. Their approach incorporates fuzzified input variables and a rule-based decision-making system that mimics human reasoning for load behavior, making it suitable for complex environments with vague and uncertain data. These models offer transparency in reasoning and high adaptability to varying load patterns but may suffer from increased computational complexity due to fuzzification and defuzzification stages.

### 5.4.2. Fuzzy-RNN-based models

Wen et al. [78] introduced a hybrid model called the TS-type recurrent fuzzy neural network (TS-RFNN), which integrates fuzzy inference mechanisms with recurrent neural networks. This model is particularly suited for long-term forecasting as it manages time-dependence through feedback connections and linguistic uncertainty via Takagi-Sugeno (TS) fuzzy rules. As Figure 5.5 illustrates, the architecture consists of five layers, including fuzzification, rule evaluation, normalization, consequent generation, and defuzzification. Experimental results demonstrate that the TS-RFNN effectively captures the temporal and uncertain characteristics of load data, achieving higher accuracy than standard RNNs or fuzzy systems alone.

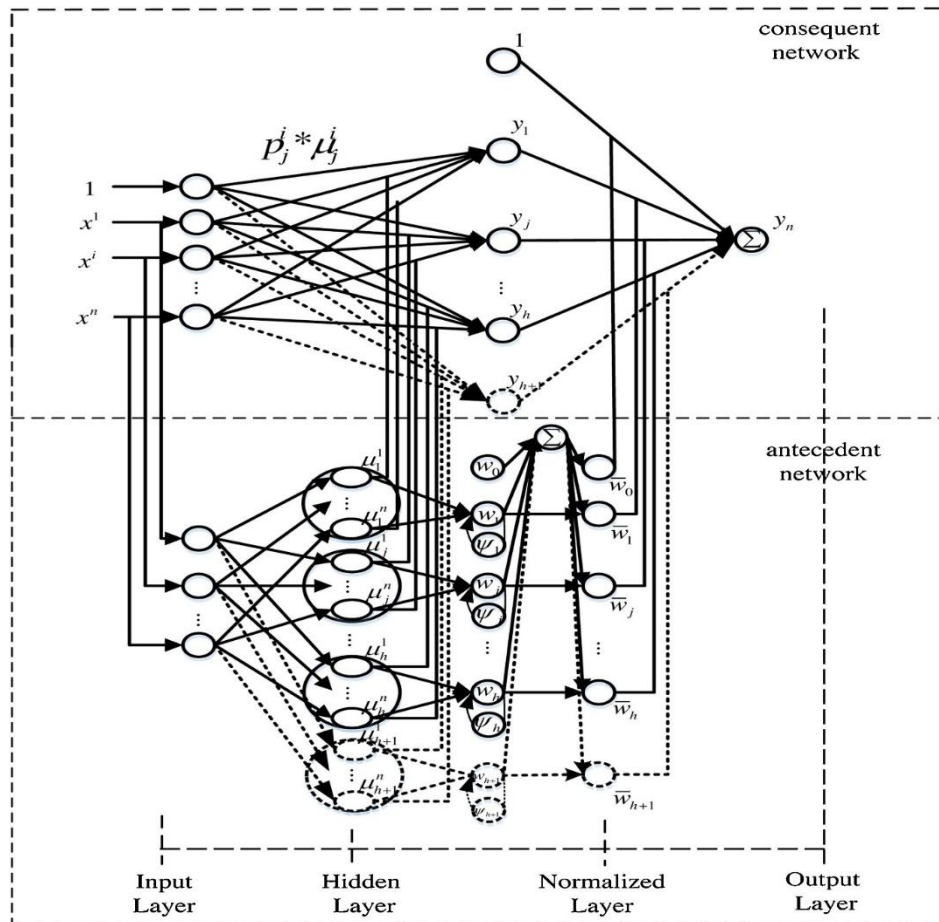


Figure 5.5: Schematic structure of ISO-TS-RBF-RFNN model by Wen et al. [78].



### 5.4.3. Fuzzy-SVM and grey system hybrid models

Chen et al. [79] proposed a hybrid model that combines fuzzy support vector machines (FSVM) with a grey forecasting mechanism that flowchart of the method is shown in Figure 5.6. The grey system model captures the overall trend in sparse and incomplete data, while the FSVM provides fine-grained learning of nonlinear patterns. This hybrid is particularly robust in uncertain or data-deficient environments. By optimizing the kernel parameters of the SVM under a fuzzy rule base and integrating trend correction from the grey model, the method improves prediction stability and generalization. It outperforms single SVM and grey models in terms of MAPE and RMSE, demonstrating its effectiveness in complex LTLF scenarios.

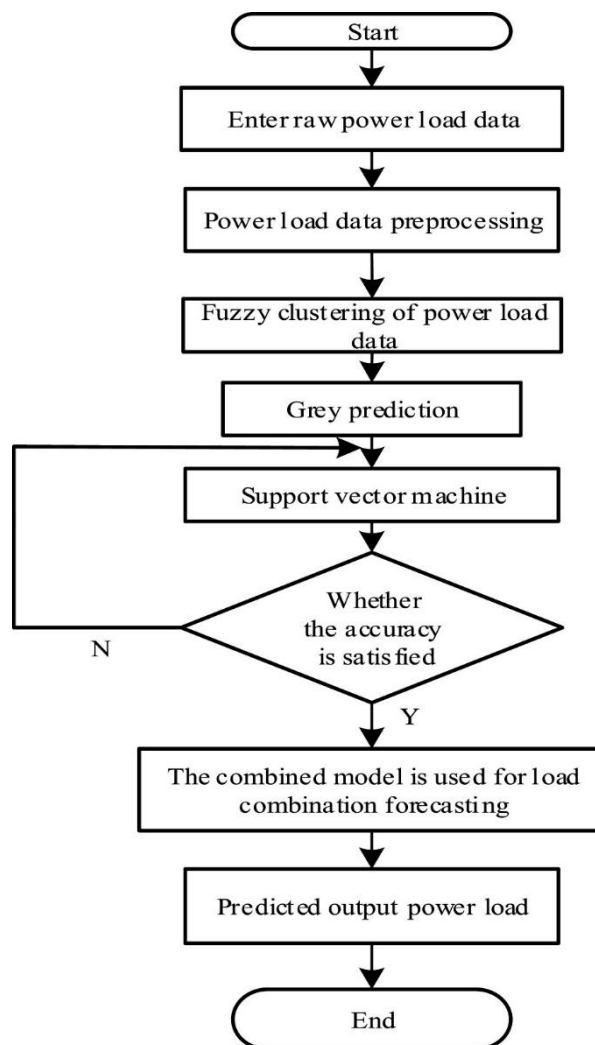


Figure 5.6: Flow chart of long-term power load grey combination forecast by Chen et al. [79].

#### 5.4.4. LSTM-based hybrid deep learning models

Ma et al. [80] developed a deep hybrid architecture that combines PSO, LSTM, and XGBoost to enhance long-term renewable energy forecasting. As Figure 5.7 illustrates, in this model, PSO optimizes LSTM's hyperparameters to avoid manual tuning, while XGBoost fine-tunes the final predictions by correcting residual errors. The integration of these methods allows for robust learning from nonlinear and seasonally fluctuating datasets. Comparative evaluations show that the hybrid outperforms standalone LSTM and statistical methods in both accuracy and generalization. However, due to its multi-stage nature, the model requires greater computational resources and more complex integration steps.

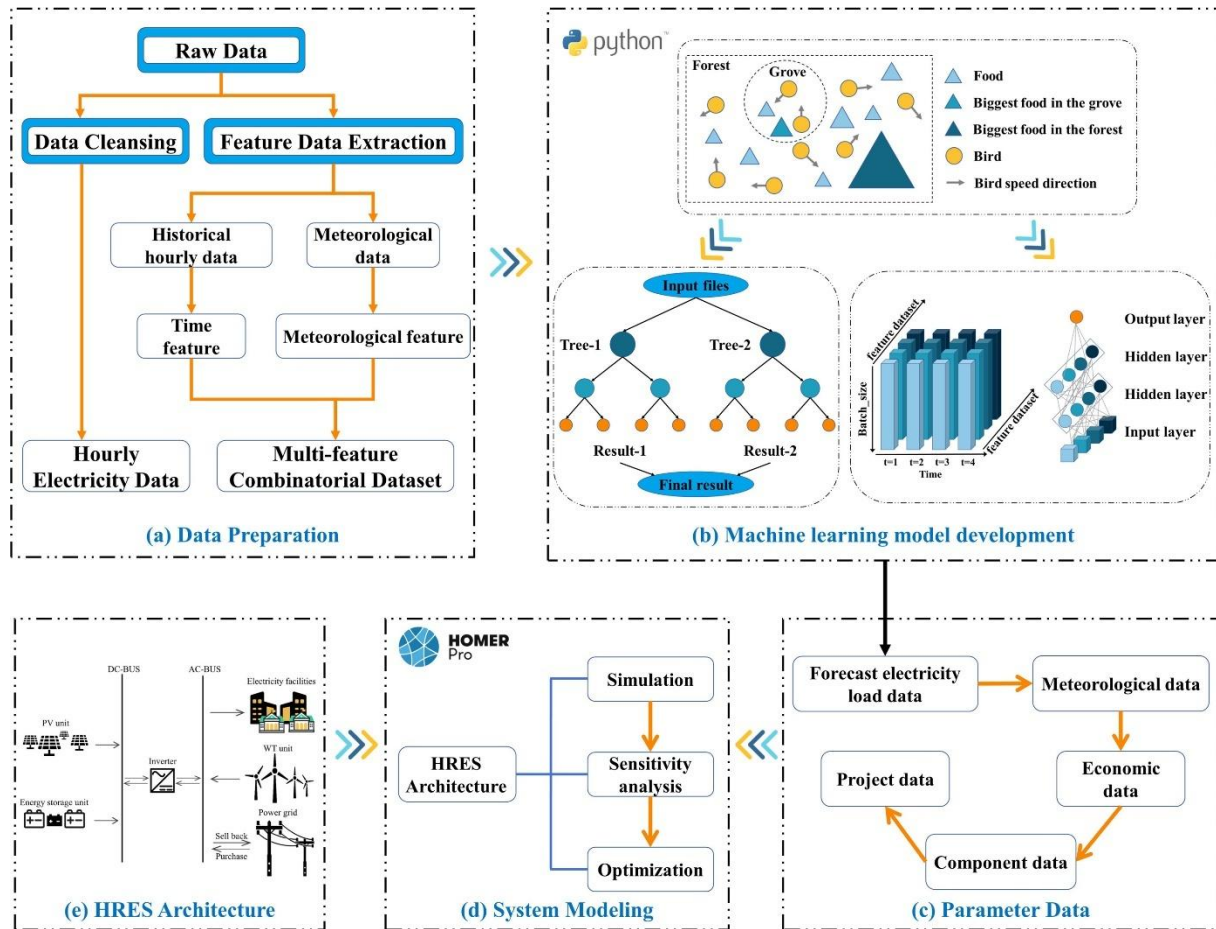


Figure 5.7: Framework for the RES design optimization by Ma et al. [80].

#### 5.4.5. Transformer-based hybrid deep learning models

Tian et al. [81] introduced LDformer, a transformer-based parallel deep neural network tailored for long-term power forecasting. LDformer fuses Informer (a lightweight transformer variant), LSTM (for long-range dependencies), convolutional layers (for local feature extraction), and UniDrop-based attention regularization. Its architecture enables simultaneous processing of long sequences with reduced redundancy and computational cost. LDformer excels in robustness and scalability

across multiple benchmark datasets, consistently outperforming classical RNNs, LSTMs, and baseline Informer models in terms of RMSE. The key advantage lies in its ability to retain deep temporal features while mitigating attention overload through ProbSparse attention mechanisms.

To provide a concise overview of the comparative strengths and limitations of the reviewed hybrid approaches, Table 5.9 summarizes the key benefits and weaknesses associated with each model type. This synthesis highlights how different combinations—ranging from metaheuristic-neural hybrids to transformer-based architectures—offer distinct advantages in handling long-term load forecasting challenges such as nonlinearity, uncertainty, and sequence length. The table serves as a reference point for evaluating trade-offs between accuracy, interpretability, computational efficiency, and practical applicability in real-world forecasting scenarios.

Table 5.9: Advantages and disadvantages of hybrid models in LTLF.

| Authors                    | Strengths                                                                   | Weaknesses                                                         |
|----------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Chen et al.</b> [79]    | Robust to sparse data, handles uncertainty and trend components             | Complex parameter tuning                                           |
| <b>Ma et al.</b> [80]      | High accuracy, residual correction, optimized structure                     | Multi-stage model increases computation and integration complexity |
| <b>Chen, Toly</b> [76]     | Collaborative expert modelling, handles peak and average forecasts together | Requires defuzzification and multiple iterations                   |
| <b>Wen et al.</b> [78]     | Captures temporal uncertainty, rule-based learning                          | Difficult to scale to high-dimensional inputs                      |
| <b>Tian et al.</b> [81]    | Fast training, strong long-range learning, reduced attention redundancy     | High model complexity and memory usage                             |
| <b>Danladi et al.</b> [77] | Transparent rule-base, effective with vague data                            | Limited adaptation if rule base is not comprehensive               |

## 5.5. Summary

Chapter 5 provided a detailed review of state-of-the-art methods used in LTLF, highlighting their theoretical foundations, practical applications, and relative strengths and weaknesses. Beginning with statistical approaches such as multiple linear regression, logistic regression, ARIMA, and SARIMA, the discussion emphasized their transparency, interpretability, and suitability for stable consumption environments, while also acknowledging their limitations in modelling nonlinear or rapidly evolving demand patterns. The chapter then explored the increasing adoption of machine learning models—including SVR, decision trees, random forests, and gradient boosting—which offer improved flexibility and accuracy but may lack explainability. Deep learning methods, such as ANN, MLP, and LSTMs, were shown to effectively capture complex temporal and nonlinear relationships, especially when supported by large, high-quality datasets or enhanced through hybridization. Finally, the chapter examined a diverse array of hybrid models that integrate fuzzy logic, optimization algorithms, deep learning, and Transformer-based architectures. These hybrids demonstrate substantial gains in accuracy, adaptability, and robustness—particularly in scenarios involving long sequences, uncertainty, or sparse data. However, they also introduce challenges related to model complexity, computational requirements, and integration costs. The comparative synthesis in tables throughout this chapter—including Table 5.9 for hybrid models—offers a practical reference for selecting appropriate forecasting techniques based on system characteristics, data availability, and planning objectives. Together, these insights provide a strong foundation for understanding and improving LTLF in evolving power systems.

## 6 Conclusion and future research directions

This thesis has undertaken a comprehensive investigation into electricity load forecasting, focusing particularly on methods suited for LTLF but contextualized within the broader landscape that includes STLF and MTLF. By systematically reviewing statistical, machine learning, deep learning, and hybrid methods, the thesis provides an organized framework to understand when and how each method should be applied, considering both forecasting horizon and data availability—two of the most critical dimensions in predictive modelling for power systems.

### 6.1. Recapitulation of research objectives and contributions

The research was driven by the increasing demand for accurate and scalable electricity load forecasting tools amid evolving energy consumption behaviors, grid decarbonization, and renewable energy integration. The review began by identifying the practical challenges utilities and planners face in making decisions under conditions of long-term uncertainty and growing data diversity. Through a layered analysis of forecasting methods, this thesis contributes:

A classification of methods by time horizon, detailing their strengths and limitations across short-, medium-, and long-term applications.

A mapping of forecasting techniques to data availability levels, from single households to national-scale energy systems.

A comparative review of hybrid approaches, which integrate models to balance accuracy, interpretability, and computational cost.

Visual tools (Figure 6.1 and Figure 6.2) to guide practitioners in selecting suitable methods based on forecast duration and data granularity.

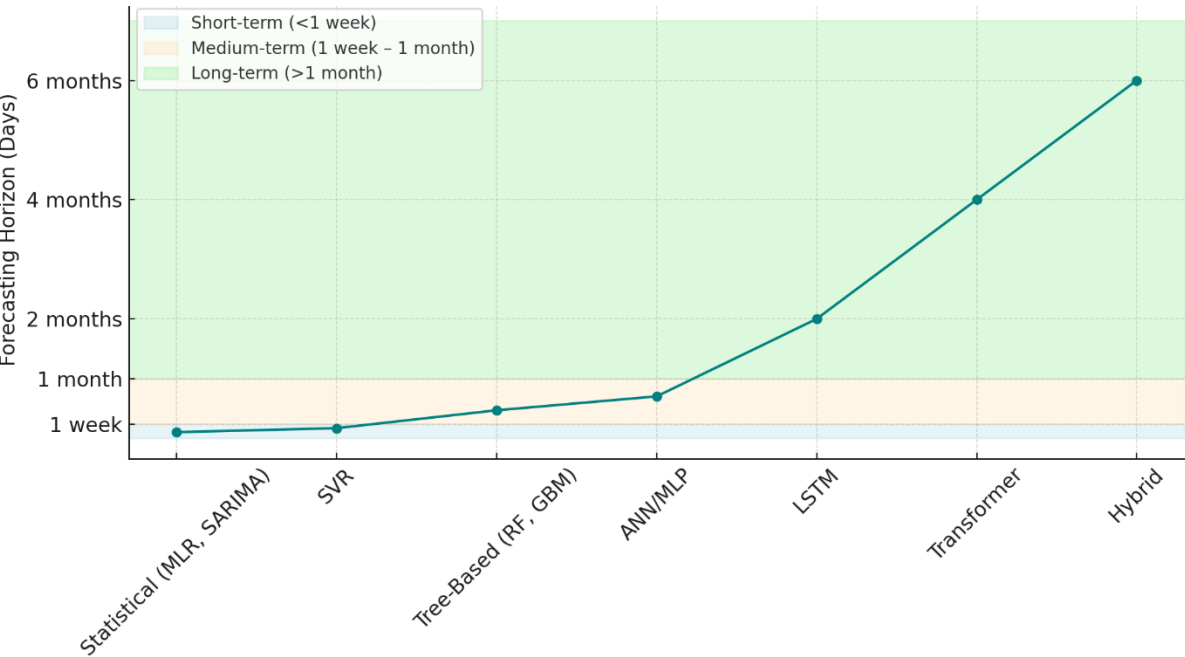


Figure 6.1: Forecasting horizon classification of different methods.

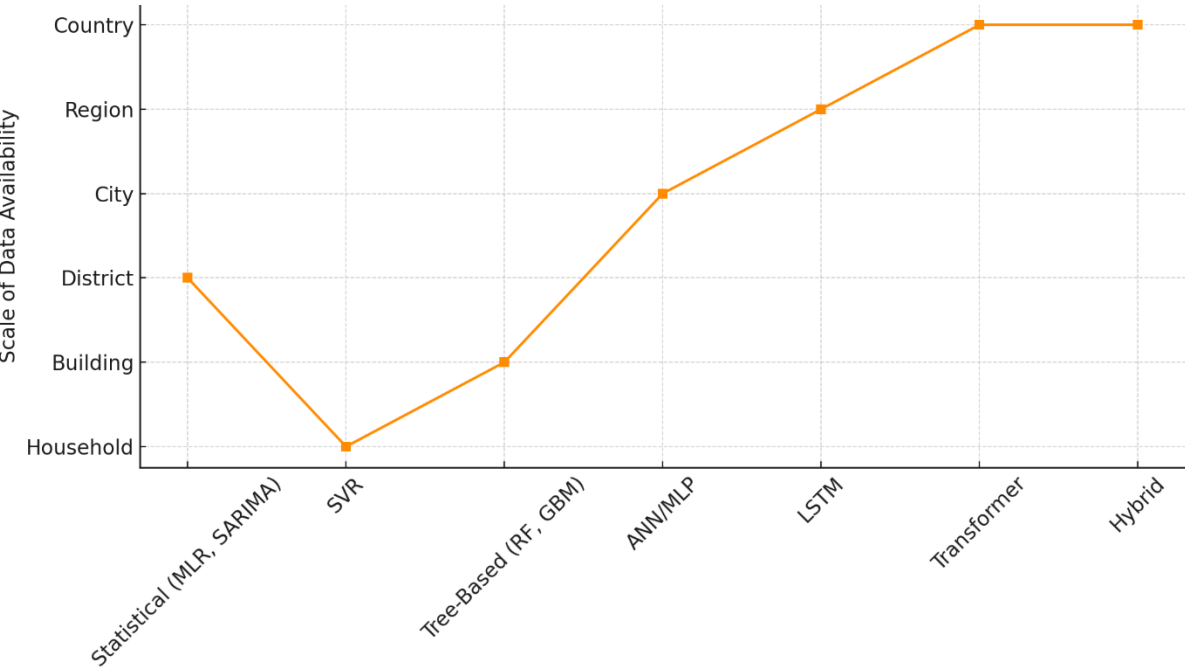


Figure 6.2: Scalability of forecasting methods based on spatial data availability.

## 6.2. Strategic alignment between forecasting horizons and methods

A core finding of this work is that forecasting accuracy and reliability are heavily dependent on temporal context:

STLF, defined as forecasting within a few minutes to 1 week, is best addressed using statistical models such as Multiple Linear Regression (MLR), ARIMA/SARIMA, or simple machine learning techniques like SVR. These methods are interpretable, fast, and data-efficient—ideal for operational decision-making.

MTLF, spanning 1 week to 1 month, benefits from tree-based models (e.g., Random Forests, Gradient Boosting Machines) and basic ANNs. These methods offer improved capacity to capture nonlinearities and complex interactions, useful for demand-side planning, tariff design, or load shifting strategies.

LTLF, beyond 1 month and up to several decades, requires models capable of handling system evolution, structural change, and socio-economic influences. Here, deep learning models such as LSTM and Transformer-based architectures, along with hybrid models integrating fuzzy logic, metaheuristics, and ensemble learning, show the best performance. These models can model intricate temporal dynamics and uncertainties but require large datasets and computational resources.

As shown in Figure 6.1, forecasting methods follow a progression from interpretable and lightweight models for short durations to more complex and data-hungry models as the prediction horizon increases.

## 6.3. Method suitability based on data scale

Another major dimension in model selection is data scale, as detailed in Figure 6.2:

For household-level and building-level applications—such as smart home control or energy efficiency management—lightweight methods (SVR, decision trees, simple ANN) are more suitable due to their ability to perform under data-scarce conditions.

District and city-level forecasting, which often includes substation, microgrid, or municipal planning data, benefits from ML models such as ensemble trees and MLPs that balance model complexity with interpretability.

For regional and national-level load forecasting—relevant to grid investment, policy simulation, and energy market design—deep learning and hybrid models are ideal, as they can process multivariate, longitudinal, and often noisy datasets with greater fidelity.



This scale-sensitive approach helps align model complexity with data richness, enhancing both efficiency and performance.

## 6.4. Implications for practice and policy

The findings of this thesis offer several implications:

Model choice must be context aware. No single technique is optimal across all conditions. Planners and operators should match forecasting models to their intended use case—be it short-term operations, medium-term tariff planning, or long-term infrastructure investment.

Explainability vs. accuracy is a trade-off. Statistical models provide clarity and confidence for policymakers but may lack precision. Conversely, DL and hybrid models offer state-of-the-art accuracy but require interpretability enhancements (e.g., attention mechanisms, post hoc analysis) for use in regulatory environments.

Data strategy is foundational. Models are only as good as the data they are built upon. Investment in smart meter infrastructure, high-resolution weather data, and socio-economic forecasting tools is essential for unleashing the full potential of advanced forecasting models.

## 6.5. Limitations and future research directions

While comprehensive, this review is subject to a few limitations: it is based on published literature and benchmark datasets and as a consequence real-world constraints such as missing data, organizational capacity, and stakeholder preferences were not modelled explicitly. Hybrid models are still evolving, and their performance may vary significantly across datasets and implementations, while probabilistic forecasting methods—which incorporate uncertainty in output—were only lightly discussed and could form a valuable extension of this work.

Future research should focus on, for instance, (i) interpretable hybrid models that combine the accuracy of deep learning with the explainability of statistical approaches, (ii) transfer learning and meta-learning to reduce the training data burden in regions or sectors with limited historical data, and (iii) probabilistic and scenario-based LTLF, which better reflect real-world uncertainty and can support resilient decision-making.

It is worthwhile discussing briefly also Spiking Neural Networks (SNNs), which more closely mimic the temporal and event-driven processing of biological neurons, represent a promising frontier for electricity load forecasting. The robustness of surrogate gradient learning algorithms enables effective training of SNNs despite the nondifferentiable nature of spike events, as demonstrated by Zenke and Vogels [82]. This makes SNNs well-suited for capturing complex temporal dynamics with sparse

and energy-efficient computations. Although SNNs have been explored for short-term load forecasting [83], their application to medium- and long-term horizons remains underdeveloped. Given the growing interest in low-power, real-time, and edge-deployable AI models in smart grid contexts, future research should investigate SNN architectures with surrogate gradient training for medium- and long-term load forecasting tasks. This includes developing benchmark datasets, refining hybrid models that integrate SNNs with classical deep learning techniques and exploring hardware implementations to leverage neuromorphic computing advantages.

Such directions could yield forecasting models that balance accuracy, efficiency, and interpretability, addressing challenges posed by increasing data volume, temporal complexity, and the need for sustainable computation in evolving power systems.

## 6.6. Final remarks

In an era where electricity systems are becoming more complex, decentralized, and data-driven, accurate and adaptive load forecasting is no longer a luxury but a necessity. This thesis provides a structured lens to evaluate forecasting models, not as standalone tools but as components of a larger decision-making ecosystem—sensitive to time, data, purpose, and context. As forecasting continues to evolve with AI, big data, and climate-aware planning, the frameworks and insights presented here can serve as a foundation for both academic research and practical implementation in future-ready power systems.

# Bibliography

- [1] S. Amoako, F. K. Andoh, and E. E. Asmah, "Household structure and electricity consumption in Ghana," *Energy Policy*, vol. 182, Nov. 2023, doi: 10.1016/j.enpol.2023.113767.
- [2] "European Commission," <https://europa.eu/>. Accessed: Mar. 18, 2025. [Online]. Available: [https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Energy\\_consumption\\_in\\_households](https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Energy_consumption_in_households)
- [3] H. Rafiq, P. Manandhar, E. Rodriguez-Ubinas, J. D. Barbosa, and O. A. Qureshi, "Analysis of residential electricity consumption patterns utilizing smart-meter data: Dubai as a case study," *Energy Build*, vol. 291, Jul. 2023, doi: 10.1016/j.enbuild.2023.113103.
- [4] C. Ragupathi, S. Dhanasekaran, N. Vijayalakshmi, and A. O. Salau, "Prediction of electricity consumption using an innovative deep energy predictor model for enhanced accuracy and efficiency," *Energy Reports*, vol. 12, pp. 5320–5337, Dec. 2024, doi: 10.1016/j.egy.2024.11.018.
- [5] H. Luo, Y. Zhang, X. Gao, Z. Liu, X. Meng, and X. Yang, "Multi-scale electricity consumption prediction model based on land use and interpretable machine learning: A case study of China," *Advances in Applied Energy*, vol. 16, p. 100197, Dec. 2024, doi: 10.1016/j.adapen.2024.100197.
- [6] D. Peteleaza *et al.*, "Electricity consumption forecasting for sustainable smart cities using machine learning methods," *Internet of Things (Netherlands)*, vol. 27, Oct. 2024, doi: 10.1016/j.iot.2024.101322.
- [7] Z. Wang, T. Hong, H. Li, and M. Ann Piette, "Predicting city-scale daily electricity consumption using data-driven models," *Advances in Applied Energy*, vol. 2, May 2021, doi: 10.1016/j.adapen.2021.100025.
- [8] S. Palaniappan, S. Karuppannan, and D. Velusamy, "Categorization of Indian residential consumers electrical energy consumption pattern using clustering and classification techniques," *Energy*, vol. 289, Feb. 2024, doi: 10.1016/j.energy.2023.129992.
- [9] J. Adinkrah *et al.*, "Artificial intelligence-based strategies for sustainable energy planning and electricity demand estimation: A systematic review," Mar. 01, 2025, *Elsevier Ltd.* doi: 10.1016/j.rser.2024.115161.
- [10] A. Najeeb, S. Sridharan, A. B. Rao, S. B. Agnihotri, and V. Mishra, "Determinants of residential electricity consumption in South, East and South East Asia: A systematic review," Jul. 01, 2024, *Elsevier Ltd.* doi: 10.1016/j.rser.2024.114400.

- [11] C. T. Nguyen, H. Nguyen, and A. D. Sakti, "Seasonal characteristics of outdoor thermal comfort and residential electricity consumption: A Snapshot in Bangkok Metropolitan Area," *Remote Sens Appl*, vol. 33, Jan. 2024, doi: 10.1016/j.rsase.2023.101106.
- [12] Y. Li *et al.*, "Efficient predictive control strategy for mitigating the overlap of EV charging demand and residential load based on distributed renewable energy," *Renew Energy*, vol. 240, Feb. 2025, doi: 10.1016/j.renene.2024.122154.
- [13] F. Lazzari *et al.*, "User behaviour models to forecast electricity consumption of residential customers based on smart metering data," *Energy Reports*, vol. 8, pp. 3680–3691, Nov. 2022, doi: 10.1016/j.egyr.2022.02.260.
- [14] Y. Huang, Z. Yao, and Q. Xu, "Classification model of electricity consumption behavior based on sparse denoising autoencoder feature dimensionality reduction and spectral clustering," *International Journal of Electrical Power and Energy Systems*, vol. 158, Jul. 2024, doi: 10.1016/j.ijepes.2024.109960.
- [15] F. Yu, Q. Yue, A. Yuniata, and H. M. A. Aljahdali, "A novel hybrid deep correction approach for electrical load demand prediction," *Sustain Cities Soc*, vol. 74, Nov. 2021, doi: 10.1016/j.scs.2021.103161.
- [16] X. Deng *et al.*, "Bagging-XGBoost algorithm based extreme weather identification and short-term load forecasting model," *Energy Reports*, vol. 8, pp. 8661–8674, Nov. 2022, doi: 10.1016/j.egyr.2022.06.072.
- [17] M. Bilgili and E. Pinar, "Gross electricity consumption forecasting using LSTM and SARIMA approaches: A case study of Türkiye," *Energy*, vol. 284, Dec. 2023, doi: 10.1016/j.energy.2023.128575.
- [18] D. Katić, H. Krstić, I. Ištoka Otković, and H. Begić Juričić, "Comparing multiple linear regression and neural network models for predicting heating energy consumption in school buildings in the Federation of Bosnia and Herzegovina," *Journal of Building Engineering*, vol. 97, Nov. 2024, doi: 10.1016/j.jobbe.2024.110728.
- [19] H. Nano, S. New, A. Cohen, and B. Ramachandran, "Chapter 16 - Load forecasting using multiple linear regression with different calendars," in *Distributed Energy Resources in Microgrids*, R. K. Chauhan and K. Chauhan, Eds., Academic Press, 2019, pp. 405–417. doi: <https://doi.org/10.1016/B978-0-12-817774-7.00016-8>.
- [20] G. Dudek, "Pattern-based local linear regression models for short-term load forecasting," *Electric Power Systems Research*, vol. 130, pp. 139–147, Jan. 2016, doi: 10.1016/j.epsr.2015.09.001.
- [21] J. C. Nsangou, J. Kenfack, U. Nzotcha, P. S. Ngohe Ekam, J. Voufo, and T. T. Tamo, "Explaining household electricity consumption using quantile

- regression, decision tree and artificial neural network," *Energy*, vol. 250, Jul. 2022, doi: 10.1016/j.energy.2022.123856.
- [22] A. Gellert, U. Fiore, A. Florea, R. Chis, and F. Palmieri, "Forecasting Electricity Consumption and Production in Smart Homes through Statistical Methods," *Sustain Cities Soc*, vol. 76, Jan. 2022, doi: 10.1016/j.scs.2021.103426.
- [23] N. M. M. Bendaoud, N. Farah, and S. Ben Ahmed, "Applying load profiles propagation to machine learning based electrical energy forecasting," *Electric Power Systems Research*, vol. 203, Feb. 2022, doi: 10.1016/j.epsr.2021.107635.
- [24] A. Haque and S. Rahman, "Short-term electrical load forecasting through heuristic configuration of regularized deep neural network," *Appl Soft Comput*, vol. 122, Jun. 2022, doi: 10.1016/j.asoc.2022.108877.
- [25] Y. Zhang, B. K. Teoh, M. Wu, J. Chen, and L. Zhang, "Data-driven estimation of building energy consumption and GHG emissions using explainable artificial intelligence," *Energy*, vol. 262, Jan. 2023, doi: 10.1016/j.energy.2022.125468.
- [26] M. S. Hasan, M. Tarequzzaman, M. Moznuzzaman, and M. A. Ahad Juel, "Prediction of energy consumption in four sectors using support vector regression optimized with genetic algorithm," *Heliyon*, vol. 11, no. 2, Jan. 2025, doi: 10.1016/j.heliyon.2025.e41765.
- [27] M. Zhou, J. Yu, F. Sun, and M. Wang, "Forecasting of short term electric power consumption for different types buildings using improved transfer learning: A case study of primary school in China," *Journal of Building Engineering*, vol. 78, Nov. 2023, doi: 10.1016/j.jobbe.2023.107618.
- [28] R. Pascanu, T. Mikolov, and Y. Bengio, "On the difficulty of training Recurrent Neural Networks," Nov. 2012, [Online]. Available: <http://arxiv.org/abs/1211.5063>
- [29] C. Wang, X. Li, Y. Shi, W. Jiang, Q. Song, and X. Li, "Load forecasting method based on CNN and extended LSTM," *Energy Reports*, vol. 12, pp. 2452–2461, Dec. 2024, doi: 10.1016/j.egyr.2024.07.030.
- [30] G. Memarzadeh and F. Keynia, "Short-term electricity load and price forecasting by a new optimal LSTM-NN based prediction algorithm," *Electric Power Systems Research*, vol. 192, p. 106995, Mar. 2021, doi: 10.1016/j.epsr.2020.106995.
- [31] X. Guo, Q. Zhao, D. Zheng, Y. Ning, and Y. Gao, "A short-term load forecasting model of multi-scale CNN-LSTM hybrid neural network considering the real-time electricity price," *Energy Reports*, vol. 6, pp. 1046–1053, 2020, doi: <https://doi.org/10.1016/j.egyr.2020.11.078>.
- [32] W. Guo, S. Liu, L. Weng, and X. Liang, "Power Grid Load Forecasting Using a CNN-LSTM Network Based on a Multi-Modal Attention Mechanism," *Applied Sciences (Switzerland)*, vol. 15, no. 5, Mar. 2025, doi: 10.3390/app15052435.

- [33] C. M. Liapis, A. Karanikola, and S. Kotsiantis, "Energy Load Forecasting: Investigating Mid-Term Predictions with Ensemble Learners," pp. 343–355, 2022, doi: 10.1007/978-3-031-08333-4\_28i.
- [34] Z. Tian, W. Liu, W. Jiang, and C. Wu, "CNNs-Transformer based day-ahead probabilistic load forecasting for weekends with limited data availability," *Energy*, vol. 293, p. 130666, 2024, doi: <https://doi.org/10.1016/j.energy.2024.130666>.
- [35] J. Wang, W. Huang, Y. Ding, Y. Dang, and L. Ye, "Forecasting the electric power load based on a novel prediction model coupled with accumulative time-delay effects and periodic fluctuation characteristics," *Energy*, vol. 317, p. 134518, 2025, doi: <https://doi.org/10.1016/j.energy.2025.134518>.
- [36] A. Mathew, R. Chikte, S. K. Sadanandan, S. Abdelaziz, S. Ijaz, and T. Ghaoud, "Medium-term feeder load forecasting and boosting peak accuracy prediction using the PWP-XGBoost model," *Electric Power Systems Research*, vol. 237, p. 111051, 2024, doi: <https://doi.org/10.1016/j.epsr.2024.111051>.
- [37] Z. Peng and X. Yang, "Short- and medium-term power load forecasting model based on a hybrid attention mechanism in the time and frequency domains," *Expert Syst Appl*, vol. 278, p. 127329, 2025, doi: <https://doi.org/10.1016/j.eswa.2025.127329>.
- [38] Candidus. U. Eya, A. O. Salau, S. L. Braide, and O. D. Chigozirim, "Improved Medium Term Approach for Load Forecasting of Nigerian Electricity Network Using Artificial Neuro-Fuzzy Inference System: A Case Study of University of Nigeria, Nsukka," *Procedia Comput Sci*, vol. 218, pp. 2585–2593, 2023, doi: <https://doi.org/10.1016/j.procs.2023.01.232>.
- [39] S. Karamolegkos and D. E. Koulouriotis, "Advancing short-term load forecasting with decomposed Fourier ARIMA: A case study on the Greek energy market," *Energy*, vol. 325, Jun. 2025, doi: 10.1016/j.energy.2025.135854.
- [40] S. Wang, S. Wang, and D. Wang, "Combined probability density model for medium term load forecasting based on quantile regression and kernel density estimation," *Energy Procedia*, vol. 158, pp. 6446–6451, 2019, doi: <https://doi.org/10.1016/j.egypro.2019.01.169>.
- [41] C. Xu, Y. Sun, A. Du, and D. Gao, "Quantile regression based probabilistic forecasting of renewable energy generation and building electrical load: A state of the art review," *Journal of Building Engineering*, vol. 79, p. 107772, 2023, doi: <https://doi.org/10.1016/j.jobbe.2023.107772>.
- [42] F. Zhu and G. Wu, "Load forecasting of the power system: An investigation based on the method of random forest regression," *Energy Engineering: Journal of the Association of Energy Engineering*, vol. 118, no. 6, pp. 1703–1712, 2021, doi: 10.32604/EE.2021.015602.



- [43] E. Royal, S. Bandyopadhyay, A. Newman, Q. Huang, and P. C. Tabares-Velasco, "A statistical framework for district energy long-term electric load forecasting," *Appl Energy*, vol. 384, p. 125445, 2025, doi: <https://doi.org/10.1016/j.apenergy.2025.125445>.
- [44] F. Dinmohammadi, Y. Han, and M. Shafiee, "Predicting Energy Consumption in Residential Buildings Using Advanced Machine Learning Algorithms," *Energies (Basel)*, vol. 16, no. 9, May 2023, doi: 10.3390/en16093748.
- [45] Y. Dong, X. Ma, and T. Fu, "Electrical load forecasting: A deep learning approach based on K-nearest neighbors," *Appl Soft Comput*, vol. 99, p. 106900, 2021, doi: <https://doi.org/10.1016/j.asoc.2020.106900>.
- [46] S. ur R. Khan *et al.*, "Enhanced Machine-Learning Techniques for Medium-Term and Short-Term Electric-Load Forecasting in Smart Grids," *Energies (Basel)*, vol. 16, no. 1, Jan. 2023, doi: 10.3390/en16010276.
- [47] G. Dudek and P. Pełka, "Pattern Similarity-based Machine Learning Methods for Mid-term Load Forecasting: A Comparative Study," Mar. 2020, [Online]. Available: <http://arxiv.org/abs/2003.01475>
- [48] R. Wazirali, E. Yaghoubi, M. S. S. Abujazar, R. Ahmad, and A. H. Vakili, "State-of-the-art review on energy and load forecasting in microgrids using artificial neural networks, machine learning, and deep learning techniques," Dec. 01, 2023, *Elsevier Ltd.* doi: 10.1016/j.epsr.2023.109792.
- [49] M. Ghiassi, D. K. Zimbra, and H. Saidane, "Medium term system load forecasting with a dynamic artificial neural network model," *Electric Power Systems Research*, vol. 76, no. 5, pp. 302–316, 2006, doi: <https://doi.org/10.1016/j.epsr.2005.06.010>.
- [50] T. Yalcinoz and U. Eminoglu, "Short term and medium term power distribution load forecasting by neural networks," *Energy Convers Manag*, vol. 46, no. 9, pp. 1393–1405, 2005, doi: <https://doi.org/10.1016/j.enconman.2004.07.005>.
- [51] S. Muzaffar and A. Afshari, "Short-term load forecasts using LSTM networks," in *Energy Procedia*, Elsevier Ltd, 2019, pp. 2922–2927. doi: 10.1016/j.egypro.2019.01.952.
- [52] J. W. Chan and C. K. Yeo, "A Transformer based approach to electricity load forecasting," *Electricity Journal*, vol. 37, no. 2, Mar. 2024, doi: 10.1016/j.tej.2024.107370.
- [53] G. Dudek, P. Pełka, and S. Smył, "A Hybrid Residual Dilated LSTM and Exponential Smoothing Model for Mid-Term Electric Load Forecasting," Mar. 2020, [Online]. Available: <http://arxiv.org/abs/2004.00508>
- [54] T. González Grandón, J. Schwenzer, T. Steens, and J. Breuing, "Electricity demand forecasting with hybrid classical statistical and machine learning



- algorithms: Case study of Ukraine," *Appl Energy*, vol. 355, Feb. 2024, doi: 10.1016/j.apenergy.2023.122249.
- [55] Z. Wang, Z. Chen, Y. Yang, C. Liu, X. Li, and J. Wu, "A hybrid Autoformer framework for electricity demand forecasting," *Energy Reports*, vol. 9, pp. 3800–3812, Dec. 2023, doi: 10.1016/j.egyr.2023.02.083.
- [56] G. Antonesi, T. Cioara, I. Anghel, I. Papias, V. Michalakopoulos, and E. Sarmaş, "Hybrid transformer model with liquid neural networks and learnable encodings for buildings' energy forecasting," *Energy and AI*, vol. 20, May 2025, doi: 10.1016/j.egyai.2025.100489.
- [57] G. Yıldız Doğan, A. Aksoy, and N. Öztürk, "A Hybrid Deep Learning Model to Estimate the Future Electricity Demand of Sustainable Cities," *Sustainability (Switzerland)*, vol. 16, no. 15, Aug. 2024, doi: 10.3390/su16156503.
- [58] S. Zhang, R. Chen, J. Cao, and J. Tan, "A CNN and LSTM-based multi-task learning architecture for short and medium-term electricity load forecasting," *Electric Power Systems Research*, vol. 222, p. 109507, 2023, doi: <https://doi.org/10.1016/j.epsr.2023.109507>.
- [59] L. Wang, X. Wang, and Z. Zhao, "Mid-term electricity demand forecasting using improved multi-mode reconstruction and particle swarm-enhanced support vector regression," *Energy*, vol. 304, Sep. 2024, doi: 10.1016/j.energy.2024.132021.
- [60] X. Huang, Y. Han, J. Yan, and X. Zhou, "Hybrid forecasting model of building cooling load based on EMD-LSTM-Markov algorithm," *Energy Build*, vol. 321, Oct. 2024, doi: 10.1016/j.enbuild.2024.114670.
- [61] M. J. A. Shohan, M. O. Faruque, and S. Y. Foo, "Forecasting of Electric Load Using a Hybrid LSTM-Neural Prophet Model," *Energies (Basel)*, vol. 15, no. 6, Mar. 2022, doi: 10.3390/en15062158.
- [62] R. Feng, J. Wang, W. Wu, S. Liu, A. Liu, and S. Xie, "Saturated load forecasting based on improved logistic regression and affinity propagation," *Electric Power Systems Research*, vol. 237, p. 110953, 2024, doi: <https://doi.org/10.1016/j.epsr.2024.110953>.
- [63] V. Kumar, R. K. Mandal, and S. Rai, "Smart metered distribution system load forecasting using statistical approaches," in *2024 4th International Conference on Emerging Frontiers in Electrical and Electronic Technologies (ICEFEET)*, 2024, pp. 1–4. doi: 10.1109/ICEFEET64463.2024.10866475.
- [64] A. Kumar Dubey, A. Kumar, V. García-Díaz, A. Kumar Sharma, and K. Kanhaiya, "Study and analysis of SARIMA and LSTM in forecasting time series data," *Sustainable Energy Technologies and Assessments*, vol. 47, p. 101474, 2021, doi: <https://doi.org/10.1016/j.seta.2021.101474>.

- [65] J. Wang, L. Li, D. Niu, and Z. Tan, "An annual load forecasting model based on support vector regression with differential evolution algorithm," *Appl Energy*, vol. 94, pp. 65–70, 2012, doi: <https://doi.org/10.1016/j.apenergy.2012.01.010>.
- [66] T. Hu, "Comparison of long-term load forecasting methods for air conditioning system of a large building," in *2021 China Automation Congress (CAC)*, 2021, pp. 4491–4496. doi: [10.1109/CAC53003.2021.9728514](https://doi.org/10.1109/CAC53003.2021.9728514).
- [67] M.-R. Kazemzadeh, A. Amjadian, and T. Amraee, "A hybrid data mining driven algorithm for long term electric peak load and energy demand forecasting," *Energy*, vol. 204, p. 117948, 2020, doi: <https://doi.org/10.1016/j.energy.2020.117948>.
- [68] H. Sangrody, N. Zhou, S. Tutun, B. Khorramdel, M. Motalleb, and M. Sarailoo, *Long Term Forecasting using Machine Learning Methods*. 2018. doi: [10.1109/PECI.2018.8334980](https://doi.org/10.1109/PECI.2018.8334980).
- [69] G. Ozdemir, "Long-term electrical energy demand forecasting by using artificial intelligence/machine learning techniques," *Electrical Engineering*, vol. 106, no. 4, pp. 5229–5251, 2024, doi: [10.1007/s00202-024-02364-1](https://doi.org/10.1007/s00202-024-02364-1).
- [70] J. Faraji, H. Hashemi-Dezaki, and A. Ketabi, "Multi-year load growth-based optimal planning of grid-connected microgrid considering long-term load demand forecasting: A case study of Tehran, Iran," *Sustainable Energy Technologies and Assessments*, vol. 42, Dec. 2020, doi: [10.1016/j.seta.2020.100827](https://doi.org/10.1016/j.seta.2020.100827).
- [71] J. H. Kim, B. S. Lee, and C. H. Kim, "A Study on the development of long-term hybrid electrical load forecasting model based on MLP and statistics using massive actual data considering field applications," *Electric Power Systems Research*, vol. 221, Aug. 2023, doi: [10.1016/j.epsr.2023.109415](https://doi.org/10.1016/j.epsr.2023.109415).
- [72] Z. Q. Dai, J. Li, Y. J. Cao, and Y. X. Zhang, "SALSTM: segmented self-attention long short-term memory for long-term forecasting," *Journal of Supercomputing*, vol. 81, no. 1, Jan. 2025, doi: [10.1007/s11227-024-06493-z](https://doi.org/10.1007/s11227-024-06493-z).
- [73] T. U. Go, Y. J. Go, J. H. Im, and D. Lee, "Long-Term Demand Forecasting: Novel Decomposition-Linear Exponential-Smoothing Algorithm with Long Short-Term Memory," *Journal of Electrical Engineering and Technology*, 2025, doi: [10.1007/s42835-025-02226-1](https://doi.org/10.1007/s42835-025-02226-1).
- [74] T. Quanwei, X. Guijun, and X. Wenju, "Cakformer: Transformer model for long-term heat load forecasting based on Cauto-correlation and KAN," *Energy*, vol. 324, p. 135460, 2025, doi: <https://doi.org/10.1016/j.energy.2025.135460>.
- [75] Y. Yang, Y. Gao, H. Zhou, J. Wu, S. Gao, and Y.-G. Wang, "Multi-Granularity Autoformer for long-term deterministic and probabilistic power load forecasting," *Neural Networks*, vol. 188, p. 107493, 2025, doi: <https://doi.org/10.1016/j.neunet.2025.107493>.

- [76] T. Chen, "A collaborative fuzzy-neural approach for long-term load forecasting in Taiwan," *Comput Ind Eng*, vol. 63, no. 3, pp. 663–670, Nov. 2012, doi: 10.1016/j.cie.2011.06.003.
- [77] D. Ali, M. Yohanna, M. I. Puwu, and B. M. Garkida, "Long-term load forecast modelling using a fuzzy logic approach," *Pacific Science Review A: Natural Science and Engineering*, vol. 18, no. 2, pp. 123–127, Jul. 2016, doi: 10.1016/j.psra.2016.09.011.
- [78] Z. Wen, L. Xie, Q. Fan, and H. Feng, "Long term electric load forecasting based on TS-type recurrent fuzzy neural network model," *Electric Power Systems Research*, vol. 179, Feb. 2020, doi: 10.1016/j.epsr.2019.106106.
- [79] Y. Chen, C. Xiao, S. Yang, Y. Yang, and W. Wang, "Research on long term power load grey combination forecasting based on fuzzy support vector machine," *Computers and Electrical Engineering*, vol. 116, May 2024, doi: 10.1016/j.compeleceng.2024.109205.
- [80] W. Ma, W. Wu, S. F. Ahmed, and G. Liu, "Techno-economic feasibility of utilizing electrical load forecasting in microgrid optimization planning," *Sustainable Energy Technologies and Assessments*, vol. 73, Jan. 2025, doi: 10.1016/j.seta.2024.104135.
- [81] R. Tian, X. Li, Z. Ma, Y. Liu, J. Wang, and C. Wang, "LDformer: a parallel neural network model for long-term power forecasting," *Frontiers of Information Technology and Electronic Engineering*, vol. 24, no. 9, pp. 1287–1301, Sep. 2023, doi: 10.1631/FITEE.2200540.
- [82] F. Zenke and T. P. Vogels, "The Remarkable Robustness of Surrogate Gradient Learning for Instilling Complex Function in Spiking Neural Networks," *Neural Comput*, vol. 33, no. 4, pp. 899–925, Mar. 2021, doi: 10.1162/neco\_a\_01367.
- [83] S. Kulkarni, S. P. Simon, and K. Sundareswaran, "A spiking neural network (SNN) forecast engine for short-term electrical load forecasting," *Appl Soft Comput*, vol. 13, no. 8, pp. 3628–3635, 2013, doi: <https://doi.org/10.1016/j.asoc.2013.04.007>.

## List of Figures

|                                                                                                                             |    |
|-----------------------------------------------------------------------------------------------------------------------------|----|
| Figure 1.1: Electricity consumption by sectors (GWh) [1].                                                                   | 1  |
| Figure 1.2: Benefits of electricity consumption prediction [4]                                                              | 2  |
| Figure 2.1: Load forecasting time horizontal.                                                                               | 4  |
| Figure 2.2: Key factors affecting load forecasting                                                                          | 5  |
| Figure 3.1: SVR model.                                                                                                      | 20 |
| Figure 3.2: Tree-based model.                                                                                               | 22 |
| Figure 3.3: Feedforward and backpropagation architecture of ANN.                                                            | 30 |
| Figure 3.4: Schematic diagram of the LSTM [29].                                                                             | 33 |
| Figure 3.5: Proposed model of GA-optimized SVR [26].                                                                        | 36 |
| Figure 3.6: Framework of extreme weather identification and STLf [16].                                                      | 38 |
| Figure 3.7: A schematic diagram of a CNN-LSTM network based on a multi-modal attention mechanism [32].                      | 39 |
| Figure 3.8: Flow chart of the deep learning framework [34].                                                                 | 41 |
| Figure 4.1: Architecture of XGBoost model [36].                                                                             | 51 |
| Figure 4.2: Framework of the proposed interval forecasting model and principles of NSGA-II, KNN, DBN and modified KDE [45]. | 56 |
| Figure 4.3: Block diagram of the ETS+RD-LSTM forecasting system [53].                                                       | 65 |
| Figure 4.4: The whole structure of HAFF [55].                                                                               | 66 |
| Figure 4.5: The architecture of the proposed hybrid model by Lei Wang et al [59].                                           | 68 |
| Figure 4.6: Block diagram of proposed hybrid model by Shohan et al [61].                                                    | 69 |
| Figure 5.1: Feedforward procedure of neural network for MLP by Kim et al. [71].                                             | 82 |
| Figure 5.2: The self-attention model and its computational process by Dai et al. [72].                                      | 83 |
| Figure 5.3: The multi-granularity Autoformer by Yang et al. [75].                                                           | 85 |
| Figure 5.4: The procedure of the proposed methodology by Toly Chen [76].                                                    | 86 |
| Figure 5.5: Schematic structure of ISO-TS-RBF-RFNN model by Wen et al. [78].                                                | 87 |

Figure 5.6: Flow chart of long-term power load grey combination forecast by Chen et al. [79]. ..... 88

Figure 5.7: Framework for the RES design optimization by Ma et al. [80]. ..... 89

Figure 6.1: Forecasting horizon classification of different methods. .... 93

Figure 6.2: Scalability of forecasting methods based on spatial data availability..... 93

## List of Tables

|                                                                                       |    |
|---------------------------------------------------------------------------------------|----|
| Table 2.1: Common error metrics used in the load forecasting literature.....          | 12 |
| Table 3.1: Strengths and limitations of regression methods in STLF.....               | 17 |
| Table 3.2: Strengths and limitations of ARIMA and SARIMA methods in STLF.....         | 19 |
| Table 3.3: Strengths and limitations of SVR method in STLF.....                       | 22 |
| Table 3.4: Strengths and Limitations of Tree-Based Methods in STLF.....               | 26 |
| Table 3.5: Strengths and limitations of k-NN method in STLF. ....                     | 27 |
| Table 3.6: Strengths and limitations of ANNs in STLF.....                             | 31 |
| Table 3.7: Strengths and limitations of RNNs and LSTMs in STLF. ....                  | 34 |
| Table 3.8: Strengths and Limitations of Transformer-Based Models in STLF.....         | 35 |
| Table 3.9: Benefits of Hybrid models in STLF.....                                     | 42 |
| Table 4.1: Strengths and limitations of regression methods in MTLF.....               | 47 |
| Table 4.2: Strengths and limitations of ARIMA and SARIMA in MTLF.....                 | 49 |
| Table 4.3: Advantages and disadvantages of Tree-based models in MTLF. ....            | 54 |
| Table 4.4: Advantages and disadvantages of K-NN in MTLF.....                          | 57 |
| Table 4.5: Advantages and disadvantages of ANN in MTLF.....                           | 60 |
| Table 4.6: Advantages and disadvantages of RNN and LSTM in MTLF.....                  | 62 |
| Table 4.7: Advantages and disadvantages of transformer-based models in MTLF. ...      | 64 |
| Table 4.8: Benefits and weakness of hybrid models in MTLF. ....                       | 69 |
| Table 5.1: Advantages and disadvantages of regression -based models in LTLF. ....     | 74 |
| Table 5.2: Advantages and disadvantages of ARIMA and SARIMA methods in LTLF.<br>..... | 76 |
| Table 5.3: Advantages and disadvantages of SVR method in LTLF.....                    | 78 |
| Table 5.4: Advantages and disadvantages of tree-based models in LTLF.....             | 79 |
| Table 5.5: Advantages and disadvantages of ANN model in LTLF.....                     | 80 |
| Table 5.6: Advantages and disadvantages of MLP model in LTLF. ....                    | 82 |
| Table 5.7: Advantages and disadvantages of LSTM model in LTLF.....                    | 84 |

|                                                                                 |    |
|---------------------------------------------------------------------------------|----|
| Table 5.8: Advantages and Disadvantages of Transformer-based models in LTLF.... | 85 |
| Table 5.9: Advantages and disadvantages of hybrid models in LTLF. ....          | 90 |



## Acknowledgments

I would like to express my deepest gratitude to my supervisors, Prof. Samuele Grillo, and Prof. Daniele Linaro and for their invaluable guidance, constructive feedback, and continuous support throughout the course of this thesis. Their expertise and encouragement have greatly shaped the quality and direction of my research.

I also extend my thanks to the faculty and staff of the Department of Electronics, Information and Bioengineering at Politecnico Di Milano for providing a stimulating academic environment and essential resources.

Finally, I am deeply grateful to my family and friends for their unwavering encouragement, patience, and love, which sustained me throughout this academic journey.

